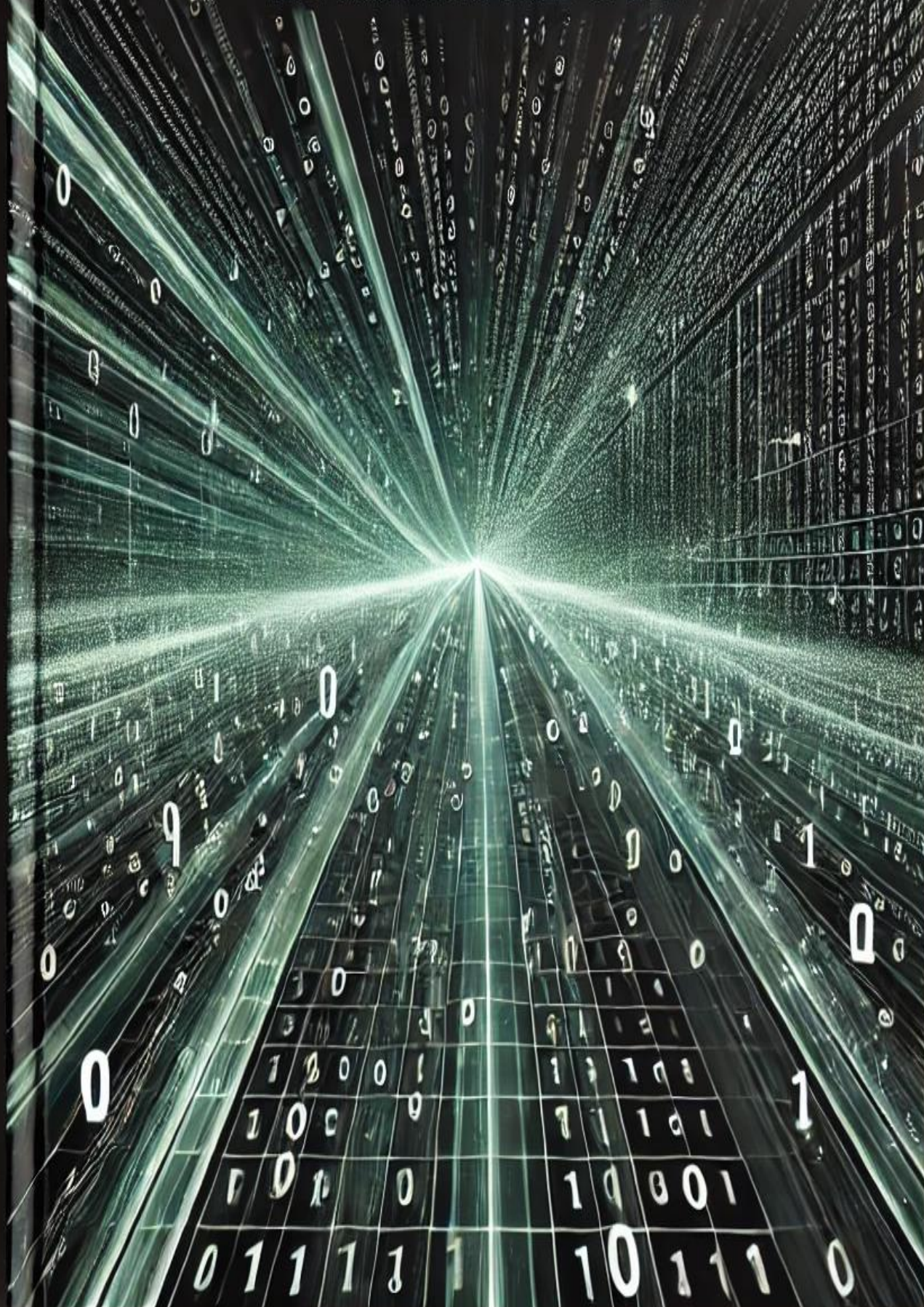


LINEAR ALGEBRA

FOUNDATIONS AND APPLICATIONS

MOAMMAR AHMED RAJEH



CHAPTER 10

Matrices

Chapter Outline

- 10.1 Introduction
- 10.2 Matrix
- 10.3 Some Definitions Associated with Matrices
- 10.4 Elementary Row Operations in Matrix
- 10.5 Row Echelon and Reduced Row Echelon Form of a Matrix
- 10.6 Rank of a Matrix
- 10.7 Inverse of a Matrix by Gauss–Jordan Method
- 10.8 System of Non-homogeneous Linear Equations
- 10.9 System of Homogeneous Linear Equations
- 10.10 Eigenvalues and Eigenvectors
- 10.11 Properties of Eigenvalues
- 10.12 Linear Dependence and Independence of Eigenvectors
- 10.13 Properties of Eigenvectors
- 10.14 Cayley-Hamilton Theorem
- 10.15 Similarity Transformation
- 10.16 Diagonalization of a Matrix

10.1 INTRODUCTION

A matrix is a rectangular table of elements which may be numbers or any abstract quantities that can be added and multiplied. Matrices are used to describe linear equations, keep track of the coefficients of linear transformation and record data that depend on multiple parameters. There are many applications of matrices in mathematics, viz. graph theory, probability theory, statistics, computer graphics, geometrical optics, etc.

10.2 MATRIX

A set of mn elements (real or complex) arranged in a rectangular array of m rows and n columns is called a matrix of order m by n , written as $m \times n$.

An $m \times n$ matrix is usually written as

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ \dots & \dots & \dots & \dots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}_{m \times n}$$

The matrix can also be expressed in the form $A = [a_{ij}]_{m \times n}$, where a_{ij} is the element in the i^{th} row and j^{th} column, written as $(i, j)^{\text{th}}$ element of the matrix.

10.3 SOME DEFINITIONS ASSOCIATED WITH MATRICES

1. Row Matrix

A matrix having only one row and any number of columns, is called a row matrix or row vector, e.g.

$$[2 \ 5 \ -3 \ 4]$$

2. Column Matrix

A matrix, having only one column and any number of rows, is called a column matrix or column vector, e.g.

$$\begin{bmatrix} 1 \\ -3 \\ 4 \end{bmatrix}$$

3. Zero or Null Matrix

A matrix, whose all the elements are zero, is called zero or null matrix and is denoted by $\mathbf{0}$, e.g.

$$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

4. Square Matrix

A matrix, in which the number of rows is equal to the number of columns, is called a square matrix, e.g.

$$\begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix}, \begin{bmatrix} 1 & 3 & 2 \\ -1 & 4 & -5 \\ 2 & 6 & 8 \end{bmatrix}$$

5. Diagonal Matrix

A square matrix, whose all non-diagonal elements are zero and at least one diagonal element is non-zero, is called a diagonal matrix. e.g.

$$\begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix}, \begin{bmatrix} 1 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 8 \end{bmatrix}$$

6. Unit or Identity Matrix

A diagonal matrix, whose all diagonal elements are unity, is called a unit or identity matrix and is denoted by I , e.g.

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

7. Scalar Matrix

A square matrix, whose all diagonal elements are equal, is called a scalar matrix, e.g.

$$\begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}, \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

8. Upper Triangular Matrix

A square matrix, in which all the elements below the diagonal are zero, is called an upper triangular matrix, e.g.

$$\begin{bmatrix} 1 & 3 & 2 \\ 0 & 4 & -5 \\ 0 & 0 & 8 \end{bmatrix}$$

9. Lower Triangular Matrix

A square matrix, in which all the elements above the diagonal are zero, is called a lower triangular matrix, e.g.

$$\begin{bmatrix} 1 & 0 & 0 \\ -1 & 4 & 0 \\ 2 & 6 & 8 \end{bmatrix}$$

10. Trace of a Matrix

The sum of all the diagonal elements of a square matrix is called the trace of a matrix,

$$\text{e.g.} \quad A = \begin{bmatrix} 2 & -1 & 0 \\ 4 & 6 & -2 \\ -1 & 0 & 3 \end{bmatrix}$$

$$\text{Trace of } A = 2 + 6 + 3 = 11$$

11. Transpose of a Matrix

A matrix obtained by interchanging rows and columns of a matrix is called transpose of a matrix and is denoted by A' or A^T , e.g.

$$\text{If } A = \begin{bmatrix} 1 & -1 & 3 \\ 0 & 2 & 6 \\ -4 & 1 & 5 \end{bmatrix} \text{ then } A^T = \begin{bmatrix} 1 & 0 & -4 \\ -1 & 2 & 1 \\ 3 & 6 & 5 \end{bmatrix}$$

i.e., if $A = [a_{ij}]_{m \times n}$, then $A^T = [a_{ji}]_{n \times m}$

12. Symmetric Matrix

A square matrix $A = [a_{ij}]_{m \times m}$ is called symmetric if $a_{ij} = a_{ji}$ for all i and j , i.e., $A = A^T$, e.g.

$$\begin{bmatrix} 2 & 4 \\ 4 & 3 \end{bmatrix}, \begin{bmatrix} 1 & i & -3i \\ i & -2 & 4 \\ -3i & 4 & 3 \end{bmatrix}$$

13. Skew-Symmetric Matrix

A square matrix $A = [a_{ij}]_{m \times m}$ is called skew-symmetric if $a_{ij} = -a_{ji}$ for all i and j , i.e., $A = -A^T$. Thus, the diagonal elements of a skew-symmetric matrix are all zero, e.g.

$$\begin{bmatrix} 0 & -3i & -4 \\ 3i & 0 & 8 \\ 4 & -8 & 0 \end{bmatrix}$$

14. Conjugate of a Matrix

A matrix obtained from any given matrix A , on replacing its elements by the corresponding conjugate complex numbers is called the conjugate of A and is denoted by $\bar{A}$, e.g.

$$A = \begin{bmatrix} 1+3i & 2+5i & 8 \\ -i & 6 & 9-i \end{bmatrix}, \quad \bar{A} = \begin{bmatrix} 1-3i & 2-5i & 8 \\ i & 6 & 9+i \end{bmatrix}$$

15. Transposed Conjugate of a Matrix

The conjugate of the transpose of a matrix A is called the conjugate transpose or transposed conjugate of A and is denoted by A^θ , e.g.

$$A^\theta = (\overline{A})^T = \overline{(A^T)}$$

For example, If $A = \begin{bmatrix} 1-2i & 2+3i & 3+4i \\ 4-5i & 5+6i & 6-7i \\ 8 & 7+8i & 7 \end{bmatrix}$, $A^T = \begin{bmatrix} 1-2i & 4-5i & 8 \\ 2+3i & 5+6i & 7+8i \\ 3+4i & 6-7i & 7 \end{bmatrix}$

Then, $A^\theta = \begin{bmatrix} 1+2i & 4+5i & 8 \\ 2-3i & 5-6i & 7-8i \\ 3-4i & 6+7i & 7 \end{bmatrix}$

16. Hermitian Matrix

A square matrix $A = [a_{ij}]$ is called Hermitian if $a_{ij} = \overline{a_{ji}}$ for all i and j , i.e., $A = A^\theta$, e.g.

$$\begin{bmatrix} 1 & 2+3i & 3-4i \\ 2-3i & 0 & 2-7i \\ 3+4i & 2+7i & 2 \end{bmatrix}$$

17. Skew-Hermitian Matrix

A square matrix $A = [a_{ij}]$ is called skew-Hermitian if $a_{ij} = -\overline{a_{ji}}$ for all i and j , i.e., $A = -A^\theta$. Hence, diagonal elements of a skew-Hermitian matrix must be either purely imaginary or zero, e.g.

$$\begin{bmatrix} i & 2+3i \\ -2+3i & 0 \end{bmatrix}$$

18. Orthogonal Matrix

A square matrix A is called orthogonal if $AA^T = I$.

19. Unitary Matrix

A square matrix A is called unitary if $AA^\theta = I$.

20. Determinant of a Matrix

If A is a square matrix, determinant of A is represented as $|A|$.

If $A = \begin{bmatrix} 2 & 3 & 1 \\ 1 & 2 & 3 \\ 1 & 1 & 0 \end{bmatrix}$, then $|A| = \begin{vmatrix} 2 & 3 & 1 \\ 1 & 2 & 3 \\ 1 & 1 & 0 \end{vmatrix}$

21. Singular and Non-Singular Matrices

A square matrix A is called singular if $|A| = 0$ and non-singular if $|A| \neq 0$.

22. Minor of an Element of a Determinant

If $|A| = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$, minor of an element of a determinant is a determinant

obtained by leaving the row and column passing through that element, e.g.,

minor of element $a_{11} = \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix}$, minor of element $a_{12} = \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix}$,

minor of element $a_{13} = \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$

23. Cofactor of an Element of a Determinant

Cofactor of an element a_{ij} of a determinant is the minor multiplied by $(-1)^{i+j}$, e.g.,

Cofactor of the element $a_{11} = (-1)^{1+1} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix}$

Cofactor of the element $a_{12} = (-1)^{1+2} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix}$

Cofactor of the element $a_{13} = (-1)^{1+3} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$

24. Inverse of a Matrix

If A is a square matrix and $|A| \neq 0$,

$$AA^{-1} = I = A^{-1}A$$

where, A^{-1} is called inverse of the matrix A .

10.4 ELEMENTARY ROW OPERATIONS IN MATRIX

Elementary transformation is any one of the following operations on a matrix.

- (i) The interchange of any two rows (or columns)
- (ii) The multiplication of the elements of any row (or column) by any non-zero number
- (iii) The addition or subtraction of k items the elements of a row (or column) to the corresponding elements of another row (or column), where $k \neq 0$

Symbols to be used for elementary transformation:

- (i) R_{ij} : Interchange of i^{th} and j^{th} row

- (ii) kR_i : Multiplication of i^{th} row by a non zero number k
 (iii) $R_i + kR_j$: Addition of k times the j^{th} row to the i^{th} row

The corresponding column transformations are denoted by C_{ij} , kC_i and $C_i + kC_j$ respectively.

10.4.1 Elementary Matrices

A matrix obtained from a unit matrix by subjecting it to any row or column transformation is called an elementary matrix.

10.4.2 Equivalence of Matrices

If B be an $m \times n$ matrix obtained from an $m \times n$ matrix by elementary transformation of A , then A is called the equivalent to B . Symbolically, we can write $A \sim B$.

10.5 ROW ECHELON AND REDUCED ROW ECHELON FORMS OF A MATRIX

A matrix A is said to be in row echelon form if it satisfies the following properties:

- Every zero row of the matrix A occurs below a non-zero row.
- The first non-zero number from the left of a non-zero row is a 1. This is called a leading 1.
- For each non-zero row, the leading 1 appears to the right and below any leading 1 in the preceding rows.

The following matrices are in row echelon form.

$$\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 2 & -1 & 3 \\ 0 & 1 & 5 & 6 \\ 0 & 0 & 1 & 4 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 3 & 5 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

A matrix A is said to be in reduced row echelon form if each column that contains a leading 1 in row echelon form of the matrix A has zeros everywhere else in that column.

The following matrices are in reduced row echelon form.

$$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 5 \\ 0 & 0 & 1 & -1 \end{bmatrix}, \begin{bmatrix} 0 & 1 & -4 & 0 & 1 \\ 0 & 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Example 1

In each part determine whether the matrix is in row echelon form, reduced row echelon form, both or neither.

$$(i) \begin{bmatrix} 1 & 2 & 0 & 3 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$(ii) \begin{bmatrix} 1 & 0 & 0 & 5 \\ 0 & 0 & 1 & 2 \\ 0 & 1 & 0 & 7 \end{bmatrix}$$

$$(iii) \begin{bmatrix} 1 & -6 & 4 & 3 \\ 0 & 1 & 3 & 2 \end{bmatrix}$$

$$(iv) \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 2 & -3 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Solution

- (i) The given matrix is in reduced row echelon form and row echelon form since it satisfies properties (i), (ii), (iii) and columns containing leading 1 have zero everywhere else.
- (ii) The given matrix is neither in row echelon form nor in reduced row echelon form since it does not satisfy the property (iii).
- (iii) The given matrix is in row echelon form since it satisfies properties (i), (ii) and (iii).
- (iv) The given matrix is neither in row echelon form nor in reduced row echelon form since it does not satisfy the property (i).

Example 2

Find a row echelon form of the following matrix:

$$\begin{bmatrix} 0 & -1 & 2 & 3 \\ 2 & 3 & 4 & 5 \\ 1 & 3 & -1 & 2 \\ 3 & 2 & 4 & 1 \end{bmatrix}$$

Solution

$$\begin{bmatrix} 0 & -1 & 2 & 3 \\ 2 & 3 & 4 & 5 \\ 1 & 3 & -1 & 2 \\ 3 & 2 & 4 & 1 \end{bmatrix} \xrightarrow{R_{13}} \begin{bmatrix} 1 & 3 & -1 & 2 \\ 2 & 3 & 4 & 5 \\ 0 & -1 & 2 & 3 \\ 3 & 2 & 4 & 1 \end{bmatrix}$$

$$R_2 - 2R_1, R_4 - 3R_1$$

$$\sim \begin{bmatrix} 1 & 3 & -1 & 2 \\ 0 & -3 & 6 & 1 \\ 0 & -1 & 2 & 3 \\ 0 & -7 & 7 & -5 \end{bmatrix}$$

$$R_{23}$$

$$\sim \begin{bmatrix} 1 & 3 & -1 & 2 \\ 0 & -1 & 2 & 3 \\ 0 & -3 & 6 & 1 \\ 0 & -7 & 7 & -5 \end{bmatrix}$$

$$(-1)R_2$$

$$\sim \begin{bmatrix} 1 & 3 & -1 & 2 \\ 0 & 1 & -2 & -3 \\ 0 & -3 & 6 & 1 \\ 0 & -7 & 7 & -5 \end{bmatrix}$$

$$R_3 + 3R_2, R_4 + 7R_2$$

$$\sim \begin{bmatrix} 1 & 3 & -1 & 2 \\ 0 & 1 & -2 & -3 \\ 0 & 0 & 0 & -8 \\ 0 & 0 & -7 & -26 \end{bmatrix}$$

$$R_{34}$$

$$\sim \begin{bmatrix} 1 & 3 & -1 & 2 \\ 0 & 1 & -2 & -3 \\ 0 & 0 & -7 & -26 \\ 0 & 0 & 0 & -8 \end{bmatrix}$$

$$\left(-\frac{1}{7}\right)R_3$$

$$\sim \begin{bmatrix} 1 & 3 & -1 & 2 \\ 0 & 1 & -2 & -3 \\ 0 & 0 & 1 & \frac{26}{7} \\ 0 & 0 & 0 & -8 \end{bmatrix}$$

$$\left(-\frac{1}{8}\right)R_4$$

$$\sim \begin{bmatrix} 1 & 3 & -1 & 2 \\ 0 & 1 & -2 & -3 \\ 0 & 0 & 1 & \frac{26}{7} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Example 3

Find a row echelon form of the following matrix:

$$\begin{bmatrix} 1 & 2 & -3 & 1 \\ -1 & 0 & 3 & 4 \\ 0 & 1 & 2 & -1 \\ 2 & 3 & 0 & -3 \end{bmatrix}$$

Solution

$$\begin{bmatrix} 1 & 2 & -3 & 1 \\ -1 & 0 & 3 & 4 \\ 0 & 1 & 2 & -1 \\ 2 & 3 & 0 & -3 \end{bmatrix}$$

$$R_2 + R_1, \quad R_4 - 2R_1$$

$$\sim \begin{bmatrix} 1 & 2 & -3 & 1 \\ 0 & 2 & 0 & 5 \\ 0 & 1 & 2 & -1 \\ 0 & -1 & 6 & -5 \end{bmatrix}$$

$$R_{23}$$

$$\sim \begin{bmatrix} 1 & 2 & -3 & 1 \\ 0 & 1 & 2 & -1 \\ 0 & 2 & 0 & 5 \\ 0 & -1 & 6 & -5 \end{bmatrix}$$

$$R_3 - 2R_2, \quad R_4 + R_2$$

$$\sim \begin{bmatrix} 1 & 2 & -3 & 1 \\ 0 & 1 & 2 & -1 \\ 0 & 0 & -4 & 7 \\ 0 & 0 & 8 & -6 \end{bmatrix}$$

$$\left(-\frac{1}{4}\right)R_3$$
$$\sim \begin{bmatrix} 1 & 2 & -3 & 1 \\ 0 & 1 & 2 & -1 \\ 0 & 0 & 1 & -\frac{7}{4} \\ 0 & 0 & 8 & -6 \end{bmatrix}$$

$$R_4 - 8R_3$$
$$\sim \begin{bmatrix} 1 & 2 & -3 & 1 \\ 0 & 1 & 2 & -1 \\ 0 & 0 & 1 & -\frac{7}{4} \\ 0 & 0 & 0 & 8 \end{bmatrix}$$

$$\left(\frac{1}{8}\right)R_4$$
$$\sim \begin{bmatrix} 1 & 2 & -3 & 1 \\ 0 & 1 & 2 & -1 \\ 0 & 0 & 1 & -\frac{7}{4} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Example 4

Find the reduced row echelon form of the following matrix:

$$\begin{bmatrix} 0 & -1 & 2 & 3 \\ 2 & 3 & 4 & 5 \\ 1 & 3 & -1 & 2 \\ 3 & 2 & 4 & 1 \end{bmatrix}$$

Solution

The row echelon form of the matrix is

$$\begin{bmatrix} 1 & 3 & -1 & 2 \\ 0 & 1 & -2 & -3 \\ 0 & 0 & 1 & \frac{26}{7} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Beginning with the last non-zero row and working upward, we add suitable multiples of each row to the rows above to introduce zeros above the leading 1's.

$$R_3 - \frac{26}{7}R_4, R_2 + 3R_4, R_1 - 2R_4$$

$$\sim \begin{bmatrix} 1 & 3 & -1 & 0 \\ 0 & 1 & -2 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$R_2 + 2R_3, R_1 + R_3$$

$$\sim \begin{bmatrix} 1 & 3 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$R_1 - 3R_2$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Example 5

Find the reduced row echelon form of the following matrix:

$$\begin{bmatrix} 1 & 2 & -3 & 1 \\ -1 & 0 & 3 & 4 \\ 0 & 1 & 2 & -1 \\ 2 & 3 & 0 & -3 \end{bmatrix}$$

Solution

The row echelon form of the matrix is

$$\begin{bmatrix} 1 & 2 & -3 & 1 \\ 0 & 1 & 2 & -1 \\ 0 & 0 & 1 & -\frac{7}{4} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Beginning with the last non-zero row and working upward, we add suitable multiples of each row to the rows above to introduce zeros above the leading 1's.

$$R_3 + \frac{7}{4}R_4, \quad R_2 + R_4, \quad R_1 - R_4$$

$$\sim \begin{bmatrix} 1 & 2 & -3 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$R_2 - 2R_3, \quad R_1 + 3R_3$$

$$\sim \begin{bmatrix} 1 & 2 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$R_1 - 2R_2$$

$$\sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

10.6 RANK OF A MATRIX

The positive integer r is said to be the rank of a matrix A if it possesses the following properties:

- (i) There is at least one minor of order r which is nonzero.
- (ii) Every minor of order greater than r is zero.

Rank of matrix A is denoted by $\rho(A)$.

Note (1): The rank of a matrix remains unchanged by elementary transformations.

Note (2): The rank of the transpose of a matrix is same as that of the original matrix.

Note (3): The rank of the product of two matrices cannot exceed the rank of either matrix.

$$\rho(AB) \leq \rho(A) \quad \text{or} \quad \rho(AB) \leq \rho(B)$$

10.6.1 Rank of a Matrix by Echelon Form

The rank of a matrix in echelon form is equal to the number of nonzero rows of the matrix,

e.g.,

$$A = \begin{bmatrix} 1 & 3 & -1 \\ 0 & 1 & 4 \\ 0 & 0 & 0 \end{bmatrix}$$

The matrix A is in echelon form and the number of nonzero rows is two. Hence, rank of the matrix is two,

i.e.,

$$\rho(A) = 2$$

Example 1

Find the rank of the following matrix by reducing it to echelon form:

$$\begin{bmatrix} 5 & 3 & 14 & 4 \\ 0 & 1 & 2 & 1 \\ 1 & -1 & 2 & 0 \end{bmatrix}$$

Solution

Let

$$A = \begin{bmatrix} 5 & 3 & 14 & 4 \\ 0 & 1 & 2 & 1 \\ 1 & -1 & 2 & 0 \end{bmatrix}$$

Reducing the matrix A to echelon form,

$$\begin{array}{c} R_{13} \\ \sim \end{array} \begin{bmatrix} 1 & -1 & 2 & 0 \\ 0 & 1 & 2 & 1 \\ 5 & 3 & 14 & 4 \end{bmatrix}$$

$$\begin{array}{c} R_3 - 5R_1 \\ \sim \end{array} \begin{bmatrix} 1 & -1 & 2 & 0 \\ 0 & 1 & 2 & 1 \\ 0 & 8 & 4 & 4 \end{bmatrix}$$

$$\begin{aligned}
 & R_3 - 8R_2 \\
 & \sim \begin{bmatrix} 1 & -1 & 2 & 0 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & -12 & -4 \end{bmatrix} \\
 & \left(-\frac{1}{12}\right)R_3 \\
 & \sim \begin{bmatrix} 1 & -1 & 2 & 0 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 1 & \frac{1}{3} \end{bmatrix}
 \end{aligned}$$

The equivalent matrix is in echelon form.

Number of nonzero rows = 3

$$\rho(A) = 3$$

Example 2

Find the rank of the following matrix by reducing it to echelon form:

$$\begin{bmatrix} 1 & 2 & 3 & -1 \\ -2 & -1 & -3 & -1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & -1 \end{bmatrix}$$

Solution

Let

$$A = \begin{bmatrix} 1 & 2 & 3 & -1 \\ -2 & -1 & -3 & -1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & -1 \end{bmatrix}$$

Reducing the matrix A to echelon form,

$$\begin{aligned}
 & R_2 + 2R_1, R_3 - R_1 \\
 & \sim \begin{bmatrix} 1 & 2 & 3 & -1 \\ 0 & 3 & 3 & -3 \\ 0 & -2 & -2 & 2 \\ 0 & 1 & 1 & -1 \end{bmatrix}
 \end{aligned}$$

$$\begin{array}{c}
R_{24} \\
\sim \begin{bmatrix} 1 & 2 & 3 & -1 \\ 0 & 1 & 1 & -1 \\ 0 & -2 & -2 & 2 \\ 0 & 3 & 3 & -3 \end{bmatrix} \\
R_3 + 2R_2, \quad R_4 - 3R_2 \\
\sim \begin{bmatrix} 1 & 2 & 3 & -1 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}
\end{array}$$

The equivalent matrix is in echelon form.

Number of nonzero rows = 2

$$\rho(A) = 2$$

Example 3

Find the rank of the following matrix by reducing it to echelon form:

$$\begin{bmatrix} 3 & -2 & 0 & -1 \\ 0 & 2 & 2 & 1 \\ 1 & -2 & -3 & 2 \\ 0 & 1 & 2 & 1 \end{bmatrix}$$

Solution

$$\text{Let } A = \begin{bmatrix} 3 & -2 & 0 & -1 \\ 0 & 2 & 2 & 1 \\ 1 & -2 & -3 & 2 \\ 0 & 1 & 2 & 1 \end{bmatrix}$$

Reducing the matrix A to echelon form,

$$\begin{array}{c}
R_{13} \\
\sim \begin{bmatrix} 1 & -2 & -3 & 2 \\ 0 & 2 & 2 & 1 \\ 3 & -2 & 0 & -1 \\ 0 & 1 & 2 & 1 \end{bmatrix} \\
R_3 - 3R_1, \quad R_{24} \\
\sim \begin{bmatrix} 1 & -2 & -3 & 2 \\ 0 & 1 & 2 & 1 \\ 0 & 4 & 9 & -7 \\ 0 & 2 & 2 & 1 \end{bmatrix}
\end{array}$$

$$R_3 - 4R_2, R_4 - 2R_2$$

$$\sim \begin{bmatrix} 1 & -2 & -3 & 2 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 1 & -11 \\ 0 & 0 & -2 & -1 \end{bmatrix}$$

$$R_4 + 2R_3$$

$$\sim \begin{bmatrix} 1 & -2 & -3 & 2 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 1 & -11 \\ 0 & 0 & 0 & -23 \end{bmatrix}$$

$$\left(-\frac{1}{23}\right)R_4$$

$$\sim \begin{bmatrix} 1 & -2 & -3 & 2 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 1 & -11 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

The equivalent matrix is in echelon form.

Number of nonzero rows = 4

$$\rho(A) = 4$$

EXERCISE 10.2

1. Find the ranks of the following matrices by reducing them to echelon forms:

$$(i) \begin{bmatrix} 1 & 1 & -1 & 1 \\ 1 & -1 & 2 & -1 \\ 3 & 1 & 0 & 1 \end{bmatrix}$$

$$(ii) \begin{bmatrix} 4 & 2 & 3 \\ 8 & 4 & 6 \\ -2 & -1 & -1.5 \end{bmatrix}$$

$$(iii) \begin{bmatrix} 1 & 2 & -2 & 3 \\ 2 & 5 & -4 & 6 \\ -1 & -3 & 2 & -2 \\ 2 & 4 & -1 & 6 \end{bmatrix}$$

$$(iv) \begin{bmatrix} 0 & 1 & -3 & -1 \\ 0 & 0 & 1 & 1 \\ 3 & 1 & 0 & 2 \\ 1 & 1 & 2 & 0 \end{bmatrix}$$

$$(v) \begin{bmatrix} 3 & -2 & 0 & -1 & -7 \\ 0 & 2 & 2 & 1 & -5 \\ 1 & -2 & -3 & -2 & 1 \\ 0 & 1 & 2 & 1 & 6 \end{bmatrix}$$

$$(vi) \begin{bmatrix} 1 & 2 & -1 & 4 \\ 2 & 4 & 3 & 5 \\ -1 & -2 & 6 & 7 \end{bmatrix}$$

[Ans. : (i) 2 (ii) 1 (iii) 4 (iv) 2 (v) 4 (vi) 2]

10.7 INVERSE OF A MATRIX BY GAUSS-JORDAN METHOD

Let A be any non-singular matrix. Then $A = IA$. Applying suitable elementary row transformation to A on the L.H.S and to I on the R.H.S, of a matrix A reduces to I and I reduces to any matrix B .

Hence, $I = BA$

$$B = A^{-1}$$

Example 1

Find the inverse of the following matrix by elementary transformations (Gauss–Jordan method):

$$\begin{bmatrix} 2 & 3 & 4 \\ 4 & 3 & 1 \\ 1 & 2 & 4 \end{bmatrix}$$

Solution

Let
$$A = \begin{bmatrix} 2 & 3 & 4 \\ 4 & 3 & 1 \\ 1 & 2 & 4 \end{bmatrix}$$

$$A = I_3 A$$

$$\begin{bmatrix} 2 & 3 & 4 \\ 4 & 3 & 1 \\ 1 & 2 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A$$

Reducing the matrix A to reduced row echelon form,

$$R_{13}$$

$$\begin{bmatrix} 1 & 2 & 4 \\ 4 & 3 & 1 \\ 2 & 3 & 4 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} A$$

$$R_2 - 4R_1, R_3 - 2R_1$$

$$\begin{bmatrix} 1 & 2 & 4 \\ 0 & -5 & -15 \\ 0 & -1 & -4 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & -4 \\ 1 & 0 & -2 \end{bmatrix} A$$

$$\left(-\frac{1}{5}\right)R_2$$

$$\begin{bmatrix} 1 & 2 & 4 \\ 0 & 1 & 3 \\ 0 & -1 & -4 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & -\frac{1}{5} & \frac{4}{5} \\ 1 & 0 & -2 \end{bmatrix} A$$

$$R_3 + R_2$$

$$\begin{bmatrix} 1 & 2 & 4 \\ 0 & 1 & 3 \\ 0 & 0 & -1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & -\frac{1}{5} & \frac{4}{5} \\ 1 & -\frac{1}{5} & -\frac{6}{5} \end{bmatrix} A$$

$$(-1)R_3$$

$$\begin{bmatrix} 1 & 2 & 4 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & -\frac{1}{5} & \frac{4}{5} \\ -1 & \frac{1}{5} & \frac{6}{5} \end{bmatrix} A$$

$$R_2 - 3R_3, R_1 - 4R_3$$

$$\begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 4 & -\frac{4}{5} & -\frac{19}{5} \\ 3 & -\frac{4}{5} & -\frac{14}{5} \\ -1 & \frac{1}{5} & \frac{6}{5} \end{bmatrix} A$$

$$R_1 - 2R_2$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -2 & \frac{4}{5} & \frac{9}{5} \\ 3 & -\frac{4}{5} & -\frac{14}{5} \\ -1 & \frac{1}{5} & \frac{6}{5} \end{bmatrix} A$$

$$I_3 = A^{-1}A$$

$$\therefore A^{-1} = \begin{bmatrix} -2 & \frac{4}{5} & \frac{9}{5} \\ 3 & -\frac{4}{5} & -\frac{14}{5} \\ -1 & \frac{1}{5} & \frac{6}{5} \end{bmatrix} = \frac{1}{5} \begin{bmatrix} -10 & 4 & 9 \\ 15 & -4 & -14 \\ -5 & 1 & 6 \end{bmatrix}$$

Example 2

Find the inverse of the following matrix by elementary transformations (Gauss–Jordan method):

$$\begin{bmatrix} 1 & -1 & 0 & 2 \\ 0 & 1 & 1 & -1 \\ 2 & 1 & 2 & 1 \\ 3 & -2 & 1 & 6 \end{bmatrix}$$

Solution

Let
$$A = \begin{bmatrix} 1 & -1 & 0 & 2 \\ 0 & 1 & 1 & -1 \\ 2 & 1 & 2 & 1 \\ 3 & -2 & 1 & 6 \end{bmatrix}$$

$$A = I_4 A$$

$$\begin{bmatrix} 1 & -1 & 0 & 2 \\ 0 & 1 & 1 & -1 \\ 2 & 1 & 2 & 1 \\ 3 & -2 & 1 & 6 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} A$$

Reducing the matrix A to reduced row echelon form,

$$R_3 - 2R_1, R_4 - 3R_1$$

$$\begin{bmatrix} 1 & -1 & 0 & 2 \\ 0 & 1 & 1 & -1 \\ 0 & 3 & 2 & -3 \\ 0 & 1 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ -2 & 0 & 1 & 0 \\ -3 & 0 & 0 & 1 \end{bmatrix} A$$

$$R_3 - 3R_2, R_4 - R_2$$

$$\begin{bmatrix} 1 & -1 & 0 & 2 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ -2 & -3 & 1 & 0 \\ -3 & -1 & 0 & 1 \end{bmatrix} A$$

$$(-1)R_3$$

$$\begin{bmatrix} 1 & -1 & 0 & 2 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 2 & 3 & -1 & 0 \\ -3 & -1 & 0 & 1 \end{bmatrix} A$$

$$R_2 + R_4, R_1 - 2R_4$$

$$\begin{bmatrix} 1 & -1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 7 & 2 & 0 & -2 \\ -3 & 0 & 0 & 1 \\ 2 & 3 & -1 & 0 \\ -3 & -1 & 0 & 1 \end{bmatrix} A$$

$$R_2 - R_3$$

$$\begin{bmatrix} 1 & -1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 7 & 2 & 0 & -2 \\ -5 & -3 & 1 & 1 \\ 2 & 3 & -1 & 0 \\ -3 & -1 & 0 & 1 \end{bmatrix} A$$

$$R_1 + R_2$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & -1 & 1 & -1 \\ -5 & -3 & 1 & 1 \\ 2 & 3 & -1 & 0 \\ -3 & -1 & 0 & 1 \end{bmatrix} A$$

$$I_4 = A^{-1}A$$

$$\therefore A^{-1} = \begin{bmatrix} 2 & -1 & 1 & -1 \\ -5 & -3 & 1 & 1 \\ 2 & 3 & -1 & 0 \\ -3 & -1 & 0 & 1 \end{bmatrix}$$

EXERCISE 10.3

1. Using elementary row transformations, find the inverses of the following matrices:

(i) $\begin{bmatrix} 8 & 4 & 3 \\ 2 & 1 & 1 \\ 1 & 2 & 1 \end{bmatrix}$

(ii) $\begin{bmatrix} 5 & -1 & 5 \\ 0 & 2 & 0 \\ -5 & 3 & -15 \end{bmatrix}$

(iii) $\begin{bmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix}$

(iv) $\begin{bmatrix} 1 & -1 & 1 \\ 2 & 2 & -3 \\ 1 & -4 & 9 \end{bmatrix}$

(v) $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 0 \\ 0 & 1 & 2 \end{bmatrix}$

(vi) $\begin{bmatrix} 2 & 4 & 3 & 2 \\ 3 & 6 & 5 & 2 \\ 2 & 5 & 2 & -3 \\ 4 & 5 & 14 & 14 \end{bmatrix}$

$$(vii) \begin{bmatrix} 0 & 1 & 2 & 2 \\ 1 & 1 & 2 & 3 \\ 2 & 2 & 2 & 3 \\ 2 & 3 & 3 & 3 \end{bmatrix}$$

$$(viii) \begin{bmatrix} -1 & -3 & 3 & -1 \\ 1 & 1 & -1 & 0 \\ 2 & -5 & 2 & -3 \\ -1 & 1 & 0 & 1 \end{bmatrix}$$

$$(ix) \begin{bmatrix} 2 & -6 & -2 & -3 \\ 5 & -13 & -4 & -7 \\ -1 & 4 & 1 & 2 \\ 0 & 1 & 0 & 1 \end{bmatrix}$$

$$(x) \begin{bmatrix} 2 & 0 & 0 & -4 \\ 2 & 6 & 0 & -16 \\ 1 & 0 & 3 & -5 \\ -2 & 0 & 0 & 10 \end{bmatrix}$$

Ans.:

$$(i) \frac{1}{3} \begin{bmatrix} 1 & -2 & -1 \\ 1 & -5 & 2 \\ -3 & 12 & 0 \end{bmatrix}$$

$$(ii) \frac{1}{5} \begin{bmatrix} -10 & 4 & 9 \\ 15 & -4 & -14 \\ -5 & 1 & 6 \end{bmatrix}$$

$$(iii) \begin{bmatrix} 1 & -1 & 0 \\ -2 & 3 & -4 \\ -2 & 3 & 3 \end{bmatrix}$$

$$(iv) \frac{1}{17} \begin{bmatrix} 6 & 5 & 1 \\ -21 & 8 & 5 \\ -10 & 3 & 4 \end{bmatrix}$$

$$(v) \frac{1}{4} \begin{bmatrix} 6 & -1 & -9 \\ -4 & 2 & 6 \\ 2 & -1 & -1 \end{bmatrix}$$

$$(vi) \begin{bmatrix} -23 & 29 & -\frac{64}{5} & -\frac{18}{5} \\ 10 & -12 & \frac{26}{5} & \frac{7}{5} \\ 1 & -2 & \frac{6}{5} & \frac{2}{5} \\ 2 & -2 & \frac{3}{5} & \frac{1}{5} \end{bmatrix}$$

$$(vii) \begin{bmatrix} -3 & 3 & -3 & 2 \\ 3 & -4 & 4 & -2 \\ -3 & 4 & -5 & 3 \\ 2 & -2 & 3 & -2 \end{bmatrix}$$

$$(viii) \begin{bmatrix} 0 & 2 & 1 & 3 \\ 1 & 1 & -1 & -2 \\ 1 & 2 & 0 & 1 \\ -1 & 1 & 2 & 6 \end{bmatrix}$$

$$(ix) \begin{bmatrix} -2 & 1 & 0 & 1 \\ 1 & 0 & 2 & -1 \\ -4 & 1 & -3 & 1 \\ -1 & 0 & -2 & 2 \end{bmatrix}$$

$$(x) \begin{bmatrix} 5 & 0 & 0 & 2 \\ 1 & 1 & 0 & 2 \\ 0 & 0 & 2 & 1 \\ 1 & 0 & 0 & 1 \end{bmatrix}$$

10.8 SYSTEM OF NON-HOMOGENEOUS LINEAR EQUATIONS

A system of m non-homogeneous linear equations in n variables $x_1, x_2, \dots, x_n$ or simply a linear system, is a set of m linear equations, each in n variables. A linear system is represented by

$$\begin{aligned}
a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\
a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\
&\vdots \quad \quad \quad \vdots \quad \quad \quad \vdots \quad \quad \quad \vdots \\
a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m
\end{aligned}$$

Writing these equations in matrix form,

$$A\mathbf{x} = B$$

where $A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$ is called coefficient matrix of order $m \times n$,

$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$ is any vector of order $n \times 1$.

$B = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$ is any vector of order $m \times 1$.

10.8.1 Solutions of System of Linear Equations: Gauss Elimination and Gauss–Jordan Methods

For a system of m linear equations in n variables, there are three possibilities of the solutions to the system:

- (i) The system has unique solution.
- (ii) The system has infinite solutions.
- (iii) The system has no solution.

When the system of linear equations has one or more solutions, the system is said to be consistent, otherwise it is inconsistent.

The matrix $[A : B] = \left[\begin{array}{cccc|c} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{array} \right]$

is called the augmented matrix of the given system of linear equations.

To solve a system of linear equations, elementary transformations are used to reduce the augmented matrix to either row echelon form or reduced row echelon form.

Reducing the augmented matrix to row echelon form is called Gauss elimination method. Reducing the augmented matrix to reduced row echelon form is called Gauss–Jordan method.

The Gauss elimination method for solving the linear system is as follows:

Step 1: Write the augmented matrix.

Step 2: Obtain the row echelon form of the augmented matrix by using elementary row operations.

Step 3: Write the corresponding linear system of equations from row echelon form.

Step 4: Solve the corresponding linear system of equations by back substitution.

The Gauss–Jordan method for solving the linear system is as follows:

Step 1: Write the augmented matrix.

Step 2: Obtain the reduced row echelon form of the augmented matrix by using elementary row operations.

Step 3: For each non-zero row of the matrix, solve the corresponding system of equations for the variables associated with the leading one in that row.

Note: The linear system has a unique solution if $\det(A) \neq 0$

Example 1

Solve the following system by Gauss elimination method:

$$\begin{aligned}x + y + 2z &= 9 \\2x + 4y - 3z &= 1 \\3x + 6y - 5z &= 0\end{aligned}$$

Solution

The matrix form of the system is

$$\begin{bmatrix} 1 & 1 & 2 \\ 2 & 4 & -3 \\ 3 & 6 & -5 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 9 \\ 1 \\ 0 \end{bmatrix}$$

The augmented matrix of the system is

$$\left[\begin{array}{ccc|c} 1 & 1 & 2 & 9 \\ 2 & 4 & -3 & 1 \\ 3 & 6 & -5 & 0 \end{array} \right]$$

Reducing the augmented matrix to row echelon form,

$$\begin{aligned}
 & R_2 - 2R_1, \quad R_3 - 3R_1 \\
 & \sim \left[\begin{array}{ccc|c} 1 & 1 & 2 & 9 \\ 0 & 2 & -7 & -17 \\ 0 & 3 & -11 & -27 \end{array} \right] \\
 & \left(\frac{1}{2} \right) R_2 \\
 & \sim \left[\begin{array}{ccc|c} 1 & 1 & 2 & 9 \\ 0 & 1 & -\frac{7}{2} & -\frac{17}{2} \\ 0 & 3 & -11 & -27 \end{array} \right] \\
 & R_3 - 3R_2 \\
 & \sim \left[\begin{array}{ccc|c} 1 & 1 & 2 & 9 \\ 0 & 1 & -\frac{7}{2} & -\frac{17}{2} \\ 0 & 0 & -\frac{1}{2} & -\frac{3}{2} \end{array} \right] \\
 & (-2)R_3 \\
 & \sim \left[\begin{array}{ccc|c} 1 & 1 & 2 & 9 \\ 0 & 1 & -\frac{7}{2} & -\frac{17}{2} \\ 0 & 0 & 1 & 3 \end{array} \right]
 \end{aligned}$$

The corresponding system of equations is

$$\begin{aligned}
 x + y + 2z &= 9 \\
 y - \frac{7}{2}z &= -\frac{17}{2} \\
 z &= 3
 \end{aligned}$$

Solving these equations,

$$x = 1, y = 2$$

Hence, $x = 1, y = 2, z = 3$ is the solution of the system.

Example 2

Solve the following system by Gauss elimination method:

$$4x - 2y + 6z = 8$$

$$x + y - 3z = -1$$

$$15x - 3y + 9z = 21$$

Solution

The matrix form of the system is

$$\begin{bmatrix} 4 & -2 & 6 \\ 1 & 1 & -3 \\ 15 & -3 & 9 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 8 \\ -1 \\ 21 \end{bmatrix}$$

The augmented matrix of the system is

$$\left[\begin{array}{ccc|c} 4 & -2 & 6 & 8 \\ 1 & 1 & -3 & -1 \\ 15 & -3 & 9 & 21 \end{array} \right]$$

Reducing the augmented matrix to row echelon form,

$$\begin{array}{l} R_{12} \\ \sim \left[\begin{array}{ccc|c} 1 & 1 & -3 & -1 \\ 4 & -2 & 6 & 8 \\ 15 & -3 & 9 & 21 \end{array} \right] \end{array}$$

$$\begin{array}{l} R_2 - 4R_1, R_3 - 15R_1 \\ \sim \left[\begin{array}{ccc|c} 1 & 1 & -3 & -1 \\ 0 & -6 & 18 & 12 \\ 0 & -18 & 54 & 36 \end{array} \right] \end{array}$$

$$\begin{array}{l} \left(-\frac{1}{6}\right)R_2, \left(-\frac{1}{18}\right)R_3 \\ \sim \left[\begin{array}{ccc|c} 1 & 1 & -3 & -1 \\ 0 & 1 & -3 & -2 \\ 0 & 1 & -3 & -2 \end{array} \right] \end{array}$$

$$\begin{array}{l} R_3 - R_2 \\ \sim \left[\begin{array}{ccc|c} 1 & 1 & -3 & -1 \\ 0 & 1 & -3 & -2 \\ 0 & 0 & 0 & 0 \end{array} \right] \end{array}$$

The corresponding system of equations is

$$\begin{aligned}x + y - 3z &= -1 \\ y - 3z &= -2\end{aligned}$$

The leading ones are in columns 1 and 2. Hence, the variables x and y are called leading variables whereas the variable z is called a free variable. Assigning the free variable z an arbitrary value t ,

$$\begin{aligned}y &= 3t - 2 \\ x &= -1 - 3t + 2 + 3t = 1\end{aligned}$$

Hence, $x = 1$, $y = 3t - 2$, $z = t$ is the solution of the system where t is a parameter.

Example 3

Solve the following system by Gauss elimination method:

$$\begin{aligned}3x + y - 3z &= 13 \\ 2x - 3y + 7z &= 5 \\ 2x + 19y - 47z &= 32\end{aligned}$$

Solution

The matrix form of the system is

$$\begin{bmatrix} 3 & 1 & -3 \\ 2 & -3 & 7 \\ 2 & 19 & -47 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 13 \\ 5 \\ 32 \end{bmatrix}$$

The augmented matrix of the system is

$$\left[\begin{array}{ccc|c} 3 & 1 & -3 & 13 \\ 2 & -3 & 7 & 5 \\ 2 & 19 & -47 & 32 \end{array} \right]$$

Reducing the augmented matrix to row echelon form,

$$\begin{aligned} & \left(\frac{1}{3} \right) R_1 \\ & \sim \left[\begin{array}{ccc|c} 1 & \frac{1}{3} & -1 & \frac{13}{3} \\ 2 & -3 & 7 & 5 \\ 2 & 19 & -47 & 32 \end{array} \right] \end{aligned}$$

$$R_2 - 2R_1, R_3 - 2R_1$$

$$\sim \left[\begin{array}{ccc|c} 1 & \frac{1}{3} & -1 & \frac{13}{3} \\ 0 & -\frac{11}{3} & 9 & -\frac{11}{3} \\ 0 & \frac{55}{3} & -45 & \frac{70}{3} \end{array} \right]$$

$$\left(-\frac{3}{11}\right)R_2$$

$$\sim \left[\begin{array}{ccc|c} 1 & \frac{1}{3} & -1 & \frac{13}{3} \\ 0 & 1 & -\frac{27}{11} & 1 \\ 0 & \frac{55}{3} & -45 & \frac{70}{3} \end{array} \right]$$

$$R_3 - \frac{55}{3}R_2$$

$$\sim \left[\begin{array}{ccc|c} 1 & \frac{1}{3} & -1 & \frac{13}{3} \\ 0 & 1 & -\frac{27}{11} & 1 \\ 0 & 0 & 0 & 5 \end{array} \right]$$

From the last row of the augmented matrix,

$$0x + 0y + 0z = 5$$

Hence, the system is inconsistent and has no solution.

Example 4

Solve the following system for x , y and z .

$$-\frac{1}{x} + \frac{3}{y} + \frac{4}{z} = 30$$

$$\frac{3}{x} + \frac{2}{y} - \frac{1}{z} = 9$$

$$\frac{2}{x} - \frac{1}{y} + \frac{2}{z} = 10$$

Solution

The matrix form of the system is

$$\begin{bmatrix} -1 & 3 & 4 \\ 3 & 2 & -1 \\ 2 & -1 & 2 \end{bmatrix} \begin{bmatrix} \frac{1}{x} \\ \frac{1}{y} \\ \frac{1}{z} \end{bmatrix} = \begin{bmatrix} 30 \\ 9 \\ 10 \end{bmatrix}$$

The augmented matrix of the system is

$$\left[\begin{array}{ccc|c} -1 & 3 & 4 & 30 \\ 3 & 2 & -1 & 9 \\ 2 & -1 & 2 & 10 \end{array} \right]$$

Reducing the augmented matrix to row echelon form,

$$\begin{array}{l} (-1)R_1 \\ \sim \left[\begin{array}{ccc|c} 1 & -3 & -4 & -30 \\ 3 & 2 & -1 & 9 \\ 2 & -1 & 2 & 10 \end{array} \right] \end{array}$$

$$\begin{array}{l} R_2 - 3R_1, R_3 - 2R_1 \\ \sim \left[\begin{array}{ccc|c} 1 & -3 & -4 & -30 \\ 0 & 11 & 11 & 99 \\ 0 & 5 & 10 & 70 \end{array} \right] \end{array}$$

$$\begin{array}{l} \left(\frac{1}{11}\right)R_2, \left(\frac{1}{5}\right)R_3 \\ \sim \left[\begin{array}{ccc|c} 1 & -3 & -4 & -30 \\ 0 & 1 & 1 & 9 \\ 0 & 1 & 2 & 14 \end{array} \right] \end{array}$$

$$\begin{array}{l} R_3 - R_2 \\ \sim \left[\begin{array}{ccc|c} 1 & -3 & -4 & -30 \\ 0 & 1 & 1 & 9 \\ 0 & 0 & 1 & 5 \end{array} \right] \end{array}$$

The corresponding system of equations is

$$\begin{aligned}\frac{1}{x} - \frac{3}{y} - \frac{4}{z} &= -30 \\ \frac{1}{y} + \frac{1}{z} &= 9 \\ \frac{1}{z} &= 5\end{aligned}$$

Solving these equations,

$$x = \frac{1}{2}, y = \frac{1}{4}, z = \frac{1}{5}$$

Hence, $x = \frac{1}{2}, y = \frac{1}{4}, z = \frac{1}{5}$ is the solution of the system.

Example 5

Solve the following system of non-linear equations for the unknown angles α , β and γ where $0 \leq \alpha \leq 2\pi$, $0 \leq \beta \leq 2\pi$ and $0 \leq \gamma < \pi$.

$$2 \sin \alpha - \cos \beta + 3 \tan \gamma = 3$$

$$4 \sin \alpha + 2 \cos \beta - 2 \tan \gamma = 2$$

$$6 \sin \alpha - 3 \cos \beta + \tan \gamma = 9$$

Solution

The matrix form of the system is

$$\begin{bmatrix} 2 & -1 & 3 \\ 4 & 2 & -2 \\ 6 & -3 & 1 \end{bmatrix} \begin{bmatrix} \sin \alpha \\ \cos \beta \\ \tan \gamma \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \\ 9 \end{bmatrix}$$

The augmented matrix of the system is

$$\left[\begin{array}{ccc|c} 2 & -1 & 3 & 3 \\ 4 & 2 & -2 & 2 \\ 6 & -3 & 1 & 9 \end{array} \right]$$

Reducing the augmented matrix to row echelon form,

$$\begin{aligned} & \left(\frac{1}{2} \right) R_1 \\ & \sim \left[\begin{array}{ccc|c} 1 & -\frac{1}{2} & \frac{3}{2} & \frac{3}{2} \\ 4 & 2 & -2 & 2 \\ 6 & -3 & 1 & 9 \end{array} \right] \end{aligned}$$

$$\begin{array}{l}
 R_2 - 4R_1, \quad R_3 - 6R_1 \\
 \sim \left[\begin{array}{ccc|c} 1 & -\frac{1}{2} & \frac{3}{2} & \frac{3}{2} \\ 0 & 4 & -8 & -4 \\ 0 & 0 & -8 & 0 \end{array} \right] \\
 \left(\frac{1}{4} \right) R_2, \quad \left(-\frac{1}{8} \right) R_3 \\
 \sim \left[\begin{array}{ccc|c} 1 & -\frac{1}{2} & \frac{3}{2} & \frac{3}{2} \\ 0 & 1 & -2 & -1 \\ 0 & 0 & 1 & 0 \end{array} \right]
 \end{array}$$

The corresponding system of equations is

$$\begin{aligned}
 \sin \alpha - \frac{1}{2} \cos \beta + \frac{3}{2} \tan \gamma &= \frac{3}{2} \\
 \cos \beta - 2 \tan \gamma &= -1 \\
 \tan \gamma &= 0
 \end{aligned}$$

Solving these equations,

$$\begin{aligned}
 \gamma &= 0 \\
 \cos \beta &= -1 \Rightarrow \beta = \pi \\
 \sin \alpha &= \frac{1}{2} \cos \beta - \frac{3}{2} \tan \gamma + \frac{3}{2} \\
 &= \frac{1}{2}(-1) - \frac{3}{2}(0) + \frac{3}{2} = 1 \\
 \alpha &= \frac{\pi}{2}
 \end{aligned}$$

Hence, $\alpha = \frac{\pi}{2}, \beta = \pi, \gamma = 0$ is the solution of the system.

Example 6

Investigate for what values of λ and μ the equations

$$\begin{aligned}
 x + 2y + z &= 8 \\
 2x + 2y + 2z &= 13 \\
 3x + 4y + \lambda z &= \mu
 \end{aligned}$$

have (i) no solution, (ii) a unique solution, and (iii) many solutions.

Solution

The matrix form of the system is

$$\begin{bmatrix} 1 & 2 & 1 \\ 2 & 2 & 2 \\ 3 & 4 & \lambda \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 8 \\ 13 \\ \mu \end{bmatrix}$$

The augmented matrix of the system is

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 8 \\ 2 & 2 & 2 & 13 \\ 3 & 4 & \lambda & \mu \end{array} \right]$$

Reducing the augmented matrix to row echelon form,

$$\begin{aligned} & R_2 - 2R_1, R_3 - 3R_1 \\ & \sim \left[\begin{array}{ccc|c} 1 & 2 & 1 & 8 \\ 0 & -2 & 0 & -3 \\ 0 & -2 & \lambda-3 & \mu-24 \end{array} \right] \\ & \left(-\frac{1}{2} \right) R_2 \\ & \sim \left[\begin{array}{ccc|c} 1 & 2 & 1 & 8 \\ 0 & 1 & 0 & \frac{3}{2} \\ 0 & -2 & \lambda-3 & \mu-24 \end{array} \right] \\ & R_3 + 2R_2 \\ & \sim \left[\begin{array}{ccc|c} 1 & 2 & 1 & 8 \\ 0 & 1 & 0 & \frac{3}{2} \\ 0 & 0 & \lambda-3 & \mu-21 \end{array} \right] \end{aligned}$$

- (i) If $\lambda = 3$ and $\mu \neq 21$, the system is inconsistent and has no solution.
- (ii) If $\lambda \neq 3$ and μ has any value, the system is consistent and has a unique solution.
- (iii) If $\lambda = 3$ and $\mu = 21$, the system is consistent and has infinite (many) solutions.

Example 7

Determine the values of λ for which the following equations are consistent. Also, solve the system for these values of λ .

$$\begin{aligned}x + 2y + z &= 3 \\x + y + z &= \lambda \\3x + y + 3z &= \lambda^2\end{aligned}$$

Solution

The matrix form of the system is

$$\begin{bmatrix} 1 & 2 & 1 \\ 1 & 1 & 1 \\ 3 & 1 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3 \\ \lambda \\ \lambda^2 \end{bmatrix}$$

The augmented matrix of the system is

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 3 \\ 1 & 1 & 1 & \lambda \\ 3 & 1 & 3 & \lambda^2 \end{array} \right]$$

Reducing the augmented matrix to row echelon form,

$$\begin{aligned}& R_2 - R_1, \quad R_3 - 3R_1 \\& \sim \left[\begin{array}{ccc|c} 1 & 2 & 1 & 3 \\ 0 & -1 & 0 & \lambda - 3 \\ 0 & -5 & 0 & \lambda^2 - 9 \end{array} \right] \\& (-1)R_2 \\& \sim \left[\begin{array}{ccc|c} 1 & 2 & 1 & 3 \\ 0 & 1 & 0 & 3 - \lambda \\ 0 & -5 & 0 & \lambda^2 - 9 \end{array} \right] \\& R_3 + 5R_2 \\& \sim \left[\begin{array}{ccc|c} 1 & 2 & 1 & 3 \\ 0 & 1 & 0 & 3 - \lambda \\ 0 & 0 & 0 & \lambda^2 - 5\lambda + 6 \end{array} \right]\end{aligned}$$

The equations will be consistent if $\lambda^2 - 5\lambda + 6 = 0$, i.e. $\lambda = 3$ or $\lambda = 2$.

Case I: When $\lambda = 3$,

$$\begin{aligned}x + 2y + z &= 3 \\y &= 0\end{aligned}$$

Assigning the free variable z any arbitrary value t ,

$$x = 3 - 2(0) - t = 3 - t$$

Hence, $x = 3 - t$, $y = 0$, $z = t$ is the solution of the system where t is a parameter.

Case II: When $\lambda = 2$,

$$\begin{aligned}x + 2y + z &= 3 \\ y &= 1\end{aligned}$$

Assigning the free variable z any arbitrary value t ,

$$x = 3 - 2(1) - t = 1 - t$$

Hence, $x = 1 - t$, $y = 1$, $z = t$ is the solution of the system where t is a parameter.

Example 8

Show that the system of equations

$$3x + 4y + 5z = \alpha$$

$$4x + 5y + 6z = \beta$$

$$5x + 6y + 7z = \gamma$$

is consistent only if α , β and γ are in arithmetic progression (A.P.).

Solution

The matrix form of the system is

$$\begin{bmatrix} 3 & 4 & 5 \\ 4 & 5 & 6 \\ 5 & 6 & 7 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix}$$

The augmented matrix of the system is

$$\left[\begin{array}{ccc|c} 3 & 4 & 5 & \alpha \\ 4 & 5 & 6 & \beta \\ 5 & 6 & 7 & \gamma \end{array} \right]$$

Reducing the augmented matrix to row echelon form,

$$\begin{aligned} & R_2 - R_1, R_3 - R_1 \\ & \sim \left[\begin{array}{ccc|c} 3 & 4 & 5 & \alpha \\ 1 & 1 & 1 & \beta - \alpha \\ 2 & 2 & 2 & \gamma - \alpha \end{array} \right] \\ & R_{12} \\ & \sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & \beta - \alpha \\ 3 & 4 & 5 & \alpha \\ 2 & 2 & 2 & \gamma - \alpha \end{array} \right] \end{aligned}$$

$$R_2 - 3R_1, R_3 - 2R_1$$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & \beta - \alpha \\ 0 & 1 & 2 & 4\alpha - 3\beta \\ 0 & 0 & 0 & \alpha - 2\beta + \gamma \end{array} \right]$$

The system of equations is consistent if,

$$\alpha - 2\beta + \gamma = 0$$

$$\beta = \frac{\alpha + \gamma}{2}$$

i.e. α, β and γ are in arithmetic progression (A.P.)

Example 9

Show that if $\lambda \neq 0$, the system of equations

$$\begin{aligned} 2x + y &= a \\ x + \lambda y - z &= b \\ y + 2z &= c \end{aligned}$$

has a unique solution for every value of a, b, c . If $\lambda = 0$, determine the relation satisfied by a, b, c such that the system is consistent. Find the solution by taking $\lambda = 0, a = 1, b = 1, c = -1$.

Solution

The matrix form of the system is

$$\begin{bmatrix} 2 & 1 & 0 \\ 1 & \lambda & -1 \\ 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$$

The system has a unique solution if $\det(A) \neq 0$

$$\begin{aligned} \det(A) &= 2(2\lambda + 1) - 1(2 + 0) \neq 0 \\ 4\lambda &\neq 0 \\ \lambda &\neq 0 \end{aligned}$$

Hence, the system of equations has a unique solution if $\lambda \neq 0$ for any value of a, b, c .

If $\lambda = 0$, the system is either inconsistent or has an infinite number of solutions.

For $\lambda = 0$, the augmented matrix of the system is

$$\left[\begin{array}{ccc|c} 2 & 1 & 0 & a \\ 1 & 0 & -1 & b \\ 0 & 1 & 2 & c \end{array} \right]$$

Reducing the augmented matrix to row echelon form,

$$\begin{array}{l} R_{12} \\ \sim \left[\begin{array}{ccc|c} 1 & 0 & -1 & b \\ 2 & 1 & 0 & a \\ 0 & 1 & 2 & c \end{array} \right] \end{array}$$

$$\begin{array}{l} R_2 - 2R_1 \\ \sim \left[\begin{array}{ccc|c} 1 & 0 & -1 & b \\ 0 & 1 & 2 & a-2b \\ 0 & 1 & 2 & c \end{array} \right] \end{array}$$

$$\begin{array}{l} R_3 - R_2 \\ \sim \left[\begin{array}{ccc|c} 1 & 0 & -1 & b \\ 0 & 1 & 2 & a-2b \\ 0 & 0 & 0 & c-a+2b \end{array} \right] \end{array}$$

The system is consistent if $c - a + 2b = 0$.

The corresponding system of equations is

$$\begin{aligned} x - z &= b \\ y + 2z &= a - 2b \end{aligned}$$

Assigning the free variable z any arbitrary value t ,

$$\begin{aligned} y &= a - 2b - 2t \\ x &= b + t \end{aligned}$$

Hence, $x = b + t$, $y = a - 2b - 2t$, $z = t$ is the solution of the system where t is a parameter.

When $a = 1$, $b = 1$, $c = -1$

$$\begin{aligned} x &= 1 + t \\ y &= -1 - 2t \\ z &= t \end{aligned}$$

Example 10

Solve the following system by Gauss–Jordan method:

$$\begin{aligned}x_1 + x_2 + 2x_3 &= 8 \\ -x_1 - 2x_2 + 3x_3 &= 1 \\ 3x_1 - 7x_2 + 4x_3 &= 10\end{aligned}$$

Solution

The matrix form of the system is

$$\begin{bmatrix} 1 & 1 & 2 \\ -1 & -2 & 3 \\ 3 & -7 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 8 \\ 1 \\ 10 \end{bmatrix}$$

The augmented matrix of the system is

$$\left[\begin{array}{ccc|c} 1 & 1 & 2 & 8 \\ -1 & -2 & 3 & 1 \\ 3 & -7 & 4 & 10 \end{array} \right]$$

Reducing the augmented matrix to reduced row echelon form,

$$\begin{array}{l} R_2 + R_1, R_3 - 3R_1 \\ \sim \left[\begin{array}{ccc|c} 1 & 1 & 2 & 8 \\ 0 & -1 & 5 & 9 \\ 0 & -10 & -2 & -14 \end{array} \right] \end{array}$$

$$\begin{array}{l} (-1)R_2 \\ \sim \left[\begin{array}{ccc|c} 1 & 1 & 2 & 8 \\ 0 & 1 & -5 & -9 \\ 0 & -10 & -2 & -14 \end{array} \right] \end{array}$$

$$\begin{array}{l} R_3 + 10R_2 \\ \sim \left[\begin{array}{ccc|c} 1 & 1 & 2 & 8 \\ 0 & 1 & -5 & -9 \\ 0 & 0 & -52 & -104 \end{array} \right] \end{array}$$

$$\begin{aligned}
 & \left(-\frac{1}{52}\right)R_3 \\
 & \sim \left[\begin{array}{ccc|c} 1 & 1 & 2 & 8 \\ 0 & 1 & -5 & -9 \\ 0 & 0 & 1 & 2 \end{array} \right] \\
 & R_2 + 5R_3, R_1 - 2R_3 \\
 & \sim \left[\begin{array}{ccc|c} 1 & 1 & 0 & 4 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 2 \end{array} \right] \\
 & R_1 - R_2 \\
 & \sim \left[\begin{array}{ccc|c} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 2 \end{array} \right]
 \end{aligned}$$

The corresponding system of equations is

$$\begin{aligned}
 x_1 &= 3 \\
 x_2 &= 1 \\
 x_3 &= 2
 \end{aligned}$$

Hence, $x_1 = 3, x_2 = 1, x_3 = 2$ is the solution of the system.

Example 11

Solve the following system by Gauss–Jordan method:

$$\begin{aligned}
 2x_1 + 2x_2 + 2x_3 &= 0 \\
 -2x_1 + 5x_2 + 2x_3 &= 1 \\
 8x_1 + x_2 + 4x_3 &= -1
 \end{aligned}$$

Solution

The matrix form of the system is

$$\begin{bmatrix} 2 & 2 & 2 \\ -2 & 5 & 2 \\ 8 & 1 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}$$

The augmented matrix of the system is

$$\left[\begin{array}{ccc|c} 2 & 2 & 2 & 0 \\ -2 & 5 & 2 & 1 \\ 8 & 1 & 4 & -1 \end{array} \right]$$

Reducing the augmented matrix to reduced row echelon form,

$$\begin{aligned} & \left(\frac{1}{2} \right) R_1 \\ & \sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ -2 & 5 & 2 & 1 \\ 8 & 1 & 4 & -1 \end{array} \right] \\ & R_2 + 2R_1, R_3 - 8R_1 \\ & \sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 7 & 4 & 1 \\ 0 & -7 & -4 & -1 \end{array} \right] \\ & R_3 + R_2 \\ & \sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 7 & 4 & 1 \\ 0 & 0 & 0 & 0 \end{array} \right] \\ & \left(\frac{1}{7} \right) R_2 \\ & \sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 1 & \frac{4}{7} & \frac{1}{7} \\ 0 & 0 & 0 & 0 \end{array} \right] \\ & R_1 - R_2 \\ & \sim \left[\begin{array}{ccc|c} 1 & 0 & \frac{3}{7} & -\frac{1}{7} \\ 0 & 1 & \frac{4}{7} & \frac{1}{7} \\ 0 & 0 & 0 & 0 \end{array} \right] \end{aligned}$$

The corresponding system of equations is

$$\begin{aligned}x_1 + \frac{3}{7}x_3 &= -\frac{1}{7} \\x_2 + \frac{4}{7}x_3 &= \frac{1}{7}\end{aligned}$$

Since leading ones are in columns 1 and 2, x_1 and x_2 are called leading variables, whereas x_3 is a free variable. Assigning the free variable x_3 any arbitrary value t ,

$$\begin{aligned}x_1 &= -\frac{1}{7} - \frac{3}{7}t \\x_2 &= \frac{1}{7} - \frac{4}{7}t\end{aligned}$$

Hence, $x_1 = -\frac{1}{7} - \frac{3}{7}t$, $x_2 = \frac{1}{7} - \frac{4}{7}t$, $x_3 = t$ is the solution of the system where t is a parameter.

Example 12

Solve the following system by Gauss–Jordan method:

$$\begin{aligned}x - y + 2z - w &= -1 \\2x + y - 2z - 2w &= -2 \\-x + 2y - 4z + w &= 1 \\3x - - - 3w &= -3\end{aligned}$$

Solution

The matrix form of the system is

$$\begin{bmatrix} 1 & -1 & 2 & -1 \\ 2 & 1 & -2 & -2 \\ -1 & 2 & -4 & 1 \\ 3 & 0 & 0 & -3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} = \begin{bmatrix} -1 \\ -2 \\ 1 \\ -3 \end{bmatrix}$$

The augmented matrix of the system is

$$\left[\begin{array}{cccc|c} 1 & -1 & 2 & -1 & -1 \\ 2 & 1 & -2 & -2 & -2 \\ -1 & 2 & -4 & 1 & 1 \\ 3 & 0 & 0 & -3 & -3 \end{array} \right]$$

Reducing the augmented matrix to reduced row echelon form,

$$\begin{aligned}
 & R_2 - 2R_1, R_3 + R_1, R_4 - 3R_1 \\
 & \sim \left[\begin{array}{cccc|c} 1 & -1 & 2 & -1 & -1 \\ 0 & 3 & -6 & 0 & 0 \\ 0 & 1 & -2 & 0 & 0 \\ 0 & 3 & -6 & 0 & 0 \end{array} \right] \\
 & \left(\frac{1}{3} \right) R_2 \\
 & \sim \left[\begin{array}{cccc|c} 1 & -1 & 2 & -1 & -1 \\ 0 & 1 & -2 & 0 & 0 \\ 0 & 1 & -2 & 0 & 0 \\ 0 & 3 & -6 & 0 & 0 \end{array} \right] \\
 & R_3 - R_2, R_4 - 3R_2 \\
 & \sim \left[\begin{array}{cccc|c} 1 & -1 & 2 & -1 & -1 \\ 0 & 1 & -2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right] \\
 & R_1 + R_2 \\
 & \sim \left[\begin{array}{cccc|c} 1 & 0 & 0 & -1 & -1 \\ 0 & 1 & -2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right]
 \end{aligned}$$

The corresponding system of equations is

$$\begin{aligned}
 x - w &= -1 \\
 y - 2z &= 0
 \end{aligned}$$

The leading ones are in columns 1 and 2. Hence, the variables x and y are called leading variables whereas the variables z and w are called free variables. Assigning the free variables z and w any arbitrary values t_1 and t_2 respectively,

$$x = -1 + t_2$$

and

$$y = 2t_1$$

Hence, $x = -1 + t_2, y = 2t_1, z = t_1, w = t_2$ is the solution of the system where t_1 and t_2 are parameters.

Example 13

Solve the following system by Gauss–Jordan method:

$$\begin{aligned} -2y + 3z &= 1 \\ 3x + 6y - 3z &= -2 \\ 6x + 6y + 3z &= 5 \end{aligned}$$

Solution

The matrix form of the system is

$$\begin{bmatrix} 0 & -2 & 3 \\ 3 & 6 & -3 \\ 6 & 6 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ -2 \\ 5 \end{bmatrix}$$

The augmented matrix of the system is

$$\left[\begin{array}{ccc|c} 0 & -2 & 3 & 1 \\ 3 & 6 & -3 & -2 \\ 6 & 6 & 3 & 5 \end{array} \right]$$

Reducing the augmented matrix to reduced row echelon form,

$$\begin{array}{l} R_{12} \\ \sim \left[\begin{array}{ccc|c} 3 & 6 & -3 & -2 \\ 0 & -2 & 3 & 1 \\ 6 & 6 & 3 & 5 \end{array} \right] \end{array}$$

$$\begin{array}{l} \left(\frac{1}{3} \right) R_1 \\ \sim \left[\begin{array}{ccc|c} 1 & 2 & -1 & -\frac{2}{3} \\ 0 & -2 & 3 & 1 \\ 6 & 6 & 3 & 5 \end{array} \right] \end{array}$$

$$\begin{array}{l} R_3 - 6R_1 \\ \sim \left[\begin{array}{ccc|c} 1 & 2 & -1 & -\frac{2}{3} \\ 0 & -2 & 3 & 1 \\ 0 & -6 & 9 & 9 \end{array} \right] \end{array}$$

$$\begin{aligned} & \left(-\frac{1}{2}\right)R_2 \\ & \sim \left[\begin{array}{ccc|c} 1 & 2 & -1 & -\frac{2}{3} \\ 0 & 1 & -\frac{3}{2} & -\frac{1}{2} \\ 0 & -6 & 9 & 9 \end{array} \right] \\ & R_3 + 6R_2 \\ & \sim \left[\begin{array}{ccc|c} 1 & 2 & -1 & -\frac{2}{3} \\ 0 & 1 & -\frac{3}{2} & -\frac{1}{2} \\ 0 & 0 & 0 & 6 \end{array} \right] \end{aligned}$$

From the last row of the augmented matrix,

$$0x + 0y + 0z = 6$$

Hence, the system is inconsistent and has no solution.

Example 14

Solve the following system by Gauss–Jordan method:

$$\begin{aligned} x_1 - 2x_2 - x_3 + 3x_4 &= 1 \\ 2x_1 - 4x_2 + x_3 &= 5 \\ x_1 - 2x_2 + 2x_3 - 3x_4 &= 4 \end{aligned}$$

Solution

The matrix form of the system is

$$\begin{bmatrix} 1 & -2 & -1 & 3 \\ 2 & -4 & 1 & 0 \\ 1 & -2 & 2 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 1 \\ 5 \\ 4 \end{bmatrix}$$

The augmented matrix of the system is

$$\left[\begin{array}{cccc|c} 1 & -2 & -1 & 3 & 1 \\ 2 & -4 & 1 & 0 & 5 \\ 1 & -2 & 2 & -3 & 4 \end{array} \right]$$

Reducing the augmented matrix to reduced row echelon form,

$$\begin{array}{l} R_2 - 2R_1, R_3 - R_1 \\ \sim \left[\begin{array}{cccc|c} 1 & -2 & -1 & 3 & 1 \\ 0 & 0 & 3 & -6 & 3 \\ 0 & 0 & 3 & -6 & 3 \end{array} \right] \end{array}$$

$$\begin{array}{l} R_3 - R_2 \\ \sim \left[\begin{array}{cccc|c} 1 & -2 & -1 & 3 & 1 \\ 0 & 0 & 3 & -6 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right] \end{array}$$

$$\begin{array}{l} \left(\frac{1}{3}\right)R_2 \\ \sim \left[\begin{array}{cccc|c} 1 & -2 & -1 & 3 & 1 \\ 0 & 0 & 1 & -2 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right] \end{array}$$

$$\begin{array}{l} R_1 + R_2 \\ \sim \left[\begin{array}{cccc|c} 1 & -2 & 0 & 1 & 2 \\ 0 & 0 & 1 & -2 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right] \end{array}$$

The corresponding system of equations is

$$\begin{array}{rcl} x_1 - 2x_2 + & & x_4 = 2 \\ & & x_3 - 2x_4 = 1 \end{array}$$

The leading ones are in columns 1 and 3. Hence, the variables x_1 and x_3 are called leading variables whereas the variables x_2 and x_4 are called free variables. Assigning the free variables x_2 and x_4 any arbitrary values t_1 and t_2 respectively,

$$x_1 = 2 + 2t_1 - t_2$$

$$x_3 = 1 + 2t_2$$

Hence, $x_1 = 2 + 2t_1 - t_2$, $x_2 = t_1$, $x_3 = 1 + 2t_2$, $x_4 = t_2$ is the solution of the system where t_1 and t_2 are the parameters.

Example 15

Solve the following system by Gauss–Jordan method:

$$2x - y + z = 9$$

$$3x - y + z = 6$$

$$4x - y + 2z = 7$$

$$-x + y - z = 4$$

Solution

The matrix form of the system is

$$\begin{bmatrix} 2 & -1 & 1 \\ 3 & -1 & 1 \\ 4 & -1 & 2 \\ -1 & 1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 9 \\ 6 \\ 7 \\ 4 \end{bmatrix}$$

The augmented matrix of the system is

$$\left[\begin{array}{ccc|c} 2 & -1 & 1 & 9 \\ 3 & -1 & 1 & 6 \\ 4 & -1 & 2 & 7 \\ -1 & 1 & -1 & 4 \end{array} \right]$$

Reducing the augmented matrix to reduced row echelon form,

$$\begin{aligned} & R_{14} \\ & \sim \left[\begin{array}{ccc|c} -1 & 1 & -1 & 4 \\ 3 & -1 & 1 & 6 \\ 4 & -1 & 2 & 7 \\ 2 & -1 & 1 & 9 \end{array} \right] \\ & (-1)R_1 \\ & \sim \left[\begin{array}{ccc|c} 1 & -1 & 1 & -4 \\ 3 & -1 & 1 & 6 \\ 4 & -1 & 2 & 7 \\ 2 & -1 & 1 & 9 \end{array} \right] \end{aligned}$$

$$R_2 - 3R_1, R_3 - 4R_1, R_4 - 2R_1$$

$$\sim \left[\begin{array}{ccc|c} 1 & -1 & 1 & -4 \\ 0 & 2 & -2 & 18 \\ 0 & 3 & -2 & 23 \\ 0 & 1 & -1 & 17 \end{array} \right]$$

$$R_{24}$$

$$\sim \left[\begin{array}{ccc|c} 1 & -1 & 1 & -4 \\ 0 & 1 & -1 & 17 \\ 0 & 3 & -2 & 23 \\ 0 & 2 & -2 & 18 \end{array} \right]$$

$$R_3 - 3R_2, R_4 - 2R_2$$

$$\sim \left[\begin{array}{ccc|c} 1 & -1 & 1 & -4 \\ 0 & 1 & -1 & 17 \\ 0 & 0 & 1 & -28 \\ 0 & 0 & 0 & -16 \end{array} \right]$$

From the last row of the augmented matrix,

$$0x + 0y + 0z = -16$$

Hence, the system is inconsistent and has no solution.

EXERCISE 10.3

1. Solve the following systems of equations by Gauss elimination method:

(i) $2x - 3y - z = 3$

$$x + 2y - z = 4$$

$$5x - 4y - 3z = -2$$

(ii) $x + 2y - z = 1$

$$x + y + 2z = 9$$

$$2x + y - z = 2$$

(iii) $6x + y + z = -4$

$$2x - 3y - z = 0$$

$$-x - 7y - 2z = 7$$

(iv) $2x - y - z = 2$

$$x + 2y + z = 2$$

$$4x - 7y - 5z = 2$$

(v) $2x_1 + x_2 + 2x_3 + x_4 = 6$

$$6x_1 - 6x_2 + 6x_3 + 12x_4 = 36$$

$$4x_1 + 3x_2 + 3x_3 - 3x_4 = 1$$

$$2x_1 + 2x_2 - x_3 + x_4 = 10$$

Ans.:	
(i) inconsistent	(ii) consistent $x = 2, y = 1, z = 3$
(iii) consistent $x = -1, y = -2, z = -4$	(iv) consistent $x = \frac{6+t}{5}, y = \frac{2-3t}{5}, z = t$
(v) consistent $x_1 = 2, x_2 = 1, x_3 = -1, x_4 = 3$	

2. Solve the following system of equations by Gauss-Jordan method:

(i) $x + 2y + z = -1$	(ii) $x + y + z = 6$
$6x + y + z = -4$	$x - 2y + 2z = 5$
$2x - 3y - z = 0$	$3x + y + z = 8$
$-x - 7y - 2z = 7$	$2x - 2y + 3z = 7$
$x - y = 1$	
(iii) $2x_1 + x_2 + 5x_4 = 4$	
$3x_1 - 2x_2 + 2x_3 = 2$	
$5x_1 - 8x_2 - 4x_3 = 1$	

Ans.:	
(i) consistent $x = -1, y = -2, z = 4$	(ii) consistent $x = -1, y = -2, z = 3$
(iii) inconsistent	

3. Investigate for what values of λ and μ , the system of simultaneous equations

$$\begin{aligned} x + y + z &= 6 \\ x + 2y + 3z &= 10 \\ x + 2y + \lambda z &= \mu \end{aligned}$$

has (i) no solution, (ii) a unique solution, and (iii) infinite number of solutions.

Ans.:		
(i) $\lambda = 3, \mu \neq 10$	(ii) $\lambda \neq 3$, any value of μ	(iii) $\lambda = 3, \mu = 10$

4. Investigate for what values of
- k
- the equations

$$\begin{aligned}x + y + z &= 1 \\2x + y + 4z &= k \\4x + y + 10z &= k^2\end{aligned}$$

have infinite number of solutions.

[Ans.: $k = 1, 2$]

5. Determine the values of
- λ
- for which the following system of equations.

$$\begin{aligned}3x - y + \lambda z &= 0 \\2x + y + z &= 2 \\x - 2y - \lambda z &= -1\end{aligned}$$

will fail to have a unique solution. For this value of λ , are the equations consistent?

[Ans.: $\lambda = -\frac{7}{2}$, no solution]

6. Find for what values
- λ
- , the set of equations

$$\begin{aligned}2x - 3y + 6z - 5t &= 3 \\y - 4z + t &= 1 \\4x - 5y + 8z - 9t &= \lambda\end{aligned}$$

has (i) no solution, and (ii) infinite number of solutions and find the solutions of the equations when they are consistent.

[Ans.: (i) $\lambda \neq 7$,
(ii) $\lambda = 7$, $x = 3k_1 + k_2 + 3$,
 $y = 4k_1 - k_2 + 1$, $z = k_1$,
 $t = k_2$]

10.9 SYSTEM OF HOMOGENEOUS LINEAR EQUATIONS

A system of m homogeneous linear equations in n variables $x_1, x_2, \dots, x_n$ or simply a linear system, is a set of m linear equations each in n variables. A linear system is represented by

$$\begin{aligned}a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= 0 \\a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= 0 \\\vdots &\quad \vdots \quad \quad \quad \vdots \quad \quad \vdots \\a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= 0\end{aligned}$$

Writing these equations in matrix form,

$$A\mathbf{x} = \mathbf{0}$$

where A is any matrix of order $m \times n$, $\mathbf{x}$ is a vector of order $n \times 1$ and $\mathbf{0}$ is a null vector of order $m \times 1$. The matrix A is called coefficient matrix of the system of equations.

10.9.1 Solutions of a System of Linear Equations

For a system of m linear equations in n variables, there are two possibilities of the solutions to the system.

- (i) The system has exactly one solution, i.e. $x_1 = 0, x_2 = 0 \dots, x_n = 0$. This solution is called the trivial solution.
- (ii) The system has infinite solutions.

Note: The system of equations has a non-trivial solution if $\det(A) = 0$.

Example 1

Solve the following system of equations by the Gauss–Jordan method.

$$3x - y - z = 0$$

$$x + y + 2z = 0$$

$$5x + y + 3z = 0$$

Solution

The matrix form of the system is

$$\begin{bmatrix} 3 & -1 & -1 \\ 1 & 1 & 2 \\ 5 & 1 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

The augmented matrix of the system is

$$\left[\begin{array}{ccc|c} 3 & -1 & -1 & 0 \\ 1 & 1 & 2 & 0 \\ 5 & 1 & 3 & 0 \end{array} \right]$$

Reducing the augmented matrix to reduced row echelon form,

$$\begin{array}{c} R_{12} \\ \sim \left[\begin{array}{ccc|c} 1 & 1 & 2 & 0 \\ 3 & -1 & -1 & 0 \\ 5 & 1 & 3 & 0 \end{array} \right] \end{array}$$

$$R_2 - 3R_1, R_3 - 5R_1$$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 2 & 0 \\ 0 & -4 & -7 & 0 \\ 0 & -4 & -7 & 0 \end{array} \right]$$

$$\left(-\frac{1}{4}\right)R_2, \left(-\frac{1}{4}\right)R_3$$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 2 & 0 \\ 0 & 1 & \frac{7}{4} & 0 \\ 0 & 1 & \frac{7}{4} & 0 \end{array} \right]$$

$$R_3 - R_2$$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 2 & 0 \\ 0 & 1 & \frac{7}{4} & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$$R_1 - R_2$$

$$\sim \left[\begin{array}{ccc|c} 1 & 0 & \frac{1}{4} & 0 \\ 0 & 1 & \frac{7}{4} & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

The corresponding system of equations is

$$x + \frac{1}{4}z = 0$$

$$y + \frac{7}{4}z = 0$$

Solving for the leading variables,

$$x = -\frac{1}{4}z$$

$$y = -\frac{7}{4}z$$

Assigning the free variable z an arbitrary value t ,

$$x = -\frac{1}{4}t$$

$$y = -\frac{7}{4}t$$

Hence, $x = -\frac{1}{4}t$, $y = -\frac{7}{4}t$ is the non-trivial solution of the system where t is a parameter.

Example 2

Solve the following system of equations by the Gauss–Jordan method.

$$\begin{aligned}x + y - z + w &= 0 \\x - y + 2z - w &= 0 \\3x + y + w &= 0\end{aligned}$$

Solution

The matrix form of the system is

$$\begin{bmatrix} 1 & 1 & -1 & 1 \\ 1 & -1 & 2 & -1 \\ 3 & 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

The augmented matrix of the system is

$$\left[\begin{array}{cccc|c} 1 & 1 & -1 & 1 & 0 \\ 1 & -1 & 2 & -1 & 0 \\ 3 & 1 & 0 & 1 & 0 \end{array} \right]$$

Reducing the augmented matrix to the reduced row echelon form,

$$\begin{aligned} & R_2 - R_1, R_3 - 3R_1 \\ & \sim \left[\begin{array}{cccc|c} 1 & 1 & -1 & 1 & 0 \\ 0 & -2 & 3 & -2 & 0 \\ 0 & -2 & 3 & -2 & 0 \end{array} \right] \\ & \left(-\frac{1}{2} \right) R_2, \left(-\frac{1}{2} \right) R_3 \\ & \sim \left[\begin{array}{cccc|c} 1 & 1 & -1 & 1 & 0 \\ 0 & 1 & -\frac{3}{2} & 1 & 0 \\ 0 & 1 & -\frac{3}{2} & 1 & 0 \end{array} \right] \end{aligned}$$

$$\begin{array}{c} R_3 - R_2 \\ \sim \left[\begin{array}{cccc|c} 1 & 1 & -1 & 1 & 0 \\ 0 & 1 & -\frac{3}{2} & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right]
 \end{array}$$

$$\begin{array}{c} R_1 - R_2 \\ \sim \left[\begin{array}{cccc|c} 1 & 0 & \frac{1}{2} & 0 & 0 \\ 0 & 1 & -\frac{3}{2} & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right]
 \end{array}$$

The corresponding system of equations is

$$\begin{aligned}
 x + \frac{1}{2}z &= 0 \\
 y - \frac{3}{2}z + w &= 0
 \end{aligned}$$

Solving for the leading variables,

$$\begin{aligned}
 x &= -\frac{1}{2}z \\
 y &= \frac{3}{2}z - w
 \end{aligned}$$

Assigning the free variables z and w arbitrary values t_1 and t_2 respectively,

$$\begin{aligned}
 x &= -\frac{1}{2}t_1 \\
 y &= \frac{3}{2}t_1 - t_2
 \end{aligned}$$

Hence, $x = -\frac{1}{2}t_1$, $y = \frac{3}{2}t_1 - t_2$, $z = t_1$, $w = t_2$ is the non-trivial solution of the system where t_1 and t_2 are parameters.

Example 3

Solve the following system of equations by the Gauss–Jordan method.

$$2x_1 + x_2 + 3x_3 = 0$$

$$x_1 + 2x_2 = 0$$

$$x_2 + x_3 = 0$$

Solution

The matrix form of the system is

$$\begin{bmatrix} 2 & 1 & 3 \\ 1 & 2 & 0 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

The augmented matrix of the system is

$$\left[\begin{array}{ccc|c} 2 & 1 & 3 & 0 \\ 1 & 2 & 0 & 0 \\ 0 & 1 & 1 & 0 \end{array} \right]$$

Reducing the augmented matrix to reduced row echelon form,

$$\begin{array}{l} R_{12} \\ \sim \left[\begin{array}{ccc|c} 1 & 2 & 0 & 0 \\ 2 & 1 & 3 & 0 \\ 0 & 1 & 1 & 0 \end{array} \right] \end{array}$$

$$\begin{array}{l} R_2 - 2R_1 \\ \sim \left[\begin{array}{ccc|c} 1 & 2 & 0 & 0 \\ 0 & -3 & 3 & 0 \\ 0 & 1 & 1 & 0 \end{array} \right] \end{array}$$

$$\begin{array}{l} \left(-\frac{1}{3} \right) R_2 \\ \sim \left[\begin{array}{ccc|c} 1 & 2 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 1 & 1 & 0 \end{array} \right] \end{array}$$

$$\begin{array}{l} R_3 - R_2 \\ \sim \left[\begin{array}{ccc|c} 1 & 2 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 2 & 0 \end{array} \right] \end{array}$$

$$\begin{aligned}
& \left(\frac{1}{2}\right)R_3 \\
& \sim \left[\begin{array}{ccc|c} 1 & 2 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & 0 \end{array} \right] \\
& R_2 + R_3 \\
& \sim \left[\begin{array}{ccc|c} 1 & 2 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{array} \right] \\
& R_1 - 2R_2 \\
& \sim \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{array} \right]
\end{aligned}$$

The corresponding system of equations is

$$\begin{aligned}
x &= 0 \\
y &= 0 \\
z &= 0
\end{aligned}$$

Hence, the system has a trivial solution, i.e. $x = 0$, $y = 0$, $z = 0$.

Example 4

Show that the following non-linear system has 18 solutions if $0 \leq \alpha \leq 2\pi$, $0 \leq \beta \leq 2\pi$ and $0 \leq \gamma < 2\pi$.

$$\begin{aligned}
\sin \alpha + 2 \cos \beta + 3 \tan \gamma &= 0 \\
2 \sin \alpha + 5 \cos \beta + 3 \tan \gamma &= 0 \\
-\sin \alpha - 5 \cos \beta + 5 \tan \gamma &= 0
\end{aligned}$$

Solution

The matrix form of the system is

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ -1 & -5 & 5 \end{bmatrix} \begin{bmatrix} \sin \alpha \\ \cos \beta \\ \tan \gamma \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

The augmented matrix of the system is

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 2 & 5 & 3 & 0 \\ -1 & -5 & 5 & 0 \end{array} \right]$$

Reducing the augmented matrix to reduced row echelon form,

$$\begin{array}{c} R_2 - 2R_1, R_3 + R_1 \\ \sim \left[\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 0 & 1 & -3 & 0 \\ 0 & -3 & 8 & 0 \end{array} \right] \end{array}$$

$$\begin{array}{c} R_3 + 3R_2 \\ \sim \left[\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 0 & 1 & -3 & 0 \\ 0 & 0 & -1 & 0 \end{array} \right] \end{array}$$

$$\begin{array}{c} (-1)R_3 \\ \sim \left[\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 0 & 1 & -3 & 0 \\ 0 & 0 & 1 & 0 \end{array} \right] \end{array}$$

$$\begin{array}{c} R_2 + 3R_3, R_1 - 3R_3 \\ \sim \left[\begin{array}{ccc|c} 1 & 2 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{array} \right] \end{array}$$

$$\begin{array}{c} R_1 - 2R_2 \\ \sim \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{array} \right] \end{array}$$

The corresponding system of equations is

$$\sin \alpha = 0$$

$$\cos \beta = 0$$

$$\tan \gamma = 0$$

From these equations,

$$\begin{aligned}\alpha &= 0, \pi, 2\pi \\ \beta &= \frac{\pi}{2}, \frac{3\pi}{2} \quad [\because \alpha, \beta \text{ and } \gamma \text{ lie between } 0 \text{ and } 2\pi] \\ \gamma &= 0, \pi, 2\pi\end{aligned}$$

Hence, there are $3 \cdot 2 \cdot 3 = 18$ possible solutions which satisfy the system of equations.

Example 5

For what value of λ does the following system of equations possess a non-trivial solution? Obtain the solution for real values of λ .

$$\begin{aligned}x + 2y + 3z &= \lambda x \\ 3x + y + 2z &= \lambda y \\ 2x + 3y + z &= \lambda z\end{aligned}$$

Solution

The system of equations is

$$\begin{aligned}(1 - \lambda)x + 2y + 3z &= 0 \\ 3x + (1 - \lambda)y + 2z &= 0 \\ 2x + 3y + (1 - \lambda)z &= 0\end{aligned}$$

The matrix form of the system is

$$\begin{bmatrix} 1-\lambda & 2 & 3 \\ 3 & 1-\lambda & 2 \\ 2 & 3 & 1-\lambda \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

The system will possess a non-trivial solution if $\det(A) = 0$.

$$\begin{aligned}\begin{vmatrix} 1-\lambda & 2 & 3 \\ 3 & 1-\lambda & 2 \\ 2 & 3 & 1-\lambda \end{vmatrix} &= 0 \\ (1-\lambda)[(1-\lambda)^2 - 6] - 2(3-3\lambda-4) + 3(9-2+2\lambda) &= 0 \\ (1-\lambda)(\lambda^2 - 2\lambda - 5) + 2 + 6\lambda + 21 + 6\lambda &= 0 \\ \lambda^2 - 2\lambda - 5 - \lambda^3 + 2\lambda^2 + 5\lambda + 12\lambda + 23 &= 0 \\ -\lambda^3 + 3\lambda^2 + 15\lambda + 18 &= 0 \\ \lambda = 6, \quad \lambda = -1.5 \pm 0.866i\end{aligned}$$

For real value of λ , i.e. $\lambda = 6$, the augmented matrix of the system is

$$\left[\begin{array}{ccc|c} -5 & 2 & 3 & 0 \\ 3 & -5 & 2 & 0 \\ 2 & 3 & -5 & 0 \end{array} \right]$$

Reducing the augmented matrix to reduced row echelon form,

$$\begin{array}{l} R_2 - R_3 \\ \sim \end{array} \left[\begin{array}{ccc|c} -5 & 2 & 3 & 0 \\ 1 & -8 & 7 & 0 \\ 2 & 3 & -5 & 0 \end{array} \right]$$

$$\begin{array}{l} R_{12} \\ \sim \end{array} \left[\begin{array}{ccc|c} 1 & -8 & 7 & 0 \\ -5 & 2 & 3 & 0 \\ 2 & 3 & -5 & 0 \end{array} \right]$$

$$\begin{array}{l} R_2 + 5R_1, R_3 - 2R_1 \\ \sim \end{array} \left[\begin{array}{ccc|c} 1 & -8 & 7 & 0 \\ 0 & -38 & 38 & 0 \\ 0 & 19 & -19 & 0 \end{array} \right]$$

$$\begin{array}{l} \left(-\frac{1}{38}\right)R_2, \left(\frac{1}{19}\right)R_3 \\ \sim \end{array} \left[\begin{array}{ccc|c} 1 & -8 & 7 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 1 & -1 & 0 \end{array} \right]$$

$$\begin{array}{l} R_3 - R_2 \\ \sim \end{array} \left[\begin{array}{ccc|c} 1 & -8 & 7 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$$\begin{array}{l} R_1 + 8R_2 \\ \sim \end{array} \left[\begin{array}{ccc|c} 1 & 0 & -1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

The corresponding system of equations is

$$x - z = 0$$

$$y - z = 0$$

Solving for the leading variables,

$$x = z$$

$$y = z$$

Assigning the free variable z an arbitrary value t ,

$$x = t$$

$$y = t$$

Hence, $x = t, y = t, z = t$ is the non-trivial solution of the system where t is a parameter.

Example 6

If the following system has a non-trivial solution, then prove that $a + b + c = 0$ or $a = b = c$ and hence find the solution in each case.

$$ax + by + cz = 0$$

$$bx + cy + az = 0$$

$$cx + ay + bz = 0$$

Solution

The matrix form of the system is

$$\begin{bmatrix} a & b & c \\ b & c & a \\ c & a & b \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

The system has a non-trivial solution if $\det(A) = 0$.

$$\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} = 0$$

$$a(bc - a^2) - b(b^2 - ac) + c(ab - c^2) = 0$$

$$-a^3 + b^3 + c^3 - 3abc = 0$$

$$-(a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca) = 0$$

$$\begin{aligned}
& a + b + c = 0 \\
\text{or } & a^2 + b^2 + c^2 - ab - bc - ca = 0 \\
& \frac{1}{2}[(a-b)^2 + (b-c)^2 + (c-a)^2] = 0 \\
& a - b = 0, b - c = 0, c - a = 0 \\
& a = b, b = c, c = a \\
& a = b = c
\end{aligned}$$

Hence, the system has a non-trivial solution if $a + b + c = 0$ or $a = b = c$.

The augmented matrix of the system is

$$\begin{aligned}
& \left[\begin{array}{ccc|c} a & b & c & 0 \\ b & c & a & 0 \\ c & a & b & 0 \end{array} \right] \\
& R_3 + R_1 + R_2 \\
& \sim \left[\begin{array}{ccc|c} a & b & c & 0 \\ b & c & a & 0 \\ a+b+c & a+b+c & a+b+c & 0 \end{array} \right]
\end{aligned}$$

The corresponding system of equations is

$$\begin{aligned}
& ax + by + cz = 0 \\
& bx + cy + az = 0 \\
& (a+b+c)x + (a+b+c)y + (a+b+c)z = 0
\end{aligned}$$

(i) When $a + b + c = 0$, we have only two equations.

$$\begin{aligned}
& ax + by + cz = 0 \\
& bx + cy + az = 0
\end{aligned}$$

$$\begin{aligned}
& \frac{x}{\begin{vmatrix} b & c \\ c & a \end{vmatrix}} = -\frac{y}{\begin{vmatrix} a & c \\ b & a \end{vmatrix}} = \frac{z}{\begin{vmatrix} a & b \\ b & c \end{vmatrix}} = t \\
& \frac{x}{ab - c^2} = -\frac{y}{a^2 - bc} = \frac{z}{ac - b^2} = t
\end{aligned}$$

Hence, $x = (ab - c^2)t$, $y = (bc - a^2)t$, $z = (ac - b^2)t$ is the solution of the system where t is a parameter.

(ii) When $a = b = c$, we have only one equation.

$$x + y + z = 0$$

$$\begin{array}{l} \text{Let} \\ y = t_1 \\ z = t_2 \\ \text{Then} \\ x = -t_1 - t_2 \end{array}$$

Hence, $x = -t_1 - t_2, y = t_1, z = t_2$ is the solution of the system where t_1 and t_2 are parameters.

Example 7

Discuss for all values of k , the system of equations

$$\begin{aligned} 2x + 3ky + (3k + 4)z &= 0 \\ x + (k + 4)y + (4k + 2)z &= 0 \\ x + 2(k + 1)y + (3k + 4)z &= 0 \end{aligned}$$

Solution

The matrix form of the system is

$$\begin{bmatrix} 2 & 3k & 3k+4 \\ 1 & k+4 & 4k+2 \\ 1 & 2k+2 & 3k+4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

R_{12}

$$\begin{bmatrix} 1 & k+4 & 4k+2 \\ 2 & 3k & 3k+4 \\ 1 & 2k+2 & 3k+4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$R_2 - 2R_1, R_3 - R_1$

$$\begin{bmatrix} 1 & k+4 & 4k+2 \\ 0 & k-8 & -5k \\ 0 & k-2 & -k+2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{aligned} \det(A) &= \begin{vmatrix} 1 & k+4 & 4k+2 \\ 0 & k-8 & -5k \\ 0 & k-2 & -k+2 \end{vmatrix} \\ &= (k-8)(-k+2) + 5k(k-2) \\ &= (k-2)(-k+8+5k) \\ &= 4(k-2)(k+2) \end{aligned}$$

- (i) When $k \neq \pm 2$, $\det(A) \neq 0$, the system has a trivial solution, i.e. $x = 0$, $y = 0$, $z = 0$.
- (ii) When $k = \pm 2$, $\det(A) = 0$, the system has non-trivial solutions.

Case I: When $k = 2$, the augmented matrix of the system is

$$\left[\begin{array}{ccc|c} 1 & 6 & 10 & 0 \\ 0 & -6 & -10 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$$\begin{aligned} & \left(-\frac{1}{6} \right) R_2 \\ & \sim \left[\begin{array}{ccc|c} 1 & 6 & 10 & 0 \\ 0 & 1 & \frac{10}{6} & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] \end{aligned}$$

$$\begin{aligned} & R_1 - 6R_2 \\ & \sim \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & \frac{10}{6} & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] \end{aligned}$$

The corresponding system of equations is

$$\begin{aligned} x &= 0 \\ y + \frac{10}{6}z &= 0 \end{aligned}$$

Solving for the leading variables,

$$\begin{aligned} x &= 0 \\ y &= -\frac{10}{6}z \end{aligned}$$

Assigning the free variable z any arbitrary value t ,

$$y = -\frac{10}{6}t = -\frac{5}{3}t$$

Hence, $x = 0$, $y = -\frac{5}{3}t$, $z = t$ is the solution of the system where t is a parameter.

Case II: When $k = -2$, the augmented matrix of the system is

$$\left[\begin{array}{ccc|c} 1 & 2 & -6 & 0 \\ 0 & -10 & 10 & 0 \\ 0 & -4 & 4 & 0 \end{array} \right]$$

Reducing the augmented matrix to reduced row echelon form,

$$\left(-\frac{1}{10} \right) R_2, \left(-\frac{1}{4} \right) R_3$$

$$\sim \left[\begin{array}{ccc|c} 1 & 2 & -6 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 1 & -1 & 0 \end{array} \right]$$

$$R_3 - R_2$$

$$\sim \left[\begin{array}{ccc|c} 1 & 2 & -6 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$$R_1 - 2R_2$$

$$\sim \left[\begin{array}{ccc|c} 1 & 0 & -4 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

The corresponding system of equations is

$$x - 4z = 0$$

$$y - z = 0$$

Solving for the leading variables,

$$x = 4z$$

$$y = z$$

Assigning the free variable z any arbitrary value t ,

$$x = 4t$$

$$y = t$$

Hence, $x = 4t, y = t, z = t$ is the solution of the system where t is a parameter.

EXERCISE 10.4

1. Solve the following equations:

(i) $x - y + z = 0$

$x + 2y + z = 0$

$2x + y + 3z = 0$

(ii) $x - 2y + 3z = 0$

$2x + 5y + 6z = 0$

(iii) $2x - 2y + 5z + 3w = 0$

$4x - y + z + w = 0$

$3x - 2y + 3z + 4w = 0$

$x - 3y + 7z + 6w = 0$

(iv) $2x - y + 3z = 0$

$3x + 2y + z = 0$

$x - 4y + 5z = 0$

(v) $7x + y - 2z = 0$

$x + 5y - 4z = 0$

$3x - 2y + z = 0$

$2x - 7y + 5z = 0$

(vi) $3x + 4y - z - 9w = 0$

$2x + 3y + 2z - 3w = 0$

$2x + y - 14z - 12w = 0$

$x + 3y + 13z + 3w = 0$

(vii) $x_1 + 2x_2 + 3x_3 + x_4 = 0$

$x_1 + x_2 - x_3 - x_4 = 0$

$3x_1 - x_2 + 2x_3 + 3x_4 = 0$

(viii) $2x_1 - x_2 + 3x_3 = 0$

$3x_1 + 2x_2 + x_3 = 0$

$x_1 - 4x_2 + 5x_3 = 0$

Ans.: (i) $x = 0, y = 0, z = 0$ (ii) $x = -3t, y = 0, z = t$

(iii) $x = \frac{211}{9}t, y = 4t, z = \frac{7}{9}t, w = t$

(iv) $x = -t, y = t, z = t$ (v) $x = \frac{3}{17}t, y = \frac{13}{17}t, z = t$

(vi) $x = 11t, y = -8t, z = t, w = 0$

(vii) $x_1 = -\frac{1}{3}t, x_2 = \frac{2}{3}t, x_3 = -\frac{2}{3}t, x_4 = t$

(viii) $x_1 = -x_2 = -x_3 = t$

2. For what value of λ does the following system of equations possess a non-trivial solution? Obtain the solution for real values of λ .

(i) $3x + y - \lambda z = 0$

$4x - 2y - 3z = 0$

$2\lambda x + 4y - \lambda z = 0$

(ii) $(1 - \lambda)x_1 + 2x_2 + 3x_3 = 0$

$3x_1 + (1 - \lambda)x_2 + 2x_3 = 0$

$2x_1 + 3x_2 + (1 - \lambda)x_3 = 0$

Ans.: (i) Non-trivial solution $\lambda = 1, -9$

For $\lambda = 1, x = -t, y = -t, z = -2t$

For $\lambda = -9, x = -3t, y = -9t, z = 2t$

(ii) $\lambda = 6, x = y = z = t$

3. Show that the system of equations $2x - 2y + z = \lambda x$, $2x - 3y + 2z = \lambda y$, $-x + 2y = \lambda z$ can possess a non-trivial solution only if $\lambda = 1$, $\lambda = -3$. Obtain the general solution in each case.

$$\left[\begin{array}{l} \text{Ans.: For } \lambda = 1, x = 2t_2 - t_1, y = t_2, z = t_1 \\ \text{For } \lambda = -3, x = -t, y = -2t, z = t \end{array} \right]$$

10.10 EIGENVALUES AND EIGENVECTORS

Eigenvalues and eigenvectors are important concepts in linear algebra. They are derived from the German word '*eigen*' which means proper or characteristic. Eigenvectors are nonzero vectors that get mapped into scalar multiples of themselves under a linear operator.

Any nonzero vector $\mathbf{x}$ is said to be a characteristic vector or eigenvector of a matrix A if there exists a number λ such that

$$A\mathbf{x} = \lambda\mathbf{x}$$

where $A = [a_{ij}]_{n \times n}$ is an n -rowed square matrix and $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$ is a non-zero column vector.

Also, λ is said to be characteristic root or characteristic value or eigenvalue of the matrix A .

Depending on the sign and the magnitude of the eigenvalue λ corresponding to $\mathbf{x}$, the linear operator $A\mathbf{x} = \lambda\mathbf{x}$ compresses or stretches the eigenvector $\mathbf{x}$ by a factor λ . If λ is negative, the direction of the eigenvector reverses. (Fig. 10.1).

Now

$$A\mathbf{x} = \lambda\mathbf{x} = \lambda I\mathbf{x}$$

$$(A - \lambda I)\mathbf{x} = \mathbf{0}$$

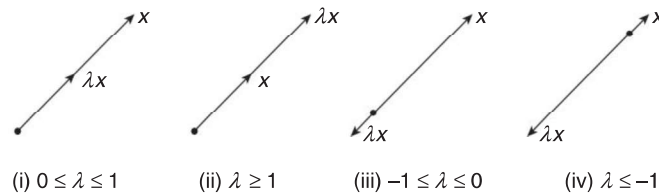


Fig. 6.1

The matrix $A - \lambda I$ is called the *characteristic matrix* of A where I is the unit matrix of order n .

$$\det(A - \lambda I) = \begin{vmatrix} a_{11} - \lambda & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} - \lambda & \cdots & a_{2n} \\ \cdots & \cdots & \cdots & \cdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} - \lambda \end{vmatrix}$$

which is an ordinary polynomial in λ of degree n , is called the *characteristic polynomial* of A .

The equation $\det(A - \lambda I) = 0$ is called the *characteristic equation* of A and the roots of this equation are called the eigenvalues of the matrix A . The set of all eigenvectors is called the *eigenspace* of A corresponding to λ . The set of all eigenvalues of A is called the *spectrum* of A .

Note: (1) The characteristic equation of the matrix A of order 2 can be obtained from

$$\lambda^2 - S_1 \lambda + S_2 = 0$$

where S_1 = sum of principal diagonal elements and

S_2 = determinant A

(2) The characteristic equation of the matrix A of order 3 can be obtained from

$$\lambda^3 - S_1 \lambda^2 + S_2 \lambda - S_3 = 0$$

where S_1 = sum of principal diagonal elements,

S_2 = sum of minors of principal diagonal elements and

S_3 = determinant A

(3) The sum of the eigenvalues of a matrix is the sum of its principal diagonal elements.

(4) The product of the eigenvalues of a matrix is the determinant of the matrix.

10.11 PROPERTIES OF EIGENVALUES

Property 1: If λ is an eigenvalue of the matrix A then λ is also an eigenvalue of A^T .

Proof: Let λ be an eigenvalue of the matrix A .

The characteristic equation of A is $\det(A - \lambda I) = 0$.

The characteristic equation of A^T is $\det(A^T - \lambda I) = 0$.

The determinant value does not change by the interchange of rows and columns.

$$\det(A - \lambda I) = \det(A^T - \lambda I)$$

The characteristic equations are same for both A and A^T .

Hence, λ is also an eigenvalue of A^T .

Property 2: If λ is an eigenvalue of the non-singular matrix A then $\frac{1}{\lambda}$ is an eigenvalue of A^{-1} .

Proof: Let λ be an eigenvalue of the non-singular matrix A .

$$A\mathbf{x} = \lambda\mathbf{x} \quad \dots(10.1)$$

Premultiplying both sides of Eq. (10.1) by A^{-1} ,

$$\begin{aligned} A^{-1}A\mathbf{x} &= A^{-1}\lambda\mathbf{x} \\ I\mathbf{x} &= \lambda A^{-1}\mathbf{x} \\ \mathbf{x} &= \lambda A^{-1}\mathbf{x} \\ A^{-1}\mathbf{x} &= \frac{1}{\lambda}\mathbf{x} \end{aligned}$$

Hence, $\frac{1}{\lambda}$ is an eigenvalue of A^{-1} .

Property 3: If λ is an eigenvalue of the matrix A then λ^k is an eigenvalue of A^k .

Proof: Let λ be an eigenvalue of the matrix A .

$$A\mathbf{x} = \lambda\mathbf{x} \quad \dots(10.2)$$

Premultiplying both sides of Eq. (10.2) by A ,

$$\begin{aligned} AA\mathbf{x} &= A\lambda\mathbf{x} \\ A^2\mathbf{x} &= \lambda(A\mathbf{x}) \\ &= \lambda(\lambda\mathbf{x}) \\ &= \lambda^2\mathbf{x} \end{aligned}$$

Similarly, $A^3\mathbf{x} = \lambda^3\mathbf{x}$

In general, $A^k\mathbf{x} = \lambda^k\mathbf{x}$

Hence, λ^k is an eigenvalue of A^k .

Property 4: If λ is an eigenvalue of the matrix A then $\lambda \pm k$ is an eigenvalue of $A \pm kI$.

Proof: Let λ be an eigenvalue of the matrix A .

$$A\mathbf{x} = \lambda\mathbf{x} \quad \dots(10.3)$$

Adding $kI\mathbf{x}$ on both sides of Eq. (10.3),

$$\begin{aligned} A\mathbf{x} + kI\mathbf{x} &= \lambda\mathbf{x} + kI\mathbf{x} \\ (A + kI)\mathbf{x} &= \lambda\mathbf{x} + k\mathbf{x} \\ &= (\lambda + k)\mathbf{x} \end{aligned}$$

Similarly, $(A - kI)\mathbf{x} = (\lambda - k)\mathbf{x}$

In general, $(A \pm kI)\mathbf{x} = (\lambda \pm k)\mathbf{x}$

Hence, $\lambda \pm k$ is an eigenvalue of $A \pm kI$.

Property 5: If λ is an eigenvalue of the matrix A then $k\lambda$ is an eigenvalue of kA .

Proof: Let λ be an eigenvalue of the matrix A .

$$A\mathbf{x} = \lambda\mathbf{x} \quad \dots(10.4)$$

Multiplying both sides of Eq. (10.4) by the scalar k ,

$$kA\mathbf{x} = k\lambda\mathbf{x}$$

Hence, $k\lambda$ is an eigenvalue of kA .

Property 6: The eigenvalues of a triangular matrix are the diagonal elements of the matrix.

Proof: Let $A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ 0 & a_{22} & \cdots & a_{2n} \\ \cdots & \cdots & \cdots & \cdots \\ \cdots & \cdots & \cdots & \cdots \\ 0 & 0 & \cdots & a_{nn} \end{bmatrix}$ be a triangular matrix of order n .

The characteristic equation is

$$\det(A - \lambda I) = 0$$

$$\begin{bmatrix} a_{11} - \lambda & a_{12} & \cdots & a_{1n} \\ 0 & a_{22} - \lambda & \cdots & a_{2n} \\ \cdots & \cdots & \cdots & \cdots \\ \cdots & \cdots & \cdots & \cdots \\ 0 & 0 & \cdots & a_{nn} - \lambda \end{bmatrix} = 0$$

$$(a_{11} - \lambda)(a_{22} - \lambda) \cdots (a_{nn} - \lambda) = 0$$

$$\lambda = a_{11}, a_{22}, \dots, a_{nn}$$

Hence, the eigenvalues of A are $a_{11}, a_{22}, \dots, a_{nn}$ which are the diagonal elements of the matrix.

Property 7: The eigenvalues of a real symmetric matrix are real.

Proof: Let λ be an eigenvalue of the real symmetric matrix.

$$A\mathbf{x} = \lambda\mathbf{x} \quad \dots (10.5)$$

Premultiplying both sides of Eq. (10.5) by $\bar{\mathbf{x}}^T$,

$$\bar{\mathbf{x}}^T A\mathbf{x} = \lambda \bar{\mathbf{x}}^T \mathbf{x} \quad \dots (10.6)$$

Taking complex conjugate on both sides of Eq. (10.6),

$$\mathbf{x}^T \bar{A} \bar{\mathbf{x}} = \bar{\lambda} \mathbf{x}^T \bar{\mathbf{x}}$$

$$\mathbf{x}^T A \bar{\mathbf{x}} = \bar{\lambda} \mathbf{x}^T \bar{\mathbf{x}} \quad (\because \bar{A} = A \text{ for real matrix}) \quad \dots (10.7)$$

Taking transpose on both sides of Eq. (10.7),

$$\bar{\mathbf{x}}^T A^T \mathbf{x} = \bar{\lambda} \bar{\mathbf{x}}^T \mathbf{x}$$

$$\bar{\mathbf{x}}^T A \mathbf{x} = \bar{\lambda} \bar{\mathbf{x}}^T \mathbf{x} \quad (\because A^T = A \text{ for symmetric matrix}) \quad \dots (10.8)$$

From Eqs (10.6) and (10.8),

$$\lambda \bar{\mathbf{x}}^T \mathbf{x} = \bar{\lambda} \bar{\mathbf{x}}^T \mathbf{x}$$

$$(\lambda - \bar{\lambda}) \bar{\mathbf{x}}^T \mathbf{x} = 0$$

$\bar{\mathbf{x}}^T \mathbf{x}$ is a 1×1 matrix, i.e., a single element which is positive,

$$\lambda - \bar{\lambda} = 0$$

i.e., λ is real.

Hence, the eigenvalues of a real symmetric matrix are real.

Example 1

Find the sum and product of the eigenvalues of the matrix $A = \begin{bmatrix} 2 & -3 \\ 4 & -2 \end{bmatrix}$.

Solution

Sum of the eigenvalues of A = Sum of principal diagonal elements of A

$$= 2 - 2$$

$$= 0$$

Product of the eigenvalues of A = Determinant of A

$$= \begin{vmatrix} 2 & -3 \\ 4 & -2 \end{vmatrix}$$

$$= -4 + 12$$

$$= 8$$

Example 2

Find the sum and product of the eigenvalues of the matrix $A = \begin{bmatrix} 1 & 2 & 5 \\ 2 & 3 & 4 \\ 3 & 6 & 7 \end{bmatrix}$.

Solution

Sum of the eigenvalues of A = Sum of principal diagonal elements of A

$$= 1 + 3 + 7$$

$$= 11$$

Product of the eigenvalues of A = Determinant of A

$$= \begin{vmatrix} 1 & 2 & 5 \\ 2 & 3 & 4 \\ 3 & 6 & 7 \end{vmatrix}$$

$$= 1(21 - 24) - 2(14 - 12) + 5(12 - 9)$$

$$= -3 - 4 + 15$$

$$= 8$$

Example 3

The product of two eigenvalues of the matrix $A = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$ is 16.

Find the third eigenvalue.

Solution

Let λ_1, λ_2 and λ_3 be the eigenvalues of the matrix A .

$$\lambda_1 \lambda_2 = 16$$

Product of the eigenvalues of A = Determinant of A

$$\begin{aligned} \lambda_1 \lambda_2 \lambda_3 &= \begin{vmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{vmatrix} \\ 16\lambda_3 &= 6(9-1) + 2(-6+2) + 2(2-6) \\ &= 48 - 8 - 8 \\ \lambda_3 &= 2 \end{aligned}$$

Example 4

Two eigenvalues of the matrix $A = \begin{bmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{bmatrix}$ are equal and are $\frac{1}{5}$ times to the third. Find the eigenvalues.

Solution

Let λ_1, λ_2 and λ_3 be the eigenvalues of the matrix A .

$$\lambda_1 = \lambda_2$$

$$\lambda_1 = \frac{1}{5}\lambda_3$$

$$\lambda_2 = \frac{1}{5}\lambda_3$$

Sum of the eigenvalues of A = Sum of principal diagonal elements of A

$$\lambda_1 + \lambda_2 + \lambda_3 = 2 + 3 + 2$$

$$\frac{1}{5}\lambda_3 + \frac{1}{5}\lambda_3 + \lambda_3 = 7$$

$$\frac{7}{5}\lambda_3 = 7$$

$$\lambda_3 = 5$$

$$\therefore \lambda_1 = \lambda_2 = 1$$

Hence, the eigenvalues of A are 1, 1, 5.

Example 5

If 2 is an eigenvalue of the matrix $A = \begin{bmatrix} 2 & -2 & 2 \\ 1 & 1 & 1 \\ 1 & 3 & -1 \end{bmatrix}$, find the other two eigen values.

Solution

Let λ_1, λ_2 and λ_3 be the eigenvalues of the matrix A .

$$\lambda_1 = 2$$

Sum of the eigenvalues of A = Sum of principal diagonal elements of A

$$2 + \lambda_2 + \lambda_3 = 2 + 1 - 1$$

$$\lambda_2 + \lambda_3 = 0 \quad \dots(1)$$

Product of the eigenvalues of A = Determinant of A

$$\begin{aligned} 2\lambda_2\lambda_3 &= \begin{vmatrix} 2 & -2 & 2 \\ 1 & 1 & 1 \\ 1 & 3 & -1 \end{vmatrix} \\ &= 2(-1-3) + 2(-1-1) + 2(3-1) \\ &= -8 - 4 + 4 \\ &= -8 \\ \lambda_2\lambda_3 &= -4 \quad \dots(2) \end{aligned}$$

Solving Eqs (1) and (2),

$$\lambda_2 = 2, \lambda_3 = -2$$

Hence, the other two eigenvalues are 2, -2.

Example 6

For a given matrix A of order 3, $|A| = 32$ and two of its eigenvalues are 8 and 2. Find the sum of the eigenvalues.

Solution

Let λ_1, λ_2 and λ_3 be the eigenvalues of the matrix A .

$$\lambda_1 = 8, \lambda_2 = 2$$

Product of the eigenvalues of A = Determinant of A

$$\begin{aligned} \lambda_1\lambda_2\lambda_3 &= |A| \\ 8 \times 2 \times \lambda_3 &= 32 \\ \lambda_3 &= 2 \end{aligned}$$

Hence, the sum of the eigenvalues = $8 + 2 + 2 = 12$

Example 7

For a singular matrix of order three, 2 and 3 are the eigenvalues. Find its third eigenvalue.

Solution

Let λ_1, λ_2 and λ_3 be the eigenvalues of the matrix A .

$$\lambda_1 = 2, \lambda_2 = 3$$

For a singular matrix, $|A| = 0$

Product of the eigenvalues of A = Determinant of A

$$\begin{aligned}\lambda_1 \lambda_2 \lambda_3 &= |A| \\ 2 \times 3 \times \lambda_3 &= 0 \\ \lambda_3 &= 0\end{aligned}$$

Hence, the third eigenvalue = 0

Example 8

If 2, 3 are the eigenvalues of $\begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ a & 0 & 2 \end{bmatrix}$, find the value of a .

Solution

Let λ_1, λ_2 and λ_3 be the eigenvalues of the matrix A .

$$\lambda_1 = 2, \lambda_2 = 3$$

Sum of the eigenvalues of A = Sum of principal diagonal elements of A

$$\begin{aligned}2 + 3 + \lambda_3 &= 2 + 2 + 2 \\ \lambda_3 &= 1\end{aligned}$$

Product of the eigenvalues of A = Determinant of A

$$\begin{aligned}\lambda_1 \lambda_2 \lambda_3 &= |A| \\ 2 \times 3 \times 1 &= \begin{vmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ a & 0 & 2 \end{vmatrix} \\ 6 &= 2(4 - 0) - 0(0 - 0) + 1(0 - 2a) \\ &= 8 - 2a \\ 2a &= 2 \\ a &= 1\end{aligned}$$

Example 9

If 3 and 15 are the two eigenvalues of $A = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$, find $|A|$ without expanding the determinant.

Solution

Let λ_1, λ_2 and λ_3 be the eigenvalues of the matrix A .

$$\lambda_1 = 3, \lambda_2 = 15$$

Sum of the eigenvalues of A = Sum of principal diagonal elements of A

$$3 + 15 + \lambda_3 = 8 + 7 + 3$$

$$\lambda_3 = 0$$

Product of the eigenvalues of A = Determinant of A

$$3 \times 15 \times 0 = |A|$$

$$|A| = 0$$

Example 10

Find a and b such that $A = \begin{bmatrix} a & 4 \\ 1 & b \end{bmatrix}$ has 3 and -2 as eigenvalues.

Solution

Let λ_1 and λ_2 be the eigenvalues of the matrix A .

$$\lambda_1 = 3, \lambda_2 = -2$$

Sum of the eigenvalues of A = Sum of principal diagonal elements of A

$$3 - 2 = a + b$$

$$a + b = 1 \quad \dots(1)$$

Product of the eigenvalues of A = Determinant of A

$$\lambda_1 \lambda_2 = |A|$$

$$(3)(-2) = \begin{vmatrix} a & 4 \\ 1 & b \end{vmatrix}$$

$$= ab - 4$$

$$ab = -2 \quad \dots(2)$$

From Eq. (1), $a = 1 - b$

Substituting in Eq. (2),

$$\begin{aligned}(1-b)b &= -2 \\ b-b^2 &= -2 \\ b^2-b-2 &= 0 \\ b &= 2 \quad \text{or} \quad b = -1 \\ \therefore a &= -1 \quad \text{or} \quad a = 2\end{aligned}$$

Example 11

Two eigenvalues of the matrix $A = \begin{bmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{bmatrix}$ are 1 and 1. Find the eigenvalues of A^{-1} .

Solution

Let λ_1, λ_2 and λ_3 be the eigenvalues of the matrix A .

$$\lambda_1 = \lambda_2 = 1$$

Sum of the eigenvalues of A = Sum of principal diagonal elements of A

$$\begin{aligned}1 + 1 + \lambda_3 &= 2 + 3 + 2 \\ \lambda_3 &= 5\end{aligned}$$

Hence, the eigenvalues of A are 1, 1, 5.

The eigenvalues of A^{-1} are $\frac{1}{1}, \frac{1}{1}, \frac{1}{5}$, i.e., 1, 1, $\frac{1}{5}$.

Example 12

Two of the eigenvalues of the matrix $A = \begin{bmatrix} 3 & -1 & 1 \\ -1 & 5 & -1 \\ 1 & -1 & 3 \end{bmatrix}$ are 3 and 6. Find the eigenvalues of A^{-1} and A^3 .

Solution

Let λ_1, λ_2 and λ_3 be the eigenvalues of the matrix A .

$$\lambda_1 = 3, \lambda_2 = 6$$

Sum of the eigenvalues of A = Sum of the principal diagonal elements of A

$$\begin{aligned}3 + 6 + \lambda_3 &= 3 + 5 + 3 \\ \lambda_3 &= 2\end{aligned}$$

The eigenvalues of A are 3, 6, 2.

Hence, the eigenvalues of A^{-1} are $\frac{1}{3}, \frac{1}{6}, \frac{1}{2}$, and the eigenvalues of A^3 are $3^3, 6^3, 2^3$, i.e., 27, 216, 8.

Example 13

Form the matrix whose eigenvalues are $\alpha - 5, \beta - 5, \gamma - 5$ where α, β, γ are the eigenvalues of $A = \begin{bmatrix} -1 & -2 & -3 \\ 4 & 5 & -6 \\ 7 & -8 & 9 \end{bmatrix}$.

Solution

If λ_1, λ_2 and λ_3 are eigenvalues of the matrix A then $\lambda_1 - k, \lambda_2 - k$ and $\lambda_3 - k$ are the eigenvalues of $A - kI$.

$$\begin{aligned} \text{Required matrix} &= A - 5I = \begin{bmatrix} -1 & -2 & -3 \\ 4 & 5 & -6 \\ 7 & -8 & 9 \end{bmatrix} - 5 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} -6 & -2 & -3 \\ 4 & 0 & -6 \\ 7 & -8 & 4 \end{bmatrix} \end{aligned}$$

Example 14

If α and β are the eigenvalues of $\begin{bmatrix} 3 & -1 \\ -1 & 5 \end{bmatrix}$, form a matrix whose eigenvalues are α^3 and β^3 .

Solution

If λ_1 and λ_2 are the eigenvalues of the matrix A then λ_1^k and λ_2^k are the eigenvalues of A^k .

Let $A = \begin{bmatrix} 3 & -1 \\ -1 & 5 \end{bmatrix}$

$$A^2 = AA = \begin{bmatrix} 3 & -1 \\ -1 & 5 \end{bmatrix} \begin{bmatrix} 3 & -1 \\ -1 & 5 \end{bmatrix} = \begin{bmatrix} 10 & -8 \\ -8 & 26 \end{bmatrix}$$

$$\text{Required matrix} = A^3 = A^2 A = \begin{bmatrix} 10 & -8 \\ -8 & 26 \end{bmatrix} \begin{bmatrix} 3 & -1 \\ -1 & 5 \end{bmatrix} = \begin{bmatrix} 38 & -50 \\ -50 & 138 \end{bmatrix}$$

Example 15

If $A = \begin{bmatrix} 3 & -1 & 1 \\ -1 & 5 & -1 \\ 1 & -1 & 3 \end{bmatrix}$, find the eigenvalues for the following matrices:

(i) A (ii) A^T (iii) A^{-1} (iv) $4A^{-1}$ (v) A^2 (vi) $A^2 - 2A + I$ (vii) $A^3 + 2I$

Solution

$$A = \begin{bmatrix} 3 & -1 & 1 \\ -1 & 5 & -1 \\ 1 & -1 & 3 \end{bmatrix}$$

The characteristic equation is

$$\det(A - \lambda I) = 0$$

$$\begin{vmatrix} 3-\lambda & -1 & 1 \\ -1 & 5-\lambda & -1 \\ 1 & -1 & 3-\lambda \end{vmatrix} = 0$$

$$\lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0$$

where S_1 = Sum of the principal diagonal elements of $A = 3 + 5 + 3 = 11$

S_2 = Sum of the minors of principal diagonal elements of A

$$\begin{aligned} &= \begin{vmatrix} 5 & -1 \\ -1 & 3 \end{vmatrix} + \begin{vmatrix} 3 & 1 \\ 1 & 3 \end{vmatrix} + \begin{vmatrix} 3 & -1 \\ -1 & 5 \end{vmatrix} \\ &= (15 - 1) + (9 - 1) + (15 - 1) \\ &= 14 + 8 + 14 \\ &= 36 \end{aligned}$$

$$\begin{aligned} S_3 &= \det(A) = \begin{vmatrix} 3 & -1 & 1 \\ -1 & 5 & -1 \\ 1 & -1 & 3 \end{vmatrix} \\ &= 3(15 - 1) + 1(-3 + 1) + 1(1 - 5) \\ &= 42 - 2 - 4 \\ &= 36 \end{aligned}$$

Hence, the characteristic equation is

$$\lambda^3 - 11\lambda^2 + 36\lambda - 36 = 0$$

$$\lambda = 2, 3, 6$$

- | | |
|--|---|
| (i) Eigenvalues of $A = \lambda$ | : 2, 3, 6 |
| (ii) Eigenvalues of $A^T = \lambda$ | : 2, 3, 6 |
| (iii) Eigenvalues of $A^{-1} = \lambda^{-1}$ | : $\frac{1}{2}, \frac{1}{3}, \frac{1}{6}$ |

- (iv) Eigenvalues of $4A^{-1} = 4\lambda^{-1}$: $2, \frac{4}{3}, \frac{2}{3}$
 (v) Eigenvalues of $A^2 = \lambda^2$: $4, 9, 36$
 (vi) Eigenvalues of $A^2 - 2A + I = \lambda^2 - 2\lambda + 1$: $1, 4, 25$
 (vii) Eigenvalues of $A^3 + 2I = \lambda^3 + 2$: $10, 29, 218$

10.12 LINEAR DEPENDENCE AND INDEPENDENCE OF EIGENVECTORS

10.12.1 Linear Dependence

A set of r eigenvectors $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_r$ is said to be linearly dependent if there exist r scalars (numbers) $k_1, k_2, \dots, k_r$ not all zero, such that

$$k_1\mathbf{x}_1 + k_2\mathbf{x}_2 + \dots + k_r\mathbf{x}_r = \mathbf{0}$$

10.12.2 Linear Independence

A set of r eigenvectors $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_r$ is said to be linearly independent if there exist r scalars (numbers) $k_1, k_2, \dots, k_r$ such that if

$$k_1\mathbf{x}_1 + k_2\mathbf{x}_2 + \dots + k_r\mathbf{x}_r = \mathbf{0}$$

then $k_1 = k_2 = \dots = k_r = 0$

- Note:** (1) If a set of eigenvectors is linearly dependent then at least one eigenvector of the set can be expressed as a linear combination of the remaining eigenvectors.
 (2) If a set of eigenvectors is linearly independent then no eigenvector of the set can be expressed as a linear combination of the remaining eigenvectors.

10.13 PROPERTIES OF EIGENVECTORS

Property 1: If $\mathbf{x}$ is an eigenvector of a matrix A corresponding to the eigenvalue λ then $k\mathbf{x}$ is also an eigenvector of A corresponding to the same eigenvalue λ , where k is a nonzero scalar.

Proof: Let $\mathbf{x}$ be an eigenvector of a matrix A corresponding to the eigenvalue λ .

$$A\mathbf{x} = \lambda\mathbf{x}$$

and

$$\mathbf{x} \neq \mathbf{0}$$

If k is any nonzero scalar then $k\mathbf{x} \neq \mathbf{0}$.

Also,

$$A(k\mathbf{x}) = k(A\mathbf{x}) = k(\lambda\mathbf{x}) = \lambda(k\mathbf{x})$$

Hence, $k\mathbf{x}$ is an eigenvector of A corresponding to the eigenvalue λ . Thus, corresponding to an eigenvalue λ , there is more than one eigenvectors.

Property 2: If $\mathbf{x}$ is an eigenvector of a matrix A then $\mathbf{x}$ cannot correspond to more than one eigenvalue of A .

Proof: Let $\mathbf{x}$ be an eigenvector of a matrix A corresponding to two eigenvalues λ_1 and λ_2 .

$$\begin{aligned} A\mathbf{x} &= \lambda_1\mathbf{x} \\ \text{and } A\mathbf{x} &= \lambda_2\mathbf{x} \\ \therefore \lambda_1\mathbf{x} &= \lambda_2\mathbf{x} \\ (\lambda_1 - \lambda_2)\mathbf{x} &= 0 \\ \lambda_1 - \lambda_2 &= 0 & (\because \mathbf{x} \neq 0) \\ \lambda_1 &= \lambda_2 \end{aligned}$$

Hence, $\mathbf{x}$ cannot correspond to more than one eigenvalue of A .

Property 3: The eigenvectors corresponding to distinct eigenvalues of a matrix are linearly independent.

Proof: Let $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_m$ be the eigenvectors of a matrix A corresponding to distinct eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_m$.

$$A\mathbf{x}_i = \lambda_i\mathbf{x}_i, i = 1, 2, \dots, m$$

Let the eigenvectors $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_m$ be linearly dependent. Choose r such that the eigenvectors $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_r$ are linearly independent but $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_r, \mathbf{x}_{r+1}$ are linearly dependent, where $1 \leq r < m$.

$$\text{Then } k_1\mathbf{x}_1 + k_2\mathbf{x}_2 + \dots + k_r\mathbf{x}_r + k_{r+1}\mathbf{x}_{r+1} = 0 \quad \dots (10.9)$$

where $k_1, k_2, \dots, k_r, k_{r+1}$ are scalars not all zero.

$$\begin{aligned} A(k_1\mathbf{x}_1 + k_2\mathbf{x}_2 + \dots + k_r\mathbf{x}_r + k_{r+1}\mathbf{x}_{r+1}) &= 0 \\ k_1(A\mathbf{x}_1) + k_2(A\mathbf{x}_2) + \dots + k_r(A\mathbf{x}_r) + k_{r+1}(A\mathbf{x}_{r+1}) &= 0 \\ k_1(\lambda_1\mathbf{x}_1) + k_2(\lambda_2\mathbf{x}_2) + \dots + k_r(\lambda_r\mathbf{x}_r) + k_{r+1}(\lambda_{r+1}\mathbf{x}_{r+1}) &= 0 \end{aligned} \quad \dots (10.10)$$

Multiplying Eq. (10.9) by λ_{r+1} and subtracting from Eq. (10.10),

$$k_1(\lambda_1 - \lambda_{r+1})\mathbf{x}_1 + k_2(\lambda_2 - \lambda_{r+1})\mathbf{x}_2 + \dots + k_r(\lambda_r - \lambda_{r+1})\mathbf{x}_r = 0 \quad \dots (10.11)$$

Since $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_r$ are linearly independent,

$$k_1(\lambda_1 - \lambda_{r+1}) = 0, k_2(\lambda_2 - \lambda_{r+1}) = 0, \dots, k_r(\lambda_r - \lambda_{r+1}) = 0$$

But $\lambda_1, \lambda_2, \dots, \lambda_r, \lambda_{r+1}$ are distinct.

$$\therefore k_1 = 0, k_2 = 0, \dots, k_r = 0$$

Putting in Eq. (10.9),

$$k_{r+1}\mathbf{x}_{r+1} = 0$$

$$k_{r+1} = 0 \quad [\because \mathbf{x}_{r+1} \neq 0]$$

Thus, for Eq. (10.9),

$$k_1 = 0, k_2 = 0, k_r = 0, k_{r+1} = 0$$

which contradicts our assumption that $k_1, k_2, \dots, k_r, k_{r+1}$ are not all zero.

Thus, $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_{r+1}$ are linearly independent and hence $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_m$ are linearly independent.

Note: If two or more eigenvalues are equal then the corresponding eigenvectors may or may not be linearly independent.

Property 4: The eigenvectors corresponding to distinct eigenvalues of a real symmetric matrix are orthogonal.

Proof: Let λ_1 and λ_2 be the two distinct eigenvalues of a real symmetric matrix A and $\mathbf{x}_1$ and $\mathbf{x}_2$ be the corresponding eigenvectors respectively.

$$A\mathbf{x}_1 = \lambda_1\mathbf{x}_1 \quad \dots(10.12)$$

$$\text{and} \quad A\mathbf{x}_2 = \lambda_2\mathbf{x}_2 \quad \dots(10.13)$$

Premultiplying both sides of Eq. (10.12) by $\mathbf{x}_2^T$,

$$\mathbf{x}_2^T A\mathbf{x}_1 = \lambda_1 \mathbf{x}_2^T \mathbf{x}_1$$

Taking the transpose on both sides,

$$\mathbf{x}_1^T A\mathbf{x}_2 = \lambda_1 \mathbf{x}_1^T \mathbf{x}_2 \quad (\because A^T = A) \quad \dots(10.14)$$

Premultiplying both sides of Eq. (10.13) by $\mathbf{x}_1^T$,

$$\mathbf{x}_1^T A\mathbf{x}_2 = \lambda_2 \mathbf{x}_1^T \mathbf{x}_2 \quad \dots(10.15)$$

From Eqs (10.14) and (10.15),

$$\begin{aligned} \lambda_1 \mathbf{x}_1^T \mathbf{x}_2 &= \lambda_2 \mathbf{x}_1^T \mathbf{x}_2 \\ (\lambda_1 - \lambda_2) \mathbf{x}_1^T \mathbf{x}_2 &= 0 \\ \mathbf{x}_1^T \mathbf{x}_2 &= 0 \quad (\because \lambda_1 \neq \lambda_2) \end{aligned}$$

Hence, the eigenvectors $\mathbf{x}_1$ and $\mathbf{x}_2$ are orthogonal.

Working Rule for Finding the Eigenvalues and Eigenvectors

- (i) Write characteristic equation $\det(A - \lambda I) = 0$ for the given square matrix.
- (ii) Find the eigenvalues of the matrix by solving characteristic equation.
- (iii) Find eigenvectors corresponding to each eigenvalues from the equation $[A - \lambda I] \mathbf{x} = \mathbf{0}$.

Example 1

Find the eigenvalues and eigenvectors of the matrix $\begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix}$.

Solution

Let

$$A = \begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix}$$

The characteristic equation is

$$\det(A - \lambda I) = 0$$

$$\begin{vmatrix} 5-\lambda & 4 \\ 1 & 2-\lambda \end{vmatrix} = 0$$

$$\lambda^2 - S_1\lambda + S_2 = 0$$

where $S_1 = \text{Sum of the principal diagonal elements of } A = 5 + 2 = 7$

$$\begin{aligned} S_2 = \det(A) &= \begin{vmatrix} 5 & 4 \\ 1 & 2 \end{vmatrix} \\ &= 10 - 4 \\ &= 6 \end{aligned}$$

Hence, the characteristic equation is

$$\begin{aligned} \lambda^2 - 7\lambda + 6 &= 0 \\ \lambda &= 6, 1 \end{aligned}$$

(a) For $\lambda = 6$,

$$\begin{aligned} [A - \lambda I]\mathbf{x} &= \mathbf{0} \\ \begin{bmatrix} -1 & 4 \\ 1 & -4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} &= \begin{bmatrix} 0 \\ 0 \end{bmatrix} \\ -x + 4y &= 0 \end{aligned}$$

$$\begin{aligned} \text{Let} \quad y &= t \\ x &= 4t \end{aligned}$$

$$\mathbf{x} = \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 4t \\ t \end{bmatrix} = t \begin{bmatrix} 4 \\ 1 \end{bmatrix} = t\mathbf{x}_1 \text{ where } \mathbf{x}_1 \text{ is an eigenvector corresponding to } \lambda = 6.$$

(b) For $\lambda = 1$,

$$\begin{aligned} [A - \lambda I]\mathbf{x} &= \mathbf{0} \\ \begin{bmatrix} 4 & 4 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} &= \begin{bmatrix} 0 \\ 0 \end{bmatrix} \\ x + y &= 0 \end{aligned}$$

$$\begin{aligned} \text{Let} \quad y &= t \\ x &= -t \end{aligned}$$

$$\mathbf{x} = \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -t \\ t \end{bmatrix} = t \begin{bmatrix} -1 \\ 1 \end{bmatrix} = t\mathbf{x}_2 \text{ where } \mathbf{x}_2 \text{ is an eigenvector corresponding to } \lambda = 1.$$

Example 2

Find the eigenvalues and eigenvectors of the matrix $\begin{bmatrix} 4 & 6 & 6 \\ 1 & 3 & 2 \\ -1 & -4 & -3 \end{bmatrix}$.

Solution

$$\begin{aligned} \text{Let} \quad A &= \begin{bmatrix} 4 & 6 & 6 \\ 1 & 3 & 2 \\ -1 & -4 & -3 \end{bmatrix} \end{aligned}$$

The characteristic equation is

$$\begin{aligned}\det(A - \lambda I) &= 0 \\ \begin{vmatrix} 4-\lambda & 6 & 6 \\ 1 & 3-\lambda & 2 \\ -1 & -4 & -3-\lambda \end{vmatrix} &= 0 \\ \lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 &= 0\end{aligned}$$

where S_1 = Sum of the principal diagonal elements of $A = 4 + 3 - 3 = 4$

S_2 = Sum of the minors of principal diagonal elements of A

$$\begin{aligned}&= \begin{vmatrix} 3 & 2 \\ -4 & -3 \end{vmatrix} + \begin{vmatrix} 4 & 6 \\ -1 & -3 \end{vmatrix} + \begin{vmatrix} 4 & 6 \\ 1 & 3 \end{vmatrix} \\&= (-9+8) + (-12+6) + (12-6) \\&= -1-6+6 \\&= -1\end{aligned}$$

$$\begin{aligned}S_3 = \det(A) &= \begin{vmatrix} 4 & 6 & 6 \\ 1 & 3 & 2 \\ -1 & -4 & -3 \end{vmatrix} \\&= 4(-9+8) - 6(-3+2) + 6(-4+3) \\&= -4+6-6 \\&= -4\end{aligned}$$

Hence, the characteristic equation is

$$\begin{aligned}\lambda^3 - 4\lambda^2 - \lambda + 4 &= 0 \\ \lambda &= -1, 1, 4\end{aligned}$$

(a) For $\lambda = -1$, $[A - \lambda I]\mathbf{x} = \mathbf{0}$

$$\begin{aligned}\begin{bmatrix} 5 & 6 & 6 \\ 1 & 4 & 2 \\ -1 & -4 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} &= \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \\ 5x + 6y + 6z &= 0 \\ x + 4y + 2z &= 0\end{aligned}$$

By Cramer's rule,

$$\frac{x}{\begin{vmatrix} 6 & 6 \\ 4 & 2 \end{vmatrix}} = -\frac{y}{\begin{vmatrix} 5 & 6 \\ 1 & 2 \end{vmatrix}} = \frac{z}{\begin{vmatrix} 5 & 6 \\ 1 & 4 \end{vmatrix}} = t$$

$$\frac{x}{-12} = \frac{y}{-4} = \frac{z}{14} = t$$

$$\frac{x}{-6} = \frac{y}{-2} = \frac{z}{7} = t$$

$$\mathbf{x} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -6t \\ -2t \\ 7t \end{bmatrix} = t \begin{bmatrix} -6 \\ -2 \\ 7 \end{bmatrix} = t\mathbf{x}_1 \text{ where } \mathbf{x}_1 \text{ is an eigenvector corresponding to } \lambda = -1.$$

(b) For $\lambda = 1$,

$$[A - \lambda I]\mathbf{x} = \mathbf{0}$$

$$\begin{bmatrix} 3 & 6 & 6 \\ 1 & 2 & 2 \\ -1 & -4 & -4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$x + 2y + 2z = 0$$

$$-x - 4y - 4z = 0$$

By Cramer's rule,

$$\frac{x}{\begin{vmatrix} 2 & 2 \\ -4 & -4 \end{vmatrix}} = -\frac{y}{\begin{vmatrix} 1 & 2 \\ -1 & -4 \end{vmatrix}} = \frac{z}{\begin{vmatrix} 1 & 2 \\ -1 & -4 \end{vmatrix}} = t$$

$$\frac{x}{0} = \frac{y}{2} = \frac{z}{-2} = t$$

$$\frac{x}{0} = \frac{y}{1} = \frac{z}{-1} = t$$

$$\mathbf{x} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ t \\ -t \end{bmatrix} = t \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} = t\mathbf{x}_2 \text{ where } \mathbf{x}_2 \text{ is an eigenvector corresponding to } \lambda = 1.$$

(c) For $\lambda = 4$,

$$[A - \lambda I]\mathbf{x} = \mathbf{0}$$

$$\begin{bmatrix} 0 & 6 & 6 \\ 1 & -1 & 2 \\ -1 & -4 & -7 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$0x + 6y + 6z = 0$$

$$x - y + 2z = 0$$

$$x - 4y - 7z = 0$$

By Cramer's rule,

$$\frac{x}{\begin{vmatrix} 6 & 6 \\ -1 & 2 \end{vmatrix}} = -\frac{y}{\begin{vmatrix} 0 & 6 \\ 1 & 2 \end{vmatrix}} = \frac{z}{\begin{vmatrix} 0 & 6 \\ 1 & -1 \end{vmatrix}} = t$$

$$\frac{x}{18} = \frac{y}{6} = \frac{z}{-6} = t$$

$$\frac{x}{3} = \frac{y}{1} = \frac{z}{-1} = t$$

$$\mathbf{x} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3t \\ t \\ -t \end{bmatrix} = t \begin{bmatrix} 3 \\ 1 \\ -1 \end{bmatrix} = t\mathbf{x}_3 \text{ where } \mathbf{x}_3 \text{ is an eigenvector corresponding to } \lambda = 4.$$

Example 3

Find the eigenvalues and eigenvectors of the matrix $\begin{bmatrix} 1 & 0 & -1 \\ 1 & 2 & 1 \\ 2 & 2 & 3 \end{bmatrix}$.

Solution

Let $A = \begin{bmatrix} 1 & 0 & -1 \\ 1 & 2 & 1 \\ 2 & 2 & 3 \end{bmatrix}$

The characteristic equation is

$$\det(A - \lambda I) = 0$$

$$\begin{vmatrix} 1-\lambda & 0 & -1 \\ 1 & 2-\lambda & 1 \\ 2 & 2 & 3-\lambda \end{vmatrix} = 0$$

$$\lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0$$

where S_1 = Sum of the principal diagonal elements of $A = 1 + 2 + 3 = 6$

S_2 = Sum of the minors of principal diagonal elements of A

$$\begin{aligned} &= \begin{vmatrix} 2 & 1 \\ 2 & 3 \end{vmatrix} + \begin{vmatrix} 1 & -1 \\ 2 & 3 \end{vmatrix} + \begin{vmatrix} 1 & 0 \\ 1 & 2 \end{vmatrix} \\ &= (6-2) + (3+2) + (2-0) \\ &= 4+5+2 \\ &= 11 \end{aligned}$$

$$\begin{aligned} S_3 &= \det(A) = \begin{vmatrix} 1 & 0 & -1 \\ 1 & 2 & 1 \\ 2 & 2 & 3 \end{vmatrix} \\ &= 1(6-2) - 0 - 1(2-4) \\ &= 6 \end{aligned}$$

Hence, the characteristic equation is

$$\lambda^3 - 6\lambda^2 + 11\lambda - 6 = 0$$

$$\lambda = 1, 2, 3$$

(a) For $\lambda = 1$,

$$[A - \lambda I]\mathbf{x} = \mathbf{0}$$

$$\begin{bmatrix} 0 & 0 & -1 \\ 1 & 1 & 1 \\ 2 & 2 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$0x + 0y - z = 0$$

$$x + y + z = 0$$

By Cramer's rule,

$$\frac{x}{\begin{vmatrix} 0 & -1 \\ 1 & 1 \end{vmatrix}} = -\frac{y}{\begin{vmatrix} 0 & -1 \\ 1 & 1 \end{vmatrix}} = \frac{z}{\begin{vmatrix} 0 & 0 \\ 1 & 1 \end{vmatrix}} = t$$

$$\frac{x}{1} = \frac{y}{-1} = \frac{z}{0} = t$$

$$\mathbf{x} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} t \\ -t \\ 0 \end{bmatrix} = t \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} = t\mathbf{x}_1 \text{ where } \mathbf{x}_1 \text{ is an eigenvector corresponding to } \lambda = 1.$$

(b) For $\lambda = 2$,

$$[A - \lambda I]\mathbf{x} = \mathbf{0}$$

$$\begin{bmatrix} -1 & 0 & -1 \\ 1 & 0 & 1 \\ 2 & 2 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$x + 0y + z = 0$$

$$2x + 2y + z = 0$$

By Cramer's rule,

$$\frac{x}{\begin{vmatrix} 0 & 1 \\ 2 & 1 \end{vmatrix}} = -\frac{y}{\begin{vmatrix} 1 & 1 \\ 2 & 1 \end{vmatrix}} = \frac{z}{\begin{vmatrix} 1 & 0 \\ 2 & 2 \end{vmatrix}} = t$$

$$\frac{x}{-2} = \frac{y}{1} = \frac{z}{2} = t$$

$$\mathbf{x} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -2t \\ t \\ 2t \end{bmatrix} = t \begin{bmatrix} -2 \\ 1 \\ 2 \end{bmatrix} = t\mathbf{x}_2 \text{ where } \mathbf{x}_2 \text{ is an eigenvector corresponding to } \lambda = 2.$$

(c) For $\lambda = 3$,

$$[A - \lambda I]\mathbf{x} = \mathbf{0}$$

$$\begin{bmatrix} -2 & 0 & -1 \\ 1 & -1 & 1 \\ 2 & 2 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$-2x + 0y - z = 0$$

$$x - y + z = 0$$

$$2x + 2y + 0z = 0$$

By Cramer's rule,

$$\frac{x}{\begin{vmatrix} 0 & -1 \\ -1 & 1 \end{vmatrix}} = -\frac{y}{\begin{vmatrix} -2 & -1 \\ 1 & 1 \end{vmatrix}} = \frac{z}{\begin{vmatrix} -2 & 0 \\ 1 & -1 \end{vmatrix}} = t$$

$$\frac{x}{-1} = \frac{y}{1} = \frac{z}{2} = t$$

$$\mathbf{x} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -t \\ t \\ 2t \end{bmatrix} = t \begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix} = t\mathbf{x}_3 \text{ where } \mathbf{x}_3 \text{ is an eigenvector corresponding to } \lambda = 3.$$

Example 4

Find the eigenvalues and eigenvectors of the matrix $\begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$.

Solution

Let

$$A = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$$

The characteristic equation is

$$\det(A - \lambda I) = 0$$

$$\begin{vmatrix} 8-\lambda & -6 & -2 \\ -6 & 7-\lambda & -4 \\ 2 & -4 & 3-\lambda \end{vmatrix} = 0$$

$$\lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0$$

where S_1 = Sum of the principal diagonal elements of $A = 8 + 7 + 3 = 18$ S_2 = Sum of the minors of principal diagonal elements of A

$$\begin{aligned}
&= \begin{vmatrix} 7 & -4 \\ -4 & 3 \end{vmatrix} + \begin{vmatrix} 8 & 2 \\ 2 & 3 \end{vmatrix} + \begin{vmatrix} 8 & -6 \\ -6 & 7 \end{vmatrix} \\
&= (21-16) + (24-4) + (56-36) \\
&= 5 + 20 + 20 \\
&= 45
\end{aligned}$$

$$\begin{aligned}
S_3 = \det(A) &= \begin{vmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{vmatrix} \\
&= 8(21-16) + 6(-18+8) + 2(24-14) \\
&= 40 - 60 + 20 \\
&= 0
\end{aligned}$$

Hence, the characteristic equation is

$$\begin{aligned}
\lambda^3 - 18\lambda^2 + 45\lambda &= 0 \\
\lambda &= 0, 3, 15
\end{aligned}$$

(a) For $\lambda = 0$,

$$[A - \lambda I]\mathbf{x} = \mathbf{0}$$

$$\begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$8x - 6y + 2z = 0$$

$$-6x + 7y - 4z = 0$$

$$2x - 4y + 3z = 0$$

By Cramer's rule,

$$\frac{x}{\begin{vmatrix} -6 & 2 \\ 7 & -4 \end{vmatrix}} = -\frac{y}{\begin{vmatrix} 8 & 2 \\ -6 & -4 \end{vmatrix}} = \frac{z}{\begin{vmatrix} 8 & -6 \\ -6 & 7 \end{vmatrix}} = t$$

$$\frac{x}{10} = \frac{y}{20} = \frac{z}{20} = t$$

$$\frac{x}{1} = \frac{y}{2} = \frac{z}{2} = t$$

$$\mathbf{x} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} t \\ 2t \\ 2t \end{bmatrix} = t \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} = t\mathbf{x}_1 \text{ where } \mathbf{x}_1 \text{ is an eigenvector corresponding to } \lambda = 0.$$

(b) For $\lambda = 3$,

$$[A - \lambda I]\mathbf{x} = \mathbf{0}$$

$$\begin{bmatrix} 5 & -6 & 2 \\ -6 & 4 & -4 \\ 2 & -4 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$5x - 6y + 2z = 0$$

$$-6x + 4y - 4z = 0$$

$$2x - 4y + 0z = 0$$

By Cramer's rule,

$$\frac{x}{\begin{vmatrix} -6 & 2 \\ 4 & -4 \end{vmatrix}} = -\frac{y}{\begin{vmatrix} 5 & 2 \\ -6 & -4 \end{vmatrix}} = \frac{z}{\begin{vmatrix} 5 & -6 \\ -6 & 4 \end{vmatrix}} = t$$

$$\frac{x}{16} = \frac{y}{8} = \frac{z}{-16} = t$$

$$\frac{x}{2} = \frac{y}{1} = \frac{z}{-2} = t$$

$$\mathbf{x} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2t \\ t \\ -2t \end{bmatrix} = t \begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix} = t\mathbf{x}_2 \text{ where } \mathbf{x}_2 \text{ is an eigenvector corresponding to } \lambda = 3.$$

(c) For $\lambda = 15$,

$$[A - \lambda I]\mathbf{x} = \mathbf{0}$$

$$\begin{bmatrix} -7 & -6 & 2 \\ -6 & -8 & -4 \\ 2 & -4 & -12 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$-7x - 6y + 2z = 0$$

$$-6x - 8y - 4z = 0$$

$$2x - 4y - 12z = 0$$

By Cramer's rule,

$$\frac{x}{\begin{vmatrix} -6 & 2 \\ -8 & -4 \end{vmatrix}} = \frac{y}{\begin{vmatrix} -7 & 2 \\ -6 & -4 \end{vmatrix}} = \frac{z}{\begin{vmatrix} -7 & -6 \\ -6 & -8 \end{vmatrix}} = t$$

$$\frac{x}{40} = -\frac{y}{40} = \frac{z}{20} = t$$

$$\frac{x}{2} = \frac{y}{-2} = \frac{z}{1} = t$$

$$\mathbf{x} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2t \\ -2t \\ t \end{bmatrix} = t \begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix} = t\mathbf{x}_3 \text{ where } \mathbf{x}_3 \text{ is an eigenvector corresponding to } \lambda = 15.$$

Note: The eigenvectors corresponding to distinct eigenvalues of a real symmetric matrix are orthogonal which can be verified with this example.

$$\mathbf{x}_1^T \mathbf{x}_2 = \begin{bmatrix} 1 & 2 & 2 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix} = [0] = \mathbf{0}$$

$$\mathbf{x}_2^T \mathbf{x}_3 = \begin{bmatrix} 2 & 1 & -2 \end{bmatrix} \begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix} = [0] = \mathbf{0}$$

$$\mathbf{x}_3^T \mathbf{x}_1 = \begin{bmatrix} 2 & -2 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} = [0] = \mathbf{0}$$

Thus, $\mathbf{x}_1$, $\mathbf{x}_2$ and $\mathbf{x}_3$ are orthogonal to each other.

Example 5

Find the eigenvalues and eigenvectors of the matrix $\begin{bmatrix} 4 & 2 & -2 \\ -5 & 3 & 2 \\ -2 & 4 & 1 \end{bmatrix}$.

Solution

Let $A = \begin{bmatrix} 4 & 2 & -2 \\ -5 & 3 & 2 \\ -2 & 4 & 1 \end{bmatrix}$

The characteristic equation is

$$\begin{aligned} \det(A - \lambda I) &= 0 \\ \begin{vmatrix} 4-\lambda & 2 & -2 \\ -5 & 3-\lambda & 2 \\ -2 & 4 & 1-\lambda \end{vmatrix} &= 0 \\ \lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 &= 0 \end{aligned}$$

where S_1 = Sum of the principal diagonal elements of $A = 4 + 3 + 1 = 8$

S_2 = Sum of the minors of principal diagonal elements of A

$$\begin{aligned} &= \begin{vmatrix} 3 & 2 \\ 4 & 1 \end{vmatrix} + \begin{vmatrix} 4 & -2 \\ -2 & 1 \end{vmatrix} + \begin{vmatrix} 4 & 2 \\ -5 & 3 \end{vmatrix} \\ &= (3-8) + (4-4) + (12+10) \\ &= -5 + 0 + 22 \\ &= 17 \end{aligned}$$

$$\begin{aligned}
 S_3 = \det(A) &= \begin{vmatrix} 4 & 2 & -2 \\ -5 & 3 & 2 \\ -2 & 4 & 1 \end{vmatrix} \\
 &= 4(3-8) - 2(-5+4) - 2(-20+6) \\
 &= -20 + 2 + 28 \\
 &= 10
 \end{aligned}$$

Hence, the characteristic equation is

$$\begin{aligned}
 \lambda^3 - 8\lambda^2 + 17\lambda - 10 &= 0 \\
 \lambda &= 1, 2, 5
 \end{aligned}$$

(a) For $\lambda = 1$,

$$[A - \lambda I]\mathbf{x} = \mathbf{0}$$

$$\begin{aligned}
 \begin{bmatrix} 3 & 2 & -2 \\ -5 & 2 & 2 \\ -2 & 4 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} &= \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \\
 3x + 2y - 2z &= 0 \\
 -5x + 2y + 2z &= 0 \\
 -2x + 2y + 0z &= 0
 \end{aligned}$$

By Cramer's rule,

$$\frac{x}{\begin{vmatrix} 2 & -2 \\ 2 & 2 \end{vmatrix}} = -\frac{y}{\begin{vmatrix} 3 & -2 \\ -5 & 2 \end{vmatrix}} = \frac{z}{\begin{vmatrix} 3 & 2 \\ -5 & 2 \end{vmatrix}} = t$$

$$\frac{x}{8} = \frac{y}{4} = \frac{z}{16} = t$$

$$\frac{x}{2} = \frac{y}{1} = \frac{z}{4} = t$$

$$\mathbf{x} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2t \\ t \\ 4t \end{bmatrix} = t \begin{bmatrix} 2 \\ 1 \\ 4 \end{bmatrix} = t\mathbf{x}_1 \text{ where vector } \mathbf{x}_1 \text{ is an eigenvector corresponding}$$

to $\lambda = 1$.

(b) For $\lambda = 2$,

$$[A - \lambda I]\mathbf{x} = \mathbf{0}$$

$$\begin{aligned}
 \begin{bmatrix} 2 & 2 & -2 \\ -5 & 1 & 2 \\ -2 & 4 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} &= \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \\
 2x + 2y - 2z &= 0 \\
 -5x + y + 2z &= 0 \\
 -2x + 2y - z &= 0
 \end{aligned}$$

By Cramer's rule,

$$\frac{x}{\begin{vmatrix} 2 & -2 \\ 1 & 2 \end{vmatrix}} = -\frac{y}{\begin{vmatrix} 2 & -2 \\ -5 & 2 \end{vmatrix}} = \frac{z}{\begin{vmatrix} 2 & 2 \\ -5 & 1 \end{vmatrix}} = t$$

$$\frac{x}{6} = \frac{y}{6} = \frac{z}{12} = t$$

$$\frac{x}{1} = \frac{y}{1} = \frac{z}{2} = t$$

$$\mathbf{x} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = t \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} = t\mathbf{x}_2 \text{ where vector } \mathbf{x}_2 \text{ is an eigenvector corresponding to } \lambda = 2.$$

(c) For $\lambda = 5$,

$$[A - \lambda I]\mathbf{x} = \mathbf{0}$$

$$\begin{bmatrix} -1 & 2 & -2 \\ -5 & -2 & 2 \\ -2 & 4 & -4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$-x + 2y - 2z = 0$$

$$-5x - 2y + 2z = 0$$

By Cramer's rule,

$$\frac{x}{\begin{vmatrix} 2 & -2 \\ -2 & 2 \end{vmatrix}} = -\frac{y}{\begin{vmatrix} -1 & -2 \\ -5 & 2 \end{vmatrix}} = \frac{z}{\begin{vmatrix} -1 & 2 \\ -5 & -2 \end{vmatrix}} = t$$

$$\frac{x}{0} = \frac{y}{12} = \frac{z}{12} = t$$

$$\frac{x}{0} = \frac{y}{1} = \frac{z}{1} = t$$

$$\mathbf{x} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ t \\ t \end{bmatrix} = t \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} = t\mathbf{x}_3 \text{ where } \mathbf{x}_3 \text{ is an eigenvector corresponding to } \lambda = 5.$$

Example 6

Find the eigenvalues and eigenvectors of the matrix $\begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}$.

Solution

Let

$$A = \begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}$$

The characteristic equation is

$$\det(A - \lambda I) = 0$$

$$\begin{vmatrix} -2 - \lambda & 2 & -3 \\ 2 & 1 - \lambda & -6 \\ -1 & -2 & -\lambda \end{vmatrix} = 0$$

$$\lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0$$

where S_1 = Sum of the principal diagonal elements of $A = -2 + 1 + 0 = -1$

S_2 = Sum of the minors of principal diagonal elements of A

$$= \begin{vmatrix} 1 & -6 \\ -2 & 0 \end{vmatrix} + \begin{vmatrix} -2 & -3 \\ -1 & 0 \end{vmatrix} + \begin{vmatrix} -2 & 2 \\ 2 & 1 \end{vmatrix}$$

$$= (0 - 12) + (0 - 3) + (-2 - 4)$$

$$= -12 - 3 - 6$$

$$= -21$$

$$S_3 = \det(A) = \begin{vmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{vmatrix}$$

$$= (-2)(0 - 12) - 2(0 - 6) - 3(-4 + 1)$$

$$= 24 + 12 + 9$$

$$= 45$$

Hence, the characteristic equation is

$$\lambda^3 + \lambda^2 - 21\lambda - 45 = 0$$

$$\lambda = 5, -3, -3$$

(a) For $\lambda = 5$,

$$[A - \lambda I]\mathbf{x} = 0$$

$$\begin{bmatrix} -7 & 2 & -3 \\ 2 & -4 & -6 \\ -1 & -2 & -5 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{aligned} -7x + 2y - 3z &= 0 \\ 2x - 4y - 6z &= 0 \\ -x - 2y - 5z &= 0 \end{aligned}$$

By Cramer's rule,

$$\frac{x}{\begin{vmatrix} 2 & -3 \\ -4 & -6 \end{vmatrix}} = -\frac{y}{\begin{vmatrix} -7 & -3 \\ 2 & -6 \end{vmatrix}} = \frac{z}{\begin{vmatrix} -7 & 2 \\ 2 & -4 \end{vmatrix}} = t$$

$$\frac{x}{-24} = \frac{y}{-48} = \frac{z}{24} = t$$

$$\frac{x}{1} = \frac{y}{2} = \frac{z}{-1} = t$$

$$\mathbf{x} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} t \\ 2t \\ -t \end{bmatrix} = t \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix} = t\mathbf{x}_1 \text{ where } \mathbf{x}_1 \text{ is an eigenvector corresponding to } \lambda = 5.$$

(b) For $\lambda = -3$,

$$[A - \lambda I]\mathbf{x} = \mathbf{0}$$

$$\begin{bmatrix} 1 & 2 & -3 \\ 2 & 4 & -6 \\ -1 & -2 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$x + 2y - 3z = 0$$

Let

$$y = t_1 \text{ and } z = t_2$$

$$x = -2t_1 + 3t_2$$

$$\mathbf{x} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -2t_1 + 3t_2 \\ t_1 \\ t_2 \end{bmatrix} = \begin{bmatrix} -2t_1 \\ t_1 \\ 0 \end{bmatrix} + \begin{bmatrix} 3t_2 \\ 0 \\ t_2 \end{bmatrix} = t_1 \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} + t_2 \begin{bmatrix} 3 \\ 0 \\ 1 \end{bmatrix} = t_1\mathbf{x}_2 + t_2\mathbf{x}_3$$

where $\mathbf{x}_2$ and $\mathbf{x}_3$ are linearly independent eigenvectors corresponding to $\lambda = -3$.

Example 7

Find the eigenvalues and eigenvectors of the matrix $\begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$.

Solution

Let $A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$

The characteristic equation is

$$\det(A - \lambda I) = 0$$

$$\begin{vmatrix} -\lambda & 1 & 1 \\ 1 & -\lambda & 1 \\ 1 & 1 & -\lambda \end{vmatrix} = 0$$

$$\lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0$$

where S_1 = Sum of the principal diagonal elements of $A = 0 + 0 + 0 = 0$

S_2 = Sum of the minors of principal diagonal elements of A

$$\begin{aligned}
&= \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix} + \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix} + \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix} \\
&= (0-1) + (0-1) + (0-1) \\
&= -3
\end{aligned}$$

$$\begin{aligned}
S_3 = \det(A) &= \begin{vmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{vmatrix} \\
&= 0 - 1(0-1) + 1(1-0) \\
&= 2
\end{aligned}$$

Hence, the characteristic equation is

$$\lambda^3 - 3\lambda - 2 = 0$$

$$\lambda = 2, -1, -1$$

(a) For $\lambda = 2$,

$$\begin{aligned}
&[A - \lambda I]\mathbf{x} = \mathbf{0} \\
&\begin{bmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \\
&-2x + y + z = 0 \\
&x - 2y + z = 0 \\
&x + y - 2z = 0
\end{aligned}$$

By Cramer's rule,

$$\begin{aligned}
\frac{x}{\begin{vmatrix} 1 & 1 \\ -2 & 1 \end{vmatrix}} &= -\frac{y}{\begin{vmatrix} -2 & 1 \\ 1 & 1 \end{vmatrix}} = \frac{z}{\begin{vmatrix} -2 & 1 \\ 1 & -2 \end{vmatrix}} = t \\
\frac{x}{3} &= \frac{y}{3} = \frac{z}{3} = t \\
\frac{x}{1} &= \frac{y}{1} = \frac{z}{1} = t
\end{aligned}$$

$$\mathbf{x} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} t \\ t \\ t \end{bmatrix} = t \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = t\mathbf{x}_1 \text{ where } \mathbf{x}_1 \text{ is an eigenvector corresponding to } \lambda = 2.$$

(b) For $\lambda = -1$,

$$\begin{aligned}
&[A - \lambda I]\mathbf{x} = \mathbf{0} \\
&\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \\
&x + y + z = 0
\end{aligned}$$

Let

$$y = t_1 \text{ and } z = t_2$$

$$\mathbf{x} = -t_1 - t_2$$

$$\mathbf{x} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -t_1 - t_2 \\ t_1 \\ t_2 \end{bmatrix} = \begin{bmatrix} -t_1 \\ t_1 \\ 0 \end{bmatrix} + \begin{bmatrix} -t_2 \\ 0 \\ t_2 \end{bmatrix} = t_1 \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} + t_2 \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} = t_1 \mathbf{x}_2 + t_2 \mathbf{x}_3$$

where $\mathbf{x}_2$ and $\mathbf{x}_3$ are linearly independent eigenvectors corresponding to $\lambda = -1$.

Example 8

Find the eigenvalues and eigenvectors of the matrix $\begin{bmatrix} -9 & 4 & 4 \\ -8 & 3 & 4 \\ -16 & 8 & 7 \end{bmatrix}$.

Solution

Let

$$A = \begin{bmatrix} -9 & 4 & 4 \\ -8 & 3 & 4 \\ -16 & 8 & 7 \end{bmatrix}$$

The characteristic equation is

$$\det(A - \lambda I) = 0$$

$$\begin{vmatrix} -9-\lambda & 4 & 4 \\ -8 & 3-\lambda & 4 \\ -16 & 8 & 7-\lambda \end{vmatrix} = 0$$

$$\lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0$$

where S_1 = Sum of the principal diagonal elements of $A = -9 + 3 + 7 = 1$

S_2 = Sum of the minors of principal diagonal elements of A

$$\begin{aligned} &= \begin{vmatrix} 3 & 4 \\ 8 & 7 \end{vmatrix} + \begin{vmatrix} -9 & 4 \\ -16 & 7 \end{vmatrix} + \begin{vmatrix} -9 & 4 \\ -8 & 3 \end{vmatrix} \\ &= (21 - 32) + (-63 + 64) + (-27 + 32) \\ &= -11 + 1 + 5 \\ &= -5 \end{aligned}$$

$$\begin{aligned} S_3 &= \det(A) = \begin{vmatrix} -9 & 4 & 4 \\ -8 & 3 & 4 \\ -16 & 8 & 7 \end{vmatrix} \\ &= -9(21 - 32) - 4(-56 + 64) + 4(-64 + 48) \\ &= 99 - 32 - 64 \\ &= 3 \end{aligned}$$

Hence, the characteristic equation is

$$\lambda^3 - \lambda^2 - 5\lambda - 3 = 0$$

$$\lambda = -1, -1, 3$$

(a) For $\lambda = -1$,

$$[A - \lambda I] \mathbf{x} = \mathbf{0}$$

$$\begin{bmatrix} -8 & 4 & 4 \\ -8 & 4 & 4 \\ -16 & 8 & 8 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$-8x + 4y + 4z = 0$$

Let

$$y = t_1 \text{ and } z = t_2$$

$$x = \frac{1}{2}t_1 + \frac{1}{2}t_2$$

$$\mathbf{x} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} \frac{1}{2}t_1 + \frac{1}{2}t_2 \\ t_1 \\ t_2 \end{bmatrix} = \begin{bmatrix} \frac{1}{2}t_1 \\ t_1 \\ 0 \end{bmatrix} + \begin{bmatrix} \frac{1}{2}t_2 \\ 0 \\ t_2 \end{bmatrix} = t_1 \begin{bmatrix} \frac{1}{2} \\ 1 \\ 0 \end{bmatrix} + t_2 \begin{bmatrix} \frac{1}{2} \\ 0 \\ 1 \end{bmatrix} = t_1 \mathbf{x}_1 + t_2 \mathbf{x}_2$$

where $\mathbf{x}_1$ and $\mathbf{x}_2$ are linearly independent eigenvectors corresponding to $\lambda = -1$.(b) For $\lambda = 3$,

$$[A - \lambda I] \mathbf{x} = \mathbf{0}$$

$$\begin{bmatrix} -12 & 4 & 4 \\ -8 & 0 & 4 \\ -16 & 8 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$-12x + 4y + 4z = 0$$

$$-8x + 0y + 4z = 0$$

$$-16x + 8y + 4z = 0$$

By Cramer's rule,

$$\frac{x}{\begin{vmatrix} 4 & 4 \\ 0 & 4 \end{vmatrix}} = -\frac{y}{\begin{vmatrix} -12 & 4 \\ -8 & 4 \end{vmatrix}} = \frac{z}{\begin{vmatrix} -12 & 4 \\ -8 & 0 \end{vmatrix}} = t$$

$$\frac{x}{16} = \frac{y}{16} = \frac{z}{32} = t$$

$$\frac{x}{1} = \frac{y}{1} = \frac{z}{2} = t$$

$$\mathbf{x} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} t \\ t \\ 2t \end{bmatrix} = t \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} = t \mathbf{x}_3 \text{ where } \mathbf{x}_3 \text{ is an eigenvector corresponding to } \lambda = 3.$$

Example 9

Find the eigenvalues and eigenvectors of the matrix $\begin{bmatrix} 1 & -6 & -4 \\ 0 & 4 & 2 \\ 0 & -6 & -3 \end{bmatrix}$.

Solution

Let $A = \begin{bmatrix} 1 & -6 & -4 \\ 0 & 4 & 2 \\ 0 & -6 & -3 \end{bmatrix}$

The characteristic equation is

$$\det(A - \lambda I) = 0$$

$$\begin{vmatrix} 1-\lambda & -6 & -4 \\ 0 & 4-\lambda & 2 \\ 0 & -6 & -3-\lambda \end{vmatrix} = 0$$

$$\lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0$$

where S_1 = Sum of the principal diagonal elements of $A = 1 + 4 - 3 = 2$

S_2 = Sum of the minors of principal diagonal elements of A

$$= \begin{vmatrix} 4 & 2 \\ -6 & -3 \end{vmatrix} + \begin{vmatrix} 1 & -4 \\ 0 & -3 \end{vmatrix} + \begin{vmatrix} 1 & -6 \\ 0 & 4 \end{vmatrix}$$

$$= (-12 + 12) + (-3 + 0) + (4 + 0)$$

$$= 1$$

$$S_3 = \det(A) = \begin{vmatrix} 1 & -6 & -4 \\ 0 & 4 & 2 \\ 0 & -6 & -3 \end{vmatrix}$$

$$= 1(-12 + 12) - 6(0 - 0) - 4(0 - 0)$$

$$= 0$$

Hence, the characteristic equation is

$$\lambda^3 - 2\lambda^2 + \lambda = 0$$

$$\lambda = 0, 1, 1$$

(a) For $\lambda = 0$,

$$[A - \lambda I] \mathbf{x} = \mathbf{0}$$

$$\begin{bmatrix} 1 & -6 & -4 \\ 0 & 4 & 2 \\ 0 & -6 & -3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$x - 6y - 4z = 0$$

$$0x + 4y + 2z = 0$$

$$0x - 6y - 3z = 0$$

By Cramer's rule,

$$\frac{x}{\begin{vmatrix} -6 & -4 \\ 4 & 2 \end{vmatrix}} = -\frac{y}{\begin{vmatrix} 1 & -4 \\ 0 & 2 \end{vmatrix}} = \frac{z}{\begin{vmatrix} 1 & -6 \\ 0 & 4 \end{vmatrix}} = t$$

$$\frac{x}{4} = \frac{y}{-2} = \frac{z}{4} = t$$

$$\frac{x}{2} = \frac{y}{-1} = \frac{z}{2} = t$$

$$\mathbf{x} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2t \\ -t \\ 2t \end{bmatrix} = t \begin{bmatrix} 2 \\ -1 \\ 2 \end{bmatrix} = t\mathbf{x}_1 \text{ where } \mathbf{x}_1 \text{ is an eigenvector corresponding to } \lambda = 0.$$

(b) For $\lambda = 1$,

$$[A - \lambda I] \mathbf{x} = \mathbf{0}$$

$$\begin{bmatrix} 0 & -6 & -4 \\ 0 & 3 & 2 \\ 0 & -6 & -4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$0x + 3y + 2z = 0$$

Let

$$x = t_1 \text{ and } z = t_2$$

$$y = -\frac{2}{3}t_2$$

$$\mathbf{x} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} t_1 \\ -\frac{2}{3}t_2 \\ t_2 \end{bmatrix} = \begin{bmatrix} t_1 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ -\frac{2}{3}t_2 \\ t_2 \end{bmatrix} = t_1 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + t_2 \begin{bmatrix} 0 \\ -\frac{2}{3} \\ 1 \end{bmatrix} = t_1\mathbf{x}_2 + t_2\mathbf{x}_3$$

where $\mathbf{x}_2$ and $\mathbf{x}_3$ are linearly independent eigenvectors corresponding to $\lambda = 1$.

Example 10

Find the eigenvalues and eigenvectors of the matrix $\begin{bmatrix} 1 & 2 & 2 \\ 0 & 2 & 1 \\ -1 & 2 & 2 \end{bmatrix}$.

Solution

Let

$$A = \begin{bmatrix} 1 & 2 & 2 \\ 0 & 2 & 1 \\ -1 & 2 & 2 \end{bmatrix}$$

The characteristic equation is

$$\det(A - \lambda I) = 0$$

$$\begin{vmatrix} 1-\lambda & 2 & 2 \\ 0 & 2-\lambda & 1 \\ -1 & 2 & 2-\lambda \end{vmatrix} = 0$$

$$\lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0$$

where S_1 = Sum of the principal diagonal elements of $A = 1 + 2 + 2 = 5$

S_2 = Sum of the minors of principal diagonal elements of A

$$\begin{aligned} &= \begin{vmatrix} 2 & 1 \\ 2 & 2 \end{vmatrix} + \begin{vmatrix} 1 & 2 \\ -1 & 2 \end{vmatrix} + \begin{vmatrix} 1 & 2 \\ 0 & 2 \end{vmatrix} \\ &= (4 - 2) + (2 + 2) + (2 - 0) \\ &= 2 + 4 + 2 \\ &= 8 \end{aligned}$$

$$\begin{aligned} S_3 = \det(A) &= \begin{vmatrix} 1 & 2 & 2 \\ 0 & 2 & 1 \\ -1 & 2 & 2 \end{vmatrix} \\ &= 1(4 - 2) - 2(0 + 1) + 2(0 + 2) \\ &= 2 - 2 + 4 \\ &= 4 \end{aligned}$$

Hence, the characteristic equation is

$$\begin{aligned} \lambda^3 - 5\lambda^2 + 8\lambda - 4 &= 0 \\ \lambda &= 1, 2, 2 \end{aligned}$$

(a) For $\lambda = 1$,

$$\begin{aligned} [A - \lambda I] \mathbf{x} &= \mathbf{0} \\ \begin{bmatrix} 0 & 2 & 2 \\ 0 & 1 & 1 \\ -1 & 2 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} &= \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \\ 0x + y + z &= 0 \\ -x + 2y + z &= 0 \end{aligned}$$

By Cramer's rule,

$$\begin{aligned} \frac{x}{\begin{vmatrix} 1 & 1 \\ 2 & 1 \end{vmatrix}} &= -\frac{y}{\begin{vmatrix} 0 & 1 \\ -1 & 1 \end{vmatrix}} = \frac{z}{\begin{vmatrix} 0 & 1 \\ -1 & 2 \end{vmatrix}} = t \\ \frac{x}{-1} &= \frac{y}{-1} = \frac{z}{1} = t \\ \frac{x}{1} &= \frac{y}{1} = \frac{z}{-1} = t \end{aligned}$$

$$\mathbf{x} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} t \\ t \\ -t \end{bmatrix} = t \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} = t\mathbf{x}_1 \text{ where } \mathbf{x}_1 \text{ is an eigenvector corresponding to } \lambda = 1.$$

(b) For $\lambda = 2$,

$$[A - \lambda I] \mathbf{x} = \mathbf{0}$$

$$\begin{bmatrix} -1 & 2 & 2 \\ 0 & 0 & 1 \\ -1 & 2 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$-x + 2y + 2z = 0$$

$$0x + 0y + z = 0$$

$$-x + 2y + 0z = 0$$

By Cramer's rule,

$$\frac{x}{\begin{vmatrix} 2 & 2 \\ 0 & 1 \end{vmatrix}} = -\frac{y}{\begin{vmatrix} -1 & 2 \\ 0 & 1 \end{vmatrix}} = \frac{z}{\begin{vmatrix} -1 & 2 \\ 0 & 0 \end{vmatrix}} = t$$

$$\frac{x}{2} = \frac{y}{1} = \frac{z}{0} = t$$

$$\mathbf{x} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2t \\ t \\ 0 \end{bmatrix} = t \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} = t\mathbf{x}_2 \text{ where } \mathbf{x}_2 \text{ is an eigenvector corresponding to } \lambda = 2.$$

Hence, there is only one eigenvector corresponding to repeated root $\lambda = 2$.

Example 11

Find the eigenvalues and eigenvectors of the matrix $\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & -3 & 3 \end{bmatrix}$.

Solution

Let

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & -3 & 3 \end{bmatrix}$$

The characteristic equation is

$$\det(A - \lambda I) = 0$$

$$\begin{vmatrix} -\lambda & 1 & 0 \\ 0 & -\lambda & 1 \\ 1 & -3 & 3-\lambda \end{vmatrix} = 0$$

$$\lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0$$

where S_1 = Sum of the principal diagonal elements of $A = 0 + 0 + 3 = 3$

S_2 = Sum of the minors of principal diagonal elements of A

$$= \begin{vmatrix} 0 & 1 \\ -3 & 3 \end{vmatrix} + \begin{vmatrix} 0 & 0 \\ 1 & 3 \end{vmatrix} + \begin{vmatrix} 0 & 1 \\ 0 & 0 \end{vmatrix}$$

$$= (0 + 3) + (0) + (0)$$

$$= 3$$

$$\begin{aligned}
 S_3 = \det(A) &= \begin{vmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & -3 & 3 \end{vmatrix} \\
 &= 0 - 1(0 - 1) + 0 \\
 &= 1
 \end{aligned}$$

Hence, the characteristic equation is

$$\begin{aligned}
 \lambda^3 - 3\lambda^2 + 3\lambda - 1 &= 0 \\
 \lambda &= 1, 1, 1
 \end{aligned}$$

For $\lambda = 1$,

$$\begin{aligned}
 [A - \lambda I]\mathbf{x} &= \mathbf{0} \\
 \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \\ 1 & -3 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} &= \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \\
 -x + y + 0z &= 0 \\
 0x - y + z &= 0 \\
 x - 3y + 2z &= 0
 \end{aligned}$$

By Cramer's rule,

$$\begin{aligned}
 \frac{x}{\begin{vmatrix} 1 & 0 \\ -1 & 1 \end{vmatrix}} &= -\frac{y}{\begin{vmatrix} -1 & 0 \\ 0 & 1 \end{vmatrix}} = \frac{z}{\begin{vmatrix} -1 & 1 \\ 0 & -1 \end{vmatrix}} = t \\
 \frac{x}{1} &= \frac{y}{1} = \frac{z}{1} = t
 \end{aligned}$$

$$\mathbf{x} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} t \\ t \\ t \end{bmatrix} = t \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = t\mathbf{x}_1 \text{ where } \mathbf{x}_1 \text{ is an eigenvector corresponding to } \lambda = 1.$$

Hence, there is only one eigenvector corresponding to repeated root $\lambda = 1$.

Example 12

Find the values of μ which satisfy the equation $A^{100}\mathbf{x} = \mu\mathbf{x}$

where
$$A = \begin{bmatrix} 2 & 1 & -1 \\ 0 & -2 & -2 \\ 1 & 1 & 0 \end{bmatrix}.$$

Solution

$$A = \begin{bmatrix} 2 & 1 & -1 \\ 0 & -2 & -2 \\ 1 & 1 & 0 \end{bmatrix}$$

The characteristic equation is

$$\det(A - \lambda I) = 0$$

$$\begin{vmatrix} 2-\lambda & 1 & -1 \\ 0 & -2-\lambda & -2 \\ 1 & 1 & -\lambda \end{vmatrix} = 0$$

$$\lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0$$

where S_1 = Sum of the principal diagonal elements of $A = 2 - 2 + 0 = 0$

S_2 = Sum of the minors of principal diagonal elements of A

$$= \begin{vmatrix} -2 & -2 \\ 1 & 0 \end{vmatrix} + \begin{vmatrix} 2 & -1 \\ 1 & 0 \end{vmatrix} + \begin{vmatrix} 2 & 1 \\ 0 & -2 \end{vmatrix}$$

$$= (0 + 2) + (0 + 1) + (-4 - 0)$$

$$= -1$$

$$S_3 = \det(A) = \begin{vmatrix} 2 & 1 & -1 \\ 0 & -2 & -2 \\ 1 & 1 & 0 \end{vmatrix}$$

$$= 2(0 + 2) - 1(0 + 2) - 1(0 + 2)$$

$$= 4 - 2 - 2$$

$$= 0$$

Hence, the characteristic equation is

$$\lambda^3 - \lambda = 0$$

$$\lambda = 0, 1, -1$$

If λ is an eigen value of A , it satisfies the equation $A\mathbf{x} = \lambda\mathbf{x}$.

For equation $A^{100}\mathbf{x} = \mu\mathbf{x}$, μ represents eigen values of A^{100} . Eigenvalues of $A^{100} = \lambda^{100}$, i.e., 0, 1, 1.

Hence, values of μ are 0, 1, 1.

Example 13

Find the characteristic roots and characteristic vectors of $A = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$ and verify that characteristic roots are of unit modulus.

Solution

$$A = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

The characteristic equation is

$$\begin{aligned}\det(A - \lambda I) &= 0 \\ \begin{vmatrix} \cos \theta - \lambda & -\sin \theta \\ \sin \theta & \cos \theta - \lambda \end{vmatrix} &= 0 \\ (\cos \theta - \lambda)^2 + \sin^2 \theta &= 0 \\ (\cos \theta - \lambda)^2 &= -\sin^2 \theta \\ \cos \theta - \lambda &= \pm i \sin \theta \\ \lambda &= \cos \theta \pm i \sin \theta\end{aligned}$$

$$|\lambda| = \sqrt{\cos^2 \theta + \sin^2 \theta} = 1$$

Hence, the characteristic roots are of unit modules.

(a) For $\lambda = \cos \theta + i \sin \theta$, $[A - \lambda I]\mathbf{x} = \mathbf{0}$

$$\begin{aligned}\begin{bmatrix} -i \sin \theta & -\sin \theta \\ \sin \theta & -i \sin \theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} &= \begin{bmatrix} 0 \\ 0 \end{bmatrix} \\ -i \sin \theta x - \sin \theta y &= 0\end{aligned}$$

$$\begin{aligned}\text{Let } y &= t \\ x &= it\end{aligned}$$

$$\mathbf{x} = \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} it \\ t \end{bmatrix} = t \begin{bmatrix} i \\ 1 \end{bmatrix} = t \mathbf{x}_1 \text{ where } \mathbf{x}_1 \text{ is an eigenvector corresponding to}$$

$$\lambda = \cos \theta + i \sin \theta.$$

(b) For $\lambda = \cos \theta - i \sin \theta$, $[A - \lambda I]\mathbf{x} = \mathbf{0}$

$$\begin{aligned}\begin{bmatrix} i \sin \theta & -\sin \theta \\ \sin \theta & i \sin \theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} &= \begin{bmatrix} 0 \\ 0 \end{bmatrix} \\ i \sin \theta x - \sin \theta y &= 0\end{aligned}$$

$$\begin{aligned}\text{Let } y &= t \\ x &= -it\end{aligned}$$

$$\mathbf{x} = \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -it \\ t \end{bmatrix} = t \begin{bmatrix} -i \\ 1 \end{bmatrix} = t \mathbf{x}_2 \text{ where } \mathbf{x}_2 \text{ is an eigenvector corresponding to}$$

$$\lambda = \cos \theta - i \sin \theta.$$

Example 14

Find orthogonal eigenvectors for the matrix $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix}$.

Solution

Let

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix}$$

The characteristic equation is

$$\begin{aligned} \det(A - \lambda I) &= 0 \\ \begin{vmatrix} 1-\lambda & 2 & 3 \\ 2 & 4-\lambda & 6 \\ 3 & 6 & 9-\lambda \end{vmatrix} &= 0 \\ \lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 &= 0 \end{aligned}$$

where S_1 = Sum of the principal diagonal elements of $A = 1 + 4 + 9 = 14$ S_2 = Sum of the minors of principal diagonal elements of A

$$\begin{aligned} &= \begin{vmatrix} 4 & 6 \\ 6 & 9 \end{vmatrix} + \begin{vmatrix} 1 & 3 \\ 3 & 9 \end{vmatrix} + \begin{vmatrix} 1 & 2 \\ 2 & 4 \end{vmatrix} \\ &= (36 - 36) + (9 - 9) + (4 - 4) \\ &= 0 \end{aligned}$$

$$\begin{aligned} S_3 &= \det(A) = \begin{vmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{vmatrix} \\ &= 1(36 - 36) - 2(18 - 18) + 3(12 - 12) \\ &= 0 \end{aligned}$$

Hence, the characteristic equation is

$$\begin{aligned} \lambda^3 - 14\lambda^2 &= 0 \\ \lambda &= 0, 0, 14 \end{aligned}$$

(a) For $\lambda = 14$,

$$[A - \lambda I]\mathbf{x} = \mathbf{0}$$

$$\begin{bmatrix} -13 & 2 & 3 \\ 2 & -10 & 6 \\ 3 & 6 & -5 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{aligned} -13x + 2y + 3z &= 0 \\ 2x - 10y + 6z &= 0 \\ 3x + 6y - 5z &= 0 \end{aligned}$$

By Cramer's rule,

$$\frac{x}{\begin{vmatrix} 2 & 3 \\ -10 & 6 \end{vmatrix}} = \frac{y}{\begin{vmatrix} -13 & 3 \\ 2 & 6 \end{vmatrix}} = \frac{z}{\begin{vmatrix} -13 & 2 \\ 2 & -10 \end{vmatrix}} = t$$

$$\frac{x}{42} = \frac{y}{84} = \frac{z}{126} = t$$

$$\frac{x}{1} = \frac{y}{2} = \frac{z}{3} = t$$

$$\mathbf{x} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} t \\ 2t \\ 3t \end{bmatrix} = t \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = t \mathbf{x}_1 \text{ where } \mathbf{x}_1 \text{ is an eigenvector corresponding to } \lambda = 14.$$

(b) For $\lambda = 0$,

$$[A - \lambda I]\mathbf{x} = \mathbf{0}$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$x + 2y + 3z = 0$$

Let

$$y = t_1 \text{ and } z = t_2$$

$$x = -2t_1 - 3t_2$$

$$\mathbf{x} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -2t_1 - 3t_2 \\ t_1 \\ t_2 \end{bmatrix} = \begin{bmatrix} -2t_1 \\ t_1 \\ 0 \end{bmatrix} + \begin{bmatrix} -3t_2 \\ 0 \\ t_2 \end{bmatrix} = t_1 \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} + t_2 \begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix} = t_1 \mathbf{x}_2 + t_2 \mathbf{x}_3$$

where $\mathbf{x}_2$ and $\mathbf{x}_3$ are linearly independent eigenvectors corresponding to $\lambda = 0$.

Since $\mathbf{x}_2$ and $\mathbf{x}_3$ are not orthogonal, we must choose $\mathbf{x}_3$ such that $\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3$ are orthogonal.

Let

$$\mathbf{x}_3 = \begin{bmatrix} l \\ m \\ n \end{bmatrix}$$

For $\mathbf{x}_1$ and $\mathbf{x}_3$ to be orthogonal, $\mathbf{x}_1^T \mathbf{x}_3 = 0$

$$\begin{bmatrix} 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} l \\ m \\ n \end{bmatrix} = 0$$

$$l + 2m + 3n = 0 \quad \dots(1)$$

For $\mathbf{x}_2$ and $\mathbf{x}_3$ to be orthogonal, $\mathbf{x}_2^T \mathbf{x}_3 = 0$

$$\begin{bmatrix} -2 & 1 & 0 \end{bmatrix} \begin{bmatrix} l \\ m \\ n \end{bmatrix} = 0$$

$$-2l + m = 0 \quad \dots(2)$$

Solving Eqs (1) and (2) by Cramer's rule,

$$\frac{l}{\begin{vmatrix} 2 & 3 \\ 1 & 0 \end{vmatrix}} = -\frac{m}{\begin{vmatrix} 1 & 3 \\ -2 & 0 \end{vmatrix}} = \frac{n}{\begin{vmatrix} 1 & 2 \\ -2 & 1 \end{vmatrix}} = t$$

$$\frac{l}{-3} = \frac{m}{-6} = \frac{n}{5} = t$$

$$\frac{l}{3} = \frac{m}{6} = \frac{n}{5} = t$$

$$\mathbf{x} = \begin{bmatrix} l \\ m \\ n \end{bmatrix} = \begin{bmatrix} 3t \\ 6t \\ 5t \end{bmatrix} = t \begin{bmatrix} 3 \\ 6 \\ 5 \end{bmatrix} = t\mathbf{x}_3 \text{ where } \mathbf{x}_3 \text{ is an eigenvector corresponding to } \lambda = 14.$$

EXERCISE 10.5

1. Find the sum and product of the eigenvalues of the following matrices:

$$(i) \begin{bmatrix} -1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{bmatrix} \quad (ii) \begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}$$

[Ans.: (i) -3, 4 (ii) -1, 45]

2. The matrix $A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 0 & -2 \\ 1 & 2 & 3 \end{bmatrix}$ is singular. One of its eigenvalues is 2. Find the other two eigenvalues.

[Ans.: $1 + \sqrt{5}, 1 - \sqrt{5}$]

3. If two of the eigenvalues of $\begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$ are 2 and 8, find the third eigenvalue.

[Ans.: 2]

4. Find the eigenvalues of the matrix $\begin{bmatrix} 1 & -2 \\ -5 & 4 \end{bmatrix}$. Hence, form the matrix whose eigenvalues are $\frac{1}{6}$ and -1 .

$$\left[\text{Ans.: } 6, -1, -\frac{1}{6} \begin{bmatrix} 4 & 2 \\ 5 & 1 \end{bmatrix} \right]$$

5. If 2 and 3 are the eigenvalues of $A = \begin{bmatrix} 3 & 10 & 5 \\ -2 & -3 & -4 \\ 3 & 5 & 7 \end{bmatrix}$, find the eigenvalues of A^{-1} and A^3 .

$$\left[\text{Ans.: (i) } \frac{1}{2}, \frac{1}{2}, \frac{1}{3} \quad \text{(ii) } 2^3, 2^3, 3^3 \right]$$

6. Two eigenvalues of $A = \begin{bmatrix} 4 & 6 & 6 \\ 1 & 3 & 2 \\ -1 & -5 & -2 \end{bmatrix}$ are equal and are double the third.

Find the eigenvalues of A^2 .

$$[\text{Ans.: } 1, 4, 4]$$

7. If $A = \begin{bmatrix} 1 & 2 & -3 \\ 0 & 3 & 2 \\ 0 & 0 & -2 \end{bmatrix}$, find the eigenvalues of $3A^3 + 5A^2 - 6A + 2I$

$$[\text{Ans.: } 4, 110, 10]$$

8. If $A = \begin{bmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{bmatrix}$, find the eigenvalues of the following matrices:

$$(i) A^3 + I \quad (ii) A^{-1} \quad (iii) A^2 - 2A + I \quad (iv) A^3 - 3A^2 + A$$

$$\left[\text{Ans.: (i) } 2, 2, 126 \quad (ii) 1, 1, \frac{1}{5} \quad (iii) 0, 0, 16 \quad (iv) -1, -1, 55 \right]$$

9. Find the eigenvalues and eigenvectors for the following matrices:

$$(i) \begin{bmatrix} 9 & -1 & 9 \\ 3 & -1 & 3 \\ -7 & 1 & -7 \end{bmatrix} \quad (ii) \begin{bmatrix} 1 & 1 & -2 \\ -1 & 2 & 1 \\ 0 & 1 & -1 \end{bmatrix} \quad (iii) \begin{bmatrix} -1 & 1 & 2 \\ 0 & -2 & 1 \\ 0 & 0 & -3 \end{bmatrix}$$

$$(iv) \begin{bmatrix} 4 & 0 & 1 \\ -2 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix} \quad (v) \begin{bmatrix} 2 & 1 & 1 \\ 2 & 3 & 2 \\ 3 & 3 & 4 \end{bmatrix} \quad (vi) \begin{bmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{bmatrix}$$

$$(vii) \begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix} \quad (viii) \begin{bmatrix} 1 & 0 & 0 \\ 2 & 0 & 1 \\ 3 & 1 & 0 \end{bmatrix} \quad (ix) \begin{bmatrix} 2 & 4 & -6 \\ 4 & 2 & -6 \\ -6 & -6 & -15 \end{bmatrix}$$

$$(x) \begin{bmatrix} 7 & 0 & -2 \\ 0 & 5 & -2 \\ -2 & -2 & 6 \end{bmatrix} \quad (xi) \begin{bmatrix} 7 & -2 & -2 \\ -2 & 1 & 4 \\ -2 & 4 & 1 \end{bmatrix} \quad (xii) \begin{bmatrix} 3 & -1 & 1 \\ -1 & 5 & -1 \\ 1 & -1 & 3 \end{bmatrix}$$

$$(xiii) \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix} \quad (xiv) \begin{bmatrix} 7 & -2 & 1 \\ -2 & 10 & -2 \\ 1 & -2 & 7 \end{bmatrix} \quad (xv) \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$$

$$(xvi) \begin{bmatrix} 2 & 0 & 1 \\ 0 & 3 & 0 \\ 1 & 0 & 2 \end{bmatrix} \quad (xvii) \begin{bmatrix} 1 & 2 & 3 \\ 0 & 2 & 3 \\ 0 & 0 & 2 \end{bmatrix} \quad (xviii) \begin{bmatrix} 3 & 10 & 5 \\ -2 & -3 & -4 \\ 3 & 5 & 7 \end{bmatrix}$$

$$(xix) \begin{bmatrix} -3 & -7 & -5 \\ 2 & 4 & 3 \\ 1 & 2 & 2 \end{bmatrix} \quad (xx) \begin{bmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{bmatrix}$$

Ans. :

$$\begin{array}{ll} (i) -1, 0, 2; \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} 4 \\ 1 \\ -3 \end{bmatrix} & (ii) -1, 2, 1; \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}, \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix} \\ (iii) -1, -2, -3; \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ -2 \end{bmatrix} & (iv) 1, 2, 3; \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 2 \\ 2 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix} \\ (v) 1, 1, 7; \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} & (vi) 5, 1, 1; \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} \\ (vii) 5, -3, -3; \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}, \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 3 \\ 0 \\ 1 \end{bmatrix} & (viii) -1, 1, 1; \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \\ (ix) -2, 9, -18; \begin{bmatrix} 0 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 4 \end{bmatrix} & (x) 3, 6, 9; \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix} \end{array}$$

$$\left[\begin{array}{ll}
 \text{(xi)} & -3, 3, 9; \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix} \\
 \text{(xiii)} & 8, 2, 2; \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} \\
 \text{(xv)} & 4, 1, 1; \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix} \\
 \text{(xvii)} & 1, 2, 2; \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} \\
 \text{(xix)} & 1, 1, 1; \begin{bmatrix} -3 \\ 1 \\ 1 \end{bmatrix} \\
 \text{(xii)} & 2, 3, 6; \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} \\
 \text{(xiv)} & 12, 6, 6; \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \\
 \text{(xvi)} & 1, 3, 3; \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} \\
 \text{(xviii)} & 3, 2, 2; \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix}, \begin{bmatrix} 5 \\ 2 \\ -5 \end{bmatrix} \\
 \text{(xx)} & 2, 2, 2; \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}
 \end{array} \right]$$

10. Verify that $\mathbf{x} = [2, 3, -2, -3]^T$ is an eigenvector corresponding to the eigenvalue $\lambda = 2$ of the matrix

$$A = \begin{bmatrix} 1 & -4 & -1 & -4 \\ 2 & 0 & 5 & -4 \\ -1 & 1 & -2 & 3 \\ -1 & 4 & -1 & 6 \end{bmatrix}$$

11. If $A = \begin{bmatrix} 3 & -1 & 1 \\ -1 & 3 & -1 \\ 1 & -1 & 3 \end{bmatrix}$ then check whether eigenvectors of A are orthogonal.
12. If $A = \begin{bmatrix} 3 & 10 & 5 \\ -2 & -3 & -4 \\ 3 & 5 & 7 \end{bmatrix}$ then verify whether eigenvectors are linearly independent or not.

10.14 CAYLEY–HAMILTON THEOREM

Statement: Every square matrix satisfies its own characteristic equation.

Proof: Let A be an n -rowed square matrix. Its characteristic equation is

$$\begin{aligned} |A - \lambda I| &= (-1)^n (\lambda^n + a_1 \lambda^{n-1} + a_2 \lambda^{n-2} + \dots + a_n) \\ (A - \lambda I) \operatorname{adj}(A - \lambda I) &= |A - \lambda I| I \quad \dots (10.16) \\ &= \{A \operatorname{adj}(A) = |A| I\} \end{aligned}$$

Since $\operatorname{adj}(A - \lambda I)$ has elements as cofactors of elements of $|A - \lambda I|$, the elements of $\operatorname{adj}(A - \lambda I)$ are polynomials in λ of degree $n - 1$ or less. Hence, $\operatorname{adj}(A - \lambda I)$ can be written as a matrix polynomial in λ .

$$\operatorname{adj}(A - \lambda I) = B_0 \lambda^{n-1} + B_1 \lambda^{n-2} + \dots + B_{n-2} \lambda + B_{n-1}$$

where $B_0, B_1, \dots, B_{n-1}$ are matrices of order n .

$$\begin{aligned} (A - \lambda I) \operatorname{adj}(A - \lambda I) &= (A - \lambda I)[B_0 \lambda^{n-1} + B_1 \lambda^{n-2} + \dots + B_{n-2} \lambda + B_{n-1}] \\ |A - \lambda I| I &= (A - \lambda I)[B_0 \lambda^{n-1} + B_1 \lambda^{n-2} + \dots + B_{n-2} \lambda + B_{n-1}] \\ (-1)^n [I \lambda^n + a_1 I \lambda^{n-1} + a_2 I \lambda^{n-2} + \dots + a_{n-1} I \lambda + a_n I] \\ &= (-IB_0) \lambda^n + (AB_0 - IB_1) \lambda^{n-1} + (AB_1 - IB_2) \lambda^{n-2} + \dots + (AB_{n-2} - IB_{n-1}) \lambda + AB_{n-1} \end{aligned}$$

Equating corresponding coefficients,

$$\begin{aligned} -IB_0 &= (-1)^n I \\ AB_0 - IB_1 &= (-1)^n a_1 I \\ AB_1 - IB_2 &= (-1)^n a_2 I \\ &\vdots \\ AB_{n-2} - IB_{n-1} &= (-1)^n a_{n-1} I \\ AB_{n-1} &= (-1)^n a_n I \end{aligned}$$

Premultiplying the above equations successively by $A^n, A^{n-1}, A^{n-2}, \dots, I$ and adding,

$$(-1)^n [A^n + a_1 A^{n-1} + a_2 A^{n-2} + \dots + a_n I] = \mathbf{0}$$

$$\text{Hence, } A^n + a_1 A^{n-1} + a_2 A^{n-2} + \dots + a_n I = \mathbf{0} \quad \dots (10.17)$$

Corollary: If A is a non-singular matrix, i.e. $\det(A) \neq 0$ then premultiplying equation (1) by A^{-1} , we get

$$\begin{aligned} A^{n-1} + a_1 A^{n-2} + a_2 A^{n-3} + \dots + a_n A^{-1} &= \mathbf{0} \\ A^{-1} &= -\frac{1}{a_n} [A^{n-1} + a_1 A^{n-2} + \dots + a_{n-1} I] \end{aligned}$$

Example 1

Apply Cayley–Hamilton theorem to $A = \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix}$ and deduce that $A^8 = 625I$.

Solution

$$A = \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix}$$

The characteristic equation is

$$\begin{aligned} \det(A - \lambda I) &= 0 \\ \begin{vmatrix} 1-\lambda & 2 \\ 2 & -1-\lambda \end{vmatrix} &= 0 \\ \lambda^2 - S_1\lambda + S_2 &= 0 \end{aligned}$$

where S_1 = Sum of the principal diagonal elements of $A = 1 - 1 = 0$

$$\begin{aligned} S_2 &= \det(A) = \begin{vmatrix} 1 & 2 \\ 2 & -1 \end{vmatrix} \\ &= -1 - 4 \\ &= -5 \end{aligned}$$

Hence, the characteristic equation is

$$\lambda^2 - 5 = 0$$

By Cayley–Hamilton theorem, the matrix A satisfies its own characteristic equation.

$$\begin{aligned} A^2 - 5I &= \mathbf{0} \\ A^2 &= 5I \\ A^4 &= 25I \\ A^8 &= 625I \end{aligned}$$

Example 2

Verify Cayley–Hamilton theorem for the following matrix and hence, find A^{-1} and A^4 .

$$A = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$$

Solution

$$A = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$$

The characteristic equation is

$$\det(A - \lambda I) = 0$$

$$\begin{vmatrix} 2-\lambda & -1 & 1 \\ -1 & 2-\lambda & -1 \\ 1 & -1 & 2-\lambda \end{vmatrix} = 0$$

$$\lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0$$

where S_1 = Sum of the principal diagonal elements of $A = 2 + 2 + 2 = 6$

S_2 = Sum of the minors of principal diagonal elements of A

$$= \begin{vmatrix} 2 & -1 \\ -1 & 2 \end{vmatrix} + \begin{vmatrix} 2 & 1 \\ 1 & 2 \end{vmatrix} + \begin{vmatrix} 2 & -1 \\ -1 & 2 \end{vmatrix}$$

$$= (4-1) + (4-1) + (4-1)$$

$$= 9$$

$$S_3 = \det(A) = \begin{vmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{vmatrix}$$

$$= 2(4-1) + 1(-2+1) + 1(1-2)$$

$$= 6-1-1$$

$$= 4$$

Hence, the characteristic equation is

$$\lambda^3 - 6\lambda^2 + 9\lambda - 4 = 0$$

$$A^2 = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix} \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix} = \begin{bmatrix} 6 & -5 & 5 \\ -5 & 6 & -5 \\ 5 & -5 & 6 \end{bmatrix}$$

$$A^3 = \begin{bmatrix} 6 & -5 & 5 \\ -5 & 6 & -5 \\ 5 & -5 & 6 \end{bmatrix} \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix} = \begin{bmatrix} 22 & -21 & 21 \\ -21 & 22 & -21 \\ 21 & -21 & 22 \end{bmatrix}$$

$$A^3 - 6A^2 + 9A - 4I$$

$$= \begin{bmatrix} 22 & -21 & 21 \\ -21 & 22 & -21 \\ 21 & -21 & 22 \end{bmatrix} - \begin{bmatrix} 36 & -30 & 30 \\ -30 & 36 & -30 \\ 30 & -30 & 36 \end{bmatrix} + \begin{bmatrix} 18 & -9 & 9 \\ -9 & 18 & -9 \\ 9 & -9 & 18 \end{bmatrix} - \begin{bmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \mathbf{0} \quad \dots(1)$$

The matrix A satisfies its own characteristic equation. Hence, Cayley-Hamilton theorem is verified.

Premultiplying Eq. (1) by A^{-1} ,

$$A^{-1}(A^3 - 6A^2 + 9A - 4I) = \mathbf{0}$$

$$A^2 - 6A + 9I - 4A^{-1} = \mathbf{0}$$

$$4A^{-1} = (A^2 - 6A + 9I)$$

$$= \begin{bmatrix} 6 & -5 & 5 \\ -5 & 6 & -5 \\ 5 & -5 & 6 \end{bmatrix} - \begin{bmatrix} 12 & -6 & 6 \\ -6 & 12 & -6 \\ 6 & -6 & 12 \end{bmatrix} + \begin{bmatrix} 9 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 9 \end{bmatrix}$$

$$= \begin{bmatrix} 3 & 1 & -1 \\ 1 & 3 & 1 \\ -1 & 1 & 3 \end{bmatrix}$$

$$A^{-1} = \frac{1}{4} \begin{bmatrix} 3 & 1 & -1 \\ 1 & 3 & 1 \\ -1 & 1 & 3 \end{bmatrix}$$

Multiplying Eq. (1) by A ,

$$A(A^3 - 6A^2 + 9A - 4I) = \mathbf{0}$$

$$A^4 - 6A^3 + 9A^2 - 4A = \mathbf{0}$$

$$A^4 = 6A^3 - 9A^2 + 4A$$

$$= \begin{bmatrix} 132 & -126 & 126 \\ -126 & 132 & -126 \\ 126 & -126 & 132 \end{bmatrix} - \begin{bmatrix} 54 & -45 & 45 \\ -45 & 54 & -45 \\ 45 & -45 & 54 \end{bmatrix} + \begin{bmatrix} 8 & -4 & 4 \\ -4 & 8 & -4 \\ 4 & -4 & 8 \end{bmatrix}$$

$$= \begin{bmatrix} 86 & -85 & 85 \\ -85 & 86 & -85 \\ 85 & -85 & 86 \end{bmatrix}$$

Example 3

Show that matrix $A = \begin{bmatrix} 0 & c & -b \\ -c & 0 & a \\ b & -a & 0 \end{bmatrix}$ satisfies Cayley-Hamilton theorem

and hence find A^{-1} , if it exists.

Solution

$$A = \begin{bmatrix} 0 & c & -b \\ -c & 0 & a \\ b & -a & 0 \end{bmatrix}$$

The characteristic equation is

$$\det(A - \lambda I) = 0$$

$$\begin{vmatrix} -\lambda & c & -b \\ -c & -\lambda & a \\ b & -a & -\lambda \end{vmatrix} = 0$$

$$\lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0$$

where S_1 = Sum of the principal diagonal elements of $A = 0$

S_2 = Sum of the minors of principal diagonal elements of A

$$= \begin{vmatrix} 0 & a \\ -a & 0 \end{vmatrix} + \begin{vmatrix} 0 & -b \\ b & 0 \end{vmatrix} + \begin{vmatrix} 0 & c \\ -c & 0 \end{vmatrix}$$

$$= (0 + a^2) + (0 + b^2) + (0 + c^2)$$

$$= a^2 + b^2 + c^2$$

$$S_3 = \det(A) = \begin{vmatrix} 0 & c & -b \\ -c & 0 & a \\ b & -a & 0 \end{vmatrix}$$

$$= 0 - c(0 - ab) - b(ac - 0)$$

$$= abc - abc$$

$$= 0$$

Hence, the characteristic equation is

$$\lambda^3 + (a^2 + b^2 + c^2)\lambda = 0$$

$$A^2 = \begin{bmatrix} 0 & c & -b \\ -c & 0 & a \\ b & -a & 0 \end{bmatrix} \begin{bmatrix} 0 & c & -b \\ -c & 0 & a \\ b & -a & 0 \end{bmatrix} = \begin{bmatrix} -c^2 - b^2 & ab & ac \\ ab & -c^2 - a^2 & bc \\ ac & bc & -b^2 - a^2 \end{bmatrix}$$

$$A^3 = \begin{bmatrix} -c^2 - b^2 & ab & ac \\ ab & -c^2 - a^2 & bc \\ ac & bc & -b^2 - a^2 \end{bmatrix} \begin{bmatrix} 0 & c & -b \\ -c & 0 & a \\ b & -a & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & -c^3 - cb^2 - ca^2 & b^3 + bc^2 + ba^2 \\ c^3 + ca^2 + cb^2 & 0 & -ab^2 - ac^2 - a^3 \\ -bc^2 - b^3 - a^2b & ac^2 + ab^2 + a^3 & 0 \end{bmatrix}$$

$$= -(a^2 + b^2 + c^2) \begin{bmatrix} 0 & c & -b \\ -c & 0 & a \\ b & -a & 0 \end{bmatrix}$$

$$= -(a^2 + b^2 + c^2)A$$

$$A^3 + (a^2 + b^2 + c^2)A = \mathbf{0}$$

The matrix A satisfies its own characteristic equation. Hence, Cayley-Hamilton theorem is verified.

$$\begin{aligned}\det(A) &= \begin{vmatrix} 0 & c & -b \\ -c & 0 & a \\ b & -a & 0 \end{vmatrix} = -c(0-ab) - b(ac-0) \\ &= abc - abc = 0\end{aligned}$$

Hence, A^{-1} does not exist.

Example 4

Find the characteristic roots of the matrix $A = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}$ and verify Cayley-Hamilton theorem for this matrix. Find A^{-1} and also express $A^5 - 4A^4 - 7A^3 + 11A^2 - A - 10I$ as a linear polynomial in A .

Solution

$$A = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}$$

The characteristic equation is

$$\det(A - \lambda I) = 0$$

$$\begin{vmatrix} 1-\lambda & 4 \\ 2 & 3-\lambda \end{vmatrix} = 0$$

$$\lambda^2 - S_1\lambda + S_2 = 0$$

where S_1 = Sum of the principal diagonal elements of $A = 1 + 3 = 4$

$$S_2 = \det(A) = \begin{vmatrix} 1 & 4 \\ 2 & 3 \end{vmatrix}$$

$$= 3 - 8$$

$$= -5$$

Hence, the characteristic equation is

$$\lambda^2 - 4\lambda - 5 = 0$$

$$\lambda = -1, 5$$

$$A^2 = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix} = \begin{bmatrix} 9 & 16 \\ 8 & 17 \end{bmatrix}$$

$$A^2 - 4A - 5 = \begin{bmatrix} 9 & 16 \\ 8 & 17 \end{bmatrix} - \begin{bmatrix} 4 & 16 \\ 8 & 12 \end{bmatrix} - \begin{bmatrix} 5 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = \mathbf{0} \quad \dots (1)$$

The matrix A satisfies its own characteristic equation. Hence, Cayley-Hamilton theorem is verified.

Premultiplying Eq. (1) by A^{-1} ,

$$A^{-1}(A^2 - 4A - 5I) = 0$$

$$A - 4I - 5A^{-1} = 0$$

$$\begin{aligned} 4A^{-1} &= \frac{1}{5}(A - 4I) \\ &= \frac{1}{5} \begin{bmatrix} -3 & 4 \\ 2 & -1 \end{bmatrix} \end{aligned}$$

$$\begin{aligned} \text{Now, } A^5 - 4A^4 - 7A^3 + 11A^2 - A - 10I &= A^3(A^2 - 4A - 5I) - 2A(A^2 - 4A - 5I) \\ &\quad + 3(A^2 - 4A - 5I) + A + 5I \\ &= (A^2 - 4A - 5I)(A^3 - 2A + 3I) + (A + 5I) \\ &= 0 + (A + 5I) \quad [\text{Using Eq. (1)}] \\ &= A + 5I \end{aligned}$$

which is a linear polynomial in A .

Example 5

Find the characteristic equation of the matrix $A = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix}$ and hence

find the matrix represented by $A^8 - 5A^7 + 7A^6 - 3A^5 + A^4 - 5A^3 + 8A^2 - 2A + I$.

Solution

$$A = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix}$$

The characteristic equation is

$$\begin{aligned} \det(A - \lambda I) &= 0 \\ \begin{vmatrix} 2-\lambda & 1 & 1 \\ 0 & 1-\lambda & 0 \\ 1 & 1 & 2-\lambda \end{vmatrix} &= 0 \\ \lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 &= 0 \end{aligned}$$

where S_1 = Sum of the principal diagonal elements of $A = 2 + 1 + 2 = 5$

S_2 = Sum of the minors of principal diagonal elements of A

$$\begin{aligned} &= \begin{vmatrix} 1 & 0 \\ 1 & 2 \end{vmatrix} + \begin{vmatrix} 2 & 1 \\ 0 & 2 \end{vmatrix} + \begin{vmatrix} 2 & 1 \\ 0 & 1 \end{vmatrix} \\ &= (2 - 0) + (4 - 1) + (2 - 0) \\ &= 7 \end{aligned}$$

$$\begin{aligned}
S_3 = \det(A) &= \begin{vmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{vmatrix} \\
&= 2(2-0) - 1(0-0) + 1(0-1) \\
&= 4 - 0 - 1 \\
&= 3
\end{aligned}$$

Hence, the characteristic equation is

$$\lambda^3 - 5\lambda^2 + 7\lambda - 3 = 0$$

By Cayley-Hamilton theorem,

$$A^3 - 5A^2 + 7A - 3I = 0 \quad \dots(1)$$

$$\begin{aligned}
\text{Now, } A^8 - 5A^7 + 7A^6 - 3A^5 + A^4 - 5A^3 + 8A^2 - 2A + I & \\
&= A^5(A^3 - 5A^2 + 7A - 3I) + A(A^3 - 5A^2 + 7A - 3I) + (A^2 + A + I) \\
&= (A^3 - 5A^2 + 7A - 3I)(A^5 + A) + (A^2 + A + I) \\
&= 0 + (A^2 + A + I) \quad [\text{Using Eq.(1)}] \\
&= A^2 + A + I
\end{aligned}$$

$$\begin{aligned}
A^2 &= \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix} \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix} = \begin{bmatrix} 5 & 4 & 4 \\ 0 & 1 & 0 \\ 4 & 4 & 5 \end{bmatrix} \\
A^2 + A + I &= \begin{bmatrix} 5 & 4 & 4 \\ 0 & 1 & 0 \\ 4 & 4 & 5 \end{bmatrix} + \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix} + \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 8 & 5 & 5 \\ 0 & 3 & 0 \\ 5 & 5 & 8 \end{bmatrix} \\
A^8 - 5A^7 + 7A^6 - 3A^5 + A^4 - 5A^3 + 8A^2 - 2A + I &= \begin{bmatrix} 8 & 5 & 5 \\ 0 & 3 & 0 \\ 5 & 5 & 8 \end{bmatrix}
\end{aligned}$$

Example 6

If $A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$, prove by induction that for every integer $n \geq 3$,

$A^n = A^{n-2} + A^2 - I$. Hence, find A^{50} .

Solution

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

The characteristic equation is

$$\det(A - \lambda I) = 0$$

$$\begin{vmatrix} 1-\lambda & 0 & 0 \\ 1 & -\lambda & 1 \\ 0 & 1 & -\lambda \end{vmatrix} = 0$$

$$\lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0$$

where S_1 = Sum of the principal diagonal elements of $A = 1 + 0 + 0 = 1$

S_2 = Sum of the minors of principal diagonal elements of A

$$= \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix} + \begin{vmatrix} 1 & 0 \\ 0 & 0 \end{vmatrix} + \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix}$$

$$= (0-1) + (0-0) + (0-0)$$

$$= -1$$

$$S_3 = \det(A) = \begin{vmatrix} 1 & 0 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{vmatrix}$$

$$= 1(0-1) + 0 + 0$$

$$= -1$$

Hence, the characteristic equation is

$$\lambda^3 - \lambda^2 - \lambda + 1 = 0$$

By Cayley–Hamilton theorem,

$$A^3 - A^2 - A + I = 0$$

$$A^3 = A^2 + A - I$$

$$= A^{3-2} + A^2 - I \quad \dots(1)$$

Hence, $A^n = A^{n-2} + A^2 - I$ is true for $n = 3$.

Assuming that Eq. (1) is true for $n = k$,

$$A^k = A^{k-2} + A^2 - I$$

Multiplying both the sides by A

$$A^{k+1} = A^{k+1} + A^3 - A$$

Substituting the value of A^3

$$A^{k+1} = A^{k+1} + (A^2 + A - I) - A$$

$$A^{(k+1)-2} + A^2 - I$$

Hence, $A^n = A^{n-2} + A^2 - I$ is true for $n = k + 1$

Thus, by mathematical induction, it is true for $n \geq 3$.

We have, $A^n = A^{n-2} + A^2 - I$

$$\begin{aligned}
&= (A^{n-4} + A^2 - I) + A^2 - I \\
&= A^{n-4} + 2(A^2 - I) \\
&= (A^{n-6} + A^2 - I) + 2(A^2 - I) \\
&= A^{n-6} + 3(A^2 - I) \\
&\dots\dots\dots
\end{aligned}$$

$$A^n = A^{n-2r} + r(A^2 - I)$$

Putting $n = 50$ and $r = 24$,

$$\begin{aligned}
A^{50} &= A^{50-2(24)} + 24(A^2 - I) \\
&= A^2 + 24A^2 - 24I \\
&= 25A^2 - 24I
\end{aligned}$$

$$\begin{aligned}
A^2 &= \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \\
&= \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}
\end{aligned}$$

$$\begin{aligned}
A^{50} &= \begin{bmatrix} 25 & 0 & 0 \\ 25 & 25 & 0 \\ 25 & 0 & 25 \end{bmatrix} - \begin{bmatrix} 24 & 0 & 0 \\ 0 & 24 & 0 \\ 0 & 0 & 24 \end{bmatrix} \\
&= \begin{bmatrix} 1 & 0 & 0 \\ 25 & 1 & 0 \\ 25 & 0 & 1 \end{bmatrix}
\end{aligned}$$

EXERCISE 10.6

1. Verify Cayley-Hamilton theorem for the matrix A and hence, find A^{-1} and A^4 .

$$(i) \begin{bmatrix} 1 & 2 & -2 \\ -1 & 3 & 0 \\ 0 & -2 & 1 \end{bmatrix}$$

$$(ii) \begin{bmatrix} 1 & 2 & 3 \\ 2 & -1 & 4 \\ 3 & 1 & -1 \end{bmatrix}$$

$$(iii) \begin{bmatrix} 2 & 0 & -1 \\ 0 & 2 & 0 \\ -1 & 0 & 2 \end{bmatrix}$$

$$(iv) \begin{bmatrix} 1 & 2 & -2 \\ -1 & 3 & 0 \\ 0 & -2 & 1 \end{bmatrix}$$

$$\left[\begin{array}{l} \text{Ans.:} \\ \text{(i)} \begin{bmatrix} 3 & 2 & 6 \\ 1 & 1 & 2 \\ 2 & 2 & 5 \end{bmatrix}, \begin{bmatrix} -55 & 104 & 24 \\ -20 & -15 & 32 \\ 32 & -42 & 13 \end{bmatrix} \quad \text{(ii)} \frac{1}{40} \begin{bmatrix} -3 & 5 & 11 \\ 14 & -10 & 2 \\ 5 & 5 & -5 \end{bmatrix}, \begin{bmatrix} 248 & 101 & 218 \\ 272 & 109 & 50 \\ 104 & 98 & 204 \end{bmatrix} \\ \text{(iii)} \frac{1}{6} \begin{bmatrix} 4 & 0 & 2 \\ 0 & 3 & 0 \\ 2 & 0 & 4 \end{bmatrix}, \begin{bmatrix} 41 & 0 & -40 \\ 0 & 16 & 0 \\ -40 & 0 & 41 \end{bmatrix} \quad \text{(iv)} \begin{bmatrix} 3 & 2 & 6 \\ 1 & 1 & 2 \\ 2 & 2 & 5 \end{bmatrix}, \begin{bmatrix} -55 & 104 & 24 \\ -20 & -15 & 32 \\ 32 & -40 & -23 \end{bmatrix} \end{array} \right]$$

2. Verify that the matrix $A = \begin{bmatrix} 1 & 2 & 0 \\ 2 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$ satisfies its characteristic equation and hence, find A^{-2} .

$$\left[\text{Ans.: } A^3 + A^2 - 5A - 5I = 0, \frac{1}{5} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \right]$$

3. Use Cayley–Hamilton theorem to find $2A^5 - 3A^4 + A^2 - 4I$, where

$$A = \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix}.$$

$$\left[\text{Ans.: } 138A - 403I = \begin{bmatrix} 11 & 138 \\ -138 & 127 \end{bmatrix} \right]$$

4. If $A = \begin{bmatrix} 1 & 4 \\ 1 & 1 \end{bmatrix}$, find $A^7 - 9A^2 + I$.

$$[\text{Ans.: } 609A + 640I]$$

5. Verify Cayley–Hamilton theorem for (i) $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ (ii) $A = \begin{bmatrix} 2 & 4 \\ 1 & 2 \end{bmatrix}$ and hence, find A^{-1} and $A^3 - 5A^2$.

$$\left[\text{Ans.: (i)} \begin{bmatrix} -2 & 1 \\ 1.5 & -0.5 \end{bmatrix}, 2A \quad \text{(ii)} A^{-1} \text{ does not exist, } A^2 \right]$$

6. Compute $A^9 - 6A^8 + 10A^7 - 3A^6 + A + I$, where $A = \begin{bmatrix} 1 & 2 & 3 \\ -1 & 3 & 1 \\ 1 & 0 & 2 \end{bmatrix}$.

$$\left[\text{Ans.: } \begin{bmatrix} 2 & 2 & 3 \\ -1 & 4 & 1 \\ 1 & 0 & 3 \end{bmatrix} \right]$$

7. Verify Cayley-Hamilton theorem for $A = \begin{bmatrix} 1 & 2 \\ 2 & 2 \end{bmatrix}$ and evaluate $2A^4 - 5A^3 - 7A - 6I$.

$$\left[\text{Ans. : } \begin{bmatrix} 36 & 32 \\ 32 & 52 \end{bmatrix} \right]$$

10.15 SIMILARITY TRANSFORMATION

If A and B are two square matrices of order n then B is said to be similar to A , if there exists a nonsingular matrix P such that

$$B = P^{-1}AP$$

- Note:** (1) Similarity of matrices is an equivalence relation.
 (2) Similar matrices have the same determinant.
 (3) Similar matrices have the same characteristic polynomial and hence the same eigenvalues. If $\mathbf{x}$ is an eigenvector of A corresponding to the eigenvalue λ , then $P^{-1}\mathbf{x}$ is an eigenvector of B corresponding to the eigenvalue λ where $B = P^{-1}AP$.

10.16 DIAGONALIZATION OF A MATRIX

A matrix A is said to be *diagonalizable* if it is similar to a diagonal matrix.

A matrix A is diagonalizable if there exists an invertible matrix P such that $P^{-1}AP = D$ where D is a diagonal matrix, also known as *spectral matrix*. The matrix P is then said to diagonalize A or transform A to a diagonal form. P is known as the *modal matrix*.

- Note** (1) An $n \times n$ matrix is diagonalizable if and only if it possesses n linearly independent eigenvectors.
 (2) If the eigenvalues of an $n \times n$ matrix are all distinct then it is always similar to a diagonal matrix.
 (3) If A is similar to a diagonal matrix D , the diagonal elements of D are the eigenvalues of A .

10.16.1 Orthogonally Similar Matrices

If A and B are two square matrices of order n then B is said to be orthogonally similar to A if there exists an orthogonal matrix P such that

$$B = P^{-1}AP$$

Since P is orthogonal, $P^{-1} = P^T$

$$B = P^{-1}AP = P^TAP$$

- Note:** (1) Every real symmetric matrix is orthogonally similar to a diagonal matrix with real elements.
 (2) A real symmetric matrix of order n has n mutually orthogonal real eigenvectors.

- (3) Any two eigenvectors corresponding to two distinct eigenvalues of a real symmetric matrix are orthogonal.

Note: To find the orthogonal matrix P , each element of the eigenvector is divided by its norm (length).

Working Rule for Diagonalization of Square Matrix A

- (i) Find the eigenvalues of the square matrix A .
- (ii) Find the eigenvectors corresponding to each eigenvalue.
- (iii) Find the modal matrix P having the normalized eigenvectors as its column vectors.
- (iv) Find the diagonal matrix $D = P^T A P$. The diagonal matrix D has eigenvalues as its diagonal elements.

Example 1

Show that the matrix $\begin{bmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{bmatrix}$ is not diagonalizable.

Solution

Let $A = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{bmatrix}$

The characteristic equation is

$$\det(A - \lambda I) = 0$$

$$\begin{vmatrix} 2-\lambda & 1 & 0 \\ 0 & 2-\lambda & 1 \\ 0 & 0 & 2-\lambda \end{vmatrix} = 0$$

$$\lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0$$

where $S_1 = \text{Sum of the principal diagonal elements of } A = 2 + 2 + 2 = 6$

$S_2 = \text{Sum of the minors of principal diagonal elements of } A$

$$= \begin{vmatrix} 2 & 1 \\ 0 & 2 \end{vmatrix} + \begin{vmatrix} 2 & 0 \\ 0 & 2 \end{vmatrix} + \begin{vmatrix} 2 & 1 \\ 0 & 2 \end{vmatrix}$$

$$= (4 - 0) + (4 - 0) + (4 - 0)$$

$$= 12$$

$$S_3 = \det(A) = \begin{vmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{vmatrix}$$

$$= 2(4 - 0) - 1(0 - 0) + 0$$

$$= 8 - 0 + 0$$

$$= 8$$

Hence, the characteristic equation is

$$\lambda^3 - 6\lambda^2 + 12\lambda - 8 = 0$$

$$\lambda = 2, 2, 2$$

For $\lambda = 2$,

$$[A - \lambda I]\mathbf{x} = \mathbf{0}$$

$$\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$0x + y + 0z = 0$$

$$0x + 0y + z = 0$$

By Cramer's rule,

$$\frac{x}{\begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix}} = -\frac{y}{\begin{vmatrix} 0 & 0 \\ 0 & 1 \end{vmatrix}} = \frac{z}{\begin{vmatrix} 0 & 1 \\ 0 & 0 \end{vmatrix}} = t$$

$$\frac{x}{1} = \frac{y}{0} = \frac{z}{0} = t$$

$$\mathbf{x} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} t \\ 0 \\ 0 \end{bmatrix} = t \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = t\mathbf{x}_1 \text{ where } \mathbf{x}_1 \text{ is an eigenvector corresponding to } \lambda = 1.$$

Since the matrix A has only one linearly independent eigenvector which is less than its order 3, matrix A is not diagonalizable.

Example 2

Show that the matrix $\begin{bmatrix} 1 & -2 & 0 \\ 1 & 2 & 2 \\ 1 & 2 & 3 \end{bmatrix}$ is not diagonalizable.

Solution

Let $A = \begin{bmatrix} 1 & -2 & 0 \\ 1 & 2 & 2 \\ 1 & 2 & 3 \end{bmatrix}$

The characteristic equation is

$$\det(A - \lambda I) = 0$$

$$\begin{vmatrix} 1-\lambda & -2 & 0 \\ 1 & 2-\lambda & 2 \\ 1 & 2 & 3-\lambda \end{vmatrix} = 0$$

$$\lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0$$

where S_1 = Sum of the principal diagonal elements of $A = 1 + 2 + 3 = 6$

S_2 = Sum of the minors of principal diagonal elements of A

$$\begin{aligned}
 &= \begin{vmatrix} 2 & 2 \\ 2 & 3 \end{vmatrix} + \begin{vmatrix} 1 & 0 \\ 1 & 3 \end{vmatrix} + \begin{vmatrix} 1 & -2 \\ 1 & 2 \end{vmatrix} \\
 &= (6-4) + (3-0) + (2+2) \\
 &= 9
 \end{aligned}$$

$$\begin{aligned}
 S_3 = \det(A) &= \begin{vmatrix} 1 & -2 & 0 \\ 1 & 2 & 2 \\ 1 & 2 & 3 \end{vmatrix} \\
 &= 1(6-4) + 2(3-2) + 0 \\
 &= 4
 \end{aligned}$$

Hence, the characteristic equation is

$$\begin{aligned}
 \lambda^3 - 6\lambda^2 + 9\lambda - 4 &= 0 \\
 \lambda &= 1, 1, 4
 \end{aligned}$$

(a) For $\lambda = 1$,

$$\begin{aligned}
 [A - \lambda I]\mathbf{x} &= \mathbf{0} \\
 \begin{bmatrix} 0 & -2 & 0 \\ 1 & 1 & 2 \\ 0 & 2 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} &= \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \\
 0x - 2y + 0z &= 0 \\
 x + y + 2z &= 0 \\
 x + 2y + 2z &= 0
 \end{aligned}$$

By Cramer's rule,

$$\begin{aligned}
 \frac{x}{\begin{vmatrix} -2 & 0 \\ 1 & 2 \end{vmatrix}} &= -\frac{y}{\begin{vmatrix} 0 & 0 \\ 1 & 2 \end{vmatrix}} = \frac{z}{\begin{vmatrix} 0 & -2 \\ 1 & 1 \end{vmatrix}} = t \\
 \frac{x}{-4} &= \frac{y}{0} = \frac{z}{2} = t \\
 \frac{x}{2} &= \frac{y}{0} = \frac{z}{-1} = t
 \end{aligned}$$

$$\mathbf{x} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2t \\ 0 \\ -t \end{bmatrix} = t \begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix} = t\mathbf{x}_1 \text{ where } \mathbf{x}_1 \text{ is an eigenvector corresponding to } \lambda = 1.$$

(b) For $\lambda = 4$,

$$\begin{aligned}
 [A - \lambda I]\mathbf{x} &= \mathbf{0} \\
 \begin{bmatrix} -3 & -2 & 0 \\ 1 & -2 & 2 \\ 1 & 2 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} &= \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \\
 -3x - 2y + 0z &= 0 \\
 x - 2y + 2z &= 0 \\
 x + 2y - z &= 0
 \end{aligned}$$

By Cramer's rule,

$$\frac{x}{\begin{vmatrix} -2 & 0 \\ -2 & 2 \end{vmatrix}} = -\frac{y}{\begin{vmatrix} -3 & 0 \\ 1 & 2 \end{vmatrix}} = \frac{z}{\begin{vmatrix} -3 & -2 \\ 1 & -2 \end{vmatrix}} = t$$

$$\frac{x}{-4} = \frac{y}{6} = \frac{z}{8} = t$$

$$\frac{x}{-2} = \frac{y}{3} = \frac{z}{4} = t$$

$$\mathbf{x} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -2t \\ 3t \\ 4t \end{bmatrix} = t \begin{bmatrix} -2 \\ 3 \\ 4 \end{bmatrix} = t\mathbf{x}_2 \text{ where } \mathbf{x}_2 \text{ is an eigenvector corresponding to } \lambda = 1.$$

Since the matrix A has two linearly independent eigenvectors which is less than its order 3, the matrix A is not diagonalizable.

Example 3

Determine a diagonal matrix orthogonally similar to the real symmetric

matrix $\begin{bmatrix} 3 & -1 & 1 \\ -1 & 5 & -1 \\ 1 & -1 & 3 \end{bmatrix}$. Also find the modal matrix.

Solution

Let $A = \begin{bmatrix} 3 & -1 & 1 \\ -1 & 5 & -1 \\ 1 & -1 & 3 \end{bmatrix}$

The characteristic equation is

$$\det(A - \lambda I) = 0$$

$$\begin{vmatrix} 3-\lambda & -1 & 1 \\ -1 & 5-\lambda & -1 \\ 1 & -1 & 3-\lambda \end{vmatrix} = 0$$

$$\lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 = 0$$

where S_1 = Sum of the principal diagonal elements of $A = 3 + 5 + 3 = 11$

S_2 = Sum of the minors of principal diagonal elements of A

$$= \begin{vmatrix} 5 & -1 \\ -1 & 3 \end{vmatrix} + \begin{vmatrix} 3 & 1 \\ 1 & 3 \end{vmatrix} + \begin{vmatrix} 3 & -1 \\ -1 & 5 \end{vmatrix}$$

$$= (15 - 1) + (9 - 1) + (15 - 1)$$

$$= 36$$

$$\begin{aligned}
 S_3 = \det(A) &= \begin{vmatrix} 3 & -1 & 1 \\ -1 & 5 & -1 \\ 1 & -1 & 3 \end{vmatrix} \\
 &= 3(15-1) + 1(-3+1) + 1(1-5) \\
 &= 42 - 2 - 4 \\
 &= 36
 \end{aligned}$$

Hence, the characteristic equation is

$$\begin{aligned}
 \lambda^3 - 11\lambda^2 + 36\lambda - 36 &= 0 \\
 \lambda &= 2, 3, 6
 \end{aligned}$$

(a) For $\lambda = 2$, $[A - \lambda I]\mathbf{x} = \mathbf{0}$

$$\begin{aligned}
 \begin{bmatrix} 1 & -1 & 1 \\ -1 & 3 & -1 \\ 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} &= \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \\
 x - y + z &= 0 \\
 -x + 3y - z &= 0
 \end{aligned}$$

By Cramer's rule,

$$\begin{aligned}
 \frac{x}{\begin{vmatrix} -1 & 1 \\ 3 & -1 \end{vmatrix}} &= -\frac{y}{\begin{vmatrix} 1 & 1 \\ -1 & -1 \end{vmatrix}} = \frac{z}{\begin{vmatrix} 1 & -1 \\ -1 & 3 \end{vmatrix}} = t \\
 \frac{x}{-2} &= \frac{y}{0} = \frac{z}{2} = t \\
 \frac{x}{-1} &= \frac{y}{0} = \frac{z}{1} = t
 \end{aligned}$$

$$\mathbf{x} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -t \\ 0 \\ t \end{bmatrix} = t \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} = t\mathbf{x}_1 \text{ where } \mathbf{x}_1 \text{ is an eigenvector corresponding to } \lambda = 2.$$

(b) For $\lambda = 3$, $[A - \lambda I]\mathbf{x} = \mathbf{0}$

$$\begin{aligned}
 \begin{bmatrix} 0 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} &= \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \\
 0x - y + z &= 0 \\
 -x + 2y - z &= 0 \\
 x - y + 0z &= 0
 \end{aligned}$$

By Cramer's rule,

$$\frac{x}{\begin{vmatrix} -1 & 1 \\ 2 & -1 \end{vmatrix}} = -\frac{y}{\begin{vmatrix} 0 & 1 \\ -1 & -1 \end{vmatrix}} = \frac{z}{\begin{vmatrix} 0 & -1 \\ -1 & 2 \end{vmatrix}} = t$$

$$\frac{x}{-1} = \frac{y}{-1} = \frac{z}{-1} = t$$

$$\frac{x}{1} = \frac{y}{1} = \frac{z}{1} = t$$

$$\mathbf{x} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} t \\ t \\ t \end{bmatrix} = t \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = t\mathbf{x}_2 \text{ where } \mathbf{x}_2 \text{ is an eigenvector corresponding to } \lambda = 3.$$

(c) For $\lambda = 6$,

$$[A - \lambda I]\mathbf{x} = \mathbf{0}$$

$$\begin{bmatrix} -3 & -1 & 1 \\ -1 & -1 & -1 \\ 1 & -1 & -3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$-3x - y + z = 0$$

$$x + y - z = 0$$

$$x - y - 3z = 0$$

By Cramer's rule,

$$\frac{x}{\begin{vmatrix} -1 & 1 \\ -1 & -1 \end{vmatrix}} = -\frac{y}{\begin{vmatrix} -3 & 1 \\ -1 & -1 \end{vmatrix}} = \frac{z}{\begin{vmatrix} -3 & -1 \\ -1 & -1 \end{vmatrix}} = t$$

$$\frac{x}{2} = \frac{y}{-4} = \frac{z}{2} = t$$

$$\frac{x}{1} = \frac{y}{-2} = \frac{z}{1} = t$$

$$\mathbf{x} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} t \\ -2t \\ t \end{bmatrix} = t \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} = t\mathbf{x}_3 \text{ where } \mathbf{x}_3 \text{ is an eigenvector corresponding to } \lambda = 6.$$

Since matrix A has three linearly independent eigenvectors which is same as its order, the matrix A is diagonalizable.

$$\text{Length of eigenvector } \mathbf{x}_1 = \sqrt{(-1)^2 + 0^2 + 1^2} = \sqrt{2}$$

$$\text{Length of eigenvector } \mathbf{x}_2 = \sqrt{1^2 + 1^2 + 1^2} = \sqrt{3}$$

$$\text{Length of eigenvector } \mathbf{x}_3 = \sqrt{1^2 + (-2)^2 + 1^2} = \sqrt{6}$$

The normalized eigenvectors are

$$\mathbf{x}_1 = \begin{bmatrix} -\frac{1}{\sqrt{2}} \\ 0 \\ \frac{1}{\sqrt{2}} \end{bmatrix}, \quad \mathbf{x}_2 = \begin{bmatrix} \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \end{bmatrix}, \quad \mathbf{x}_3 = \begin{bmatrix} \frac{1}{\sqrt{6}} \\ -\frac{2}{\sqrt{6}} \\ \frac{1}{\sqrt{6}} \end{bmatrix}$$

The modal matrix P has normalized eigenvectors as its column vectors.

$$P = \begin{bmatrix} -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} \\ 0 & \frac{1}{\sqrt{3}} & -\frac{2}{\sqrt{6}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} \end{bmatrix}$$

$$P^{-1} = P^T = \begin{bmatrix} -\frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{6}} & -\frac{2}{\sqrt{6}} & \frac{1}{\sqrt{6}} \end{bmatrix}$$

$$D = P^T A P = \begin{bmatrix} -\frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{6}} & -\frac{2}{\sqrt{6}} & \frac{1}{\sqrt{6}} \end{bmatrix} \begin{bmatrix} 3 & -1 & 1 \\ -1 & 5 & -1 \\ 1 & -1 & 3 \end{bmatrix} \begin{bmatrix} -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} \\ 0 & \frac{1}{\sqrt{3}} & -\frac{2}{\sqrt{6}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} \end{bmatrix}$$

$$= \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 6 \end{bmatrix}$$

Hence, the diagonal matrix D has eigenvalues as its diagonal elements.

Example 4

Determine a diagonal matrix orthogonally similar to the real symmetric

matrix $\begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$. Also find the modal matrix.

Solution

Let
$$A = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$$

The characteristic equation is

$$\begin{aligned} \det(A - \lambda I) &= 0 \\ \begin{vmatrix} 8-\lambda & -6 & 2 \\ -6 & 7-\lambda & -4 \\ 2 & -4 & 3-\lambda \end{vmatrix} &= 0 \\ \lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 &= 0 \end{aligned}$$

where $S_1 = \text{Sum of the principal diagonal elements of } A = 8 + 7 + 3 = 18$

$S_2 = \text{Sum of the minors of principal diagonal elements of } A$

$$\begin{aligned} &= \begin{vmatrix} 7 & -4 \\ -4 & 3 \end{vmatrix} + \begin{vmatrix} 8 & 2 \\ 2 & 3 \end{vmatrix} + \begin{vmatrix} 8 & -6 \\ -6 & 7 \end{vmatrix} \\ &= (21 - 16) + (24 - 4) + (56 - 36) \\ &= 5 + 20 + 20 \\ &= 45 \end{aligned}$$

$$\begin{aligned} S_3 &= \det(A) = \begin{vmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{vmatrix} \\ &= 8(21 - 16) + 6(-18 + 8) + 2(24 - 14) \\ &= 40 - 60 + 20 \\ &= 0 \end{aligned}$$

Hence, the characteristic equation is

$$\begin{aligned} \lambda^3 - 18\lambda^2 + 45\lambda &= 0 \\ \lambda &= 0, 3, 15 \end{aligned}$$

(a) For $\lambda = 0$,

$$\begin{aligned} [A - \lambda I]\mathbf{x} &= \mathbf{0} \\ \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} &= \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \\ 8x - 6y + 2z &= 0 \\ -6x + 7y - 4z &= 0 \\ 2x - 4y + 3z &= 0 \end{aligned}$$

By Cramer's rule,

$$\frac{x}{\begin{vmatrix} -6 & 2 \\ 7 & -4 \end{vmatrix}} = -\frac{y}{\begin{vmatrix} 8 & 2 \\ -6 & -4 \end{vmatrix}} = \frac{z}{\begin{vmatrix} 8 & -6 \\ -6 & 7 \end{vmatrix}} = t$$

$$\frac{x}{10} = \frac{y}{20} = \frac{z}{20} = t$$

$$\frac{x}{1} = \frac{y}{2} = \frac{z}{2} = t$$

$$\mathbf{x} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} t \\ 2t \\ 2t \end{bmatrix} = t \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} = t\mathbf{x}_1 \text{ where } \mathbf{x}_1 \text{ is an eigenvector corresponding to } \lambda = 0.$$

(b) For $\lambda = 3$,

$$[A - \lambda I]\mathbf{x} = \mathbf{0}$$

$$\begin{bmatrix} 5 & -6 & 2 \\ -6 & 4 & -4 \\ 2 & -4 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$5x - 6y + 2z = 0$$

$$-6x + 4y - 4z = 0$$

$$2x - 4y + 0z = 0$$

By Cramer's rule,

$$\frac{x}{\begin{vmatrix} -6 & 2 \\ 4 & -4 \end{vmatrix}} = -\frac{y}{\begin{vmatrix} 5 & 2 \\ -6 & -4 \end{vmatrix}} = \frac{z}{\begin{vmatrix} 5 & -6 \\ -6 & 4 \end{vmatrix}} = t$$

$$\frac{x}{16} = \frac{y}{8} = \frac{z}{-16} = t$$

$$\frac{x}{2} = \frac{y}{1} = \frac{z}{-2} = t$$

$$\mathbf{x} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2t \\ t \\ -2t \end{bmatrix} = t \begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix} = t\mathbf{x}_2 \text{ where } \mathbf{x}_2 \text{ is an eigenvector corresponding to } \lambda = 3.$$

(c) For $\lambda = 15$,

$$[A - \lambda I]\mathbf{x} = \mathbf{0}$$

$$\begin{bmatrix} -7 & -6 & 2 \\ -6 & -8 & -4 \\ 2 & -4 & -12 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{aligned} -7x - 6y + 2z &= 0 \\ -6x - 8y - 4z &= 0 \\ 2x - 4y - 12z &= 0 \end{aligned}$$

By Cramer's rule,

$$\begin{aligned} \frac{x}{\begin{vmatrix} -6 & 2 \\ -8 & -4 \end{vmatrix}} &= -\frac{y}{\begin{vmatrix} -7 & 2 \\ -6 & -4 \end{vmatrix}} = \frac{z}{\begin{vmatrix} -7 & -6 \\ -6 & -8 \end{vmatrix}} = t \\ \frac{x}{40} &= \frac{y}{-40} = \frac{z}{20} = t \\ \frac{x}{2} &= \frac{y}{-2} = \frac{z}{1} = t \end{aligned}$$

$$\mathbf{x} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2t \\ -2t \\ t \end{bmatrix} = t \begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix} = t\mathbf{x}_3 \text{ where } \mathbf{x}_3 \text{ is an eigenvector corresponding to } \lambda = 15.$$

Since the matrix A has three linearly independent eigenvectors which is same as its order, the matrix A is diagonalizable.

$$\text{Length of eigenvector } \mathbf{x}_1 = \sqrt{1^2 + 2^2 + 2^2} = 3$$

$$\text{Length of eigenvector } \mathbf{x}_2 = \sqrt{2^2 + 1^2 + (-2)^2} = 3$$

$$\text{Length of eigenvector } \mathbf{x}_3 = \sqrt{2^2 + (-2)^2 + 1^2} = 3$$

The normalized eigenvectors are

$$\mathbf{x}_1 = \begin{bmatrix} \frac{1}{3} \\ \frac{2}{3} \\ \frac{2}{3} \end{bmatrix}, \quad \mathbf{x}_2 = \begin{bmatrix} \frac{2}{3} \\ \frac{1}{3} \\ -\frac{2}{3} \end{bmatrix}, \quad \mathbf{x}_3 = \begin{bmatrix} \frac{2}{3} \\ -\frac{2}{3} \\ \frac{1}{3} \end{bmatrix}$$

The modal matrix P has normalized eigenvectors as its column vectors.

$$P = \begin{bmatrix} \frac{1}{3} & \frac{2}{3} & \frac{2}{3} \\ \frac{2}{3} & \frac{1}{3} & -\frac{2}{3} \\ \frac{2}{3} & -\frac{2}{3} & \frac{1}{3} \end{bmatrix}$$

$$\begin{aligned}
 P^{-1} = P^T &= \begin{bmatrix} \frac{1}{3} & \frac{2}{3} & \frac{2}{3} \\ \frac{2}{3} & \frac{1}{3} & -\frac{2}{3} \\ \frac{2}{3} & -\frac{2}{3} & \frac{1}{3} \end{bmatrix} \\
 D = P^T A P &= \begin{bmatrix} \frac{1}{3} & \frac{2}{3} & \frac{2}{3} \\ \frac{2}{3} & \frac{1}{3} & -\frac{2}{3} \\ \frac{2}{3} & -\frac{2}{3} & \frac{1}{3} \end{bmatrix} \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix} \begin{bmatrix} \frac{1}{3} & \frac{2}{3} & \frac{2}{3} \\ \frac{2}{3} & \frac{1}{3} & -\frac{2}{3} \\ \frac{2}{3} & -\frac{2}{3} & \frac{1}{3} \end{bmatrix} \\
 &= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 15 \end{bmatrix}
 \end{aligned}$$

Hence, the diagonal matrix D has eigenvalues as its diagonal elements.

Example 5

Determine a diagonal matrix orthogonally similar to the real symmetric

matrix $\begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$. Also find the modal matrix.

Solution

Let $A = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$

The characteristic equation is

$$\begin{aligned}
 \det(A - \lambda I) &= 0 \\
 \lambda^3 - S_1\lambda^2 + S_2\lambda - S_3 &= 0
 \end{aligned}$$

where S_1 = Sum of the principal diagonal elements of $A = 6 + 3 + 3 = 12$

S_2 = Sum of the minors of principal diagonal elements of A

$$\begin{aligned}
 &= \begin{vmatrix} 3 & -1 \\ -1 & 3 \end{vmatrix} + \begin{vmatrix} 6 & 2 \\ 2 & 3 \end{vmatrix} + \begin{vmatrix} 6 & -2 \\ -2 & 3 \end{vmatrix} \\
 &= (9 - 1) + (18 - 4) + (18 - 4) \\
 &= 8 + 14 + 14 \\
 &= 36
 \end{aligned}$$

$$\begin{aligned}
 S_3 = \det(A) &= \begin{vmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{vmatrix} \\
 &= 6(9-1) + 2(-6+2) + 2(2-6) \\
 &= 48-8-8 \\
 &= 32
 \end{aligned}$$

Hence, the characteristic equation is

$$\begin{aligned}
 \lambda^3 - 12\lambda^2 + 36\lambda - 32 &= 0 \\
 \lambda &= 2, 2, 8
 \end{aligned}$$

(a) For $\lambda = 8$,

$$[A - \lambda I]\mathbf{x} = \mathbf{0}$$

$$\begin{bmatrix} -2 & -2 & 2 \\ -2 & -5 & -1 \\ 2 & -1 & -5 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$-2x - 2y + 2z = 0$$

$$-2x - 5y - z = 0$$

$$2x - y - 5z = 0$$

By Cramer's rule,

$$\frac{x}{\begin{vmatrix} -2 & 2 \\ -5 & -1 \end{vmatrix}} = -\frac{y}{\begin{vmatrix} -2 & 2 \\ -2 & -1 \end{vmatrix}} = \frac{z}{\begin{vmatrix} -2 & -2 \\ -2 & -5 \end{vmatrix}} = t$$

$$\frac{x}{12} = \frac{y}{-6} = \frac{z}{6} = t$$

$$\frac{x}{2} = \frac{y}{-1} = \frac{z}{1} = t$$

$$\mathbf{x} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2t \\ -t \\ t \end{bmatrix} = t \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix} = t\mathbf{x}_1 \text{ where } \mathbf{x}_1 \text{ is an eigenvector corresponding to } \lambda = 8.$$

Since matrix A has three linearly independent eigenvectors which is same as its order, matrix A is diagonalizable.

(b) For $\lambda = 2$,

$$[A - \lambda I]\mathbf{x} = \mathbf{0}$$

$$\begin{bmatrix} 4 & -2 & 2 \\ -2 & 1 & -1 \\ 2 & -1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$4x - 2y + 2z = 0$$

Let

$$y = t_1 \text{ and } z = t_2$$

$$x = \frac{1}{2}t_1 - \frac{1}{2}t_2$$

$$\mathbf{x} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} \frac{1}{2}t_1 - \frac{1}{2}t_2 \\ t_1 \\ t_2 \end{bmatrix} = \begin{bmatrix} \frac{1}{2}t_1 \\ t_1 \\ 0 \end{bmatrix} + \begin{bmatrix} -\frac{1}{2}t_2 \\ 0 \\ t_2 \end{bmatrix} = t_1 \begin{bmatrix} \frac{1}{2} \\ 1 \\ 0 \end{bmatrix} + t_2 \begin{bmatrix} -\frac{1}{2} \\ 0 \\ 1 \end{bmatrix} = t_1 \mathbf{x}_2 + t_2 \mathbf{x}_3$$

where $\mathbf{x}_2$ and $\mathbf{x}_3$ are linearly independent eigenvectors corresponding to $\lambda = 2$.

The orthogonal matrix P has mutually orthogonal eigenvectors. Since $\mathbf{x}_2$ and $\mathbf{x}_3$ are not orthogonal, we must choose $\mathbf{x}_3$ such that $\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3$ are orthogonal.

Let
$$\mathbf{x}_3 = \begin{bmatrix} l \\ m \\ n \end{bmatrix}$$

For orthogonality of eigenvectors,

$$\mathbf{x}_1^T \mathbf{x}_3 = 0 \quad \text{and} \quad \mathbf{x}_2^T \mathbf{x}_3 = 0$$

$$\begin{bmatrix} 2 & -1 & 1 \end{bmatrix} \begin{bmatrix} l \\ m \\ n \end{bmatrix} = 0 \quad \text{and} \quad \begin{bmatrix} \frac{1}{2} & 1 & 0 \end{bmatrix} \begin{bmatrix} l \\ m \\ n \end{bmatrix} = 0$$

$$2l - m + n = 0 \quad \text{and} \quad \frac{1}{2}l + m + 0n = 0$$

By Cramer's rule,

$$\frac{l}{\begin{vmatrix} -1 & 1 \\ 1 & 0 \end{vmatrix}} = -\frac{m}{\begin{vmatrix} 2 & 1 \\ \frac{1}{2} & 0 \end{vmatrix}} = \frac{n}{\begin{vmatrix} 2 & -1 \\ \frac{1}{2} & 1 \end{vmatrix}} = t$$

$$\frac{l}{-1} = \frac{m}{\frac{1}{2}} = \frac{n}{\frac{5}{2}} = t$$

$$\frac{l}{-2} = \frac{m}{1} = \frac{n}{5} = t$$

$$\mathbf{x} = \begin{bmatrix} l \\ m \\ n \end{bmatrix} = \begin{bmatrix} -2t \\ t \\ 5t \end{bmatrix} = t \begin{bmatrix} -2 \\ 1 \\ 5 \end{bmatrix} = t \mathbf{x}_3 \text{ where } \mathbf{x}_3 \text{ is an eigenvector corresponding to } \lambda = 2.$$

Length of eigenvector $\mathbf{x}_1 = \sqrt{(2)^2 + (-1)^2 + (1)^2} = \sqrt{6}$

Length of eigenvector $\mathbf{x}_2 = \sqrt{\left(\frac{1}{2}\right)^2 + 1^2 + 0^2} = \sqrt{\frac{5}{2}}$

Length of eigenvector $\mathbf{x}_3 = \sqrt{(-2)^2 + (1)^2 + (5)^2} = \sqrt{30}$

The normalized eigenvectors are

$$\mathbf{x}_1 = \begin{bmatrix} \frac{2}{\sqrt{6}} \\ -\frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{6}} \end{bmatrix}, \quad \mathbf{x}_2 = \begin{bmatrix} \frac{1}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} \\ 0 \end{bmatrix}, \quad \mathbf{x}_3 = \begin{bmatrix} -\frac{2}{\sqrt{30}} \\ \frac{1}{\sqrt{30}} \\ \frac{5}{\sqrt{30}} \end{bmatrix}$$

The modal matrix P has normalized eigenvectors as its column vectors.

$$P = \begin{bmatrix} \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{5}} & -\frac{2}{\sqrt{30}} \\ -\frac{1}{\sqrt{6}} & \frac{2}{\sqrt{5}} & \frac{1}{\sqrt{30}} \\ \frac{1}{\sqrt{6}} & 0 & \frac{5}{\sqrt{30}} \end{bmatrix}$$

$$P^{-1} = P^T = \begin{bmatrix} \frac{2}{\sqrt{6}} & -\frac{1}{\sqrt{6}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{5}} & \frac{2}{\sqrt{5}} & 0 \\ -\frac{2}{\sqrt{30}} & \frac{1}{\sqrt{30}} & \frac{5}{\sqrt{30}} \end{bmatrix}$$

$$\begin{aligned} D = P^T A P &= \begin{bmatrix} \frac{2}{\sqrt{6}} & -\frac{1}{\sqrt{6}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{5}} & \frac{2}{\sqrt{5}} & 0 \\ -\frac{2}{\sqrt{30}} & \frac{1}{\sqrt{30}} & \frac{5}{\sqrt{30}} \end{bmatrix} \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix} \begin{bmatrix} \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{5}} & -\frac{2}{\sqrt{30}} \\ -\frac{1}{\sqrt{6}} & \frac{2}{\sqrt{5}} & \frac{1}{\sqrt{30}} \\ \frac{1}{\sqrt{6}} & 0 & \frac{5}{\sqrt{30}} \end{bmatrix} \\ &= \begin{bmatrix} 8 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix} \end{aligned}$$

Hence, the diagonal matrix D has eigenvalues as its diagonal elements.

Example 6

For a symmetric matrix A , the eigenvectors are $[1, 1, 1]^T$, $[1, -2, 1]^T$ corresponding to $\lambda_1 = 6$ and $\lambda_2 = 12$. Find the eigenvector corresponding to $\lambda_3 = 6$ and find the matrix A .

Solution

Let $\mathbf{x}_3 = [x_3, y_3, z_3]^T$ be the eigenvector corresponding to $\lambda_3 = 6$.

$$\mathbf{x}_1 = [1, 1, 1]^T, \quad \mathbf{x}_2 = [1, -2, 1]^T$$

Since A is real symmetric matrix, $\mathbf{x}_1$, $\mathbf{x}_2$ and $\mathbf{x}_3$ are orthogonal.

i.e.,

$$\mathbf{x}_1^T \mathbf{x}_3 = 0 \quad \text{and} \quad \mathbf{x}_2^T \mathbf{x}_3 = 0$$

$$[1, 1, 1] \begin{bmatrix} x_3 \\ y_3 \\ z_3 \end{bmatrix} = 0 \quad \text{and} \quad [1, -2, 1] \begin{bmatrix} x_3 \\ y_3 \\ z_3 \end{bmatrix} = 0$$

$$x_3 + y_3 + z_3 = 0$$

$$x_3 + 2y_3 + z_3 = 0$$

By Cramer's rule,

$$\frac{x_3}{\begin{vmatrix} 1 & 1 \\ -2 & 1 \end{vmatrix}} = -\frac{y_3}{\begin{vmatrix} 1 & 1 \\ 1 & 1 \end{vmatrix}} = \frac{z_3}{\begin{vmatrix} 1 & 1 \\ 1 & -2 \end{vmatrix}} = t$$

$$\frac{x_3}{3} = -\frac{y_3}{0} = \frac{z_3}{-3} = t$$

$$\frac{x_3}{1} = \frac{y_3}{0} = \frac{z_3}{-1} = t$$

$$\mathbf{x} = \begin{bmatrix} x_3 \\ y_3 \\ z_3 \end{bmatrix} = \begin{bmatrix} t \\ 0 \\ -t \end{bmatrix} = t \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} = t \mathbf{x}_3 \text{ where } \mathbf{x}_3 \text{ is an eigenvector corresponding to } \lambda = 8.$$

$$\mathbf{x}_3 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$$

$$\text{Length of eigenvector } \mathbf{x}_1 = \sqrt{1^2 + 1^2 + 1^2} = \sqrt{3}$$

Length of eigenvector $\mathbf{x}_2 = \sqrt{1^2 + (-2)^2 + 1^2} = \sqrt{6}$

Length of eigenvector $\mathbf{x}_3 = \sqrt{1^2 + 0^2 + (-1)^2} = \sqrt{2}$

The normalized eigenvectors are

$$\mathbf{x}_1 = \begin{bmatrix} \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \end{bmatrix}, \quad \mathbf{x}_2 = \begin{bmatrix} \frac{1}{\sqrt{6}} \\ -\frac{2}{\sqrt{6}} \\ \frac{1}{\sqrt{6}} \end{bmatrix}, \quad \mathbf{x}_3 = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ -\frac{1}{\sqrt{2}} \end{bmatrix}$$

$$P = \begin{bmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{3}} & -\frac{2}{\sqrt{6}} & 0 \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{2}} \end{bmatrix}$$

$$D = \begin{bmatrix} 6 & 0 & 0 \\ 0 & 12 & 0 \\ 0 & 0 & 6 \end{bmatrix} = P^T A P$$

$$\begin{aligned} A &= I A I \\ &= P P^T A P P^T \\ &= P (P^T A P) P^T \\ &= P D P^T \end{aligned}$$

$$\begin{aligned} &= \begin{bmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{3}} & -\frac{2}{\sqrt{6}} & 0 \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} 6 & 0 & 0 \\ 0 & 12 & 0 \\ 0 & 0 & 6 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{3}} & -\frac{2}{\sqrt{6}} & 0 \\ \frac{1}{\sqrt{2}} & 0 & -\frac{1}{\sqrt{2}} \end{bmatrix} \\ &= \begin{bmatrix} 7 & -2 & 1 \\ -2 & 10 & -2 \\ 1 & -2 & 7 \end{bmatrix} \end{aligned}$$

EXERCISE 1.9

1. Show that the following matrices are not similar to diagonal matrices.

$$(i) \begin{bmatrix} 2 & 3 & 4 \\ 0 & 2 & -1 \\ 0 & 0 & 1 \end{bmatrix}$$

$$(ii) \begin{bmatrix} 2 & -1 & 1 \\ 2 & 2 & -1 \\ 1 & 2 & -1 \end{bmatrix}$$

$$(iii) \begin{bmatrix} 1 & 2 & 3 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

$$(iv) \begin{bmatrix} 3 & 10 & 5 \\ -2 & -3 & -4 \\ 3 & 5 & 7 \end{bmatrix}$$

2. Determine diagonal matrices orthogonally similar to the following real symmetric matrices. Also, find the modal matrix in each case.

$$(i) \begin{bmatrix} 7 & 4 & -4 \\ 4 & -8 & -1 \\ -4 & -1 & 8 \end{bmatrix}$$

$$(ii) \begin{bmatrix} 7 & 0 & -2 \\ 0 & 5 & -2 \\ -2 & -2 & 6 \end{bmatrix}$$

$$\left[\begin{array}{l} \text{Ans. :} \\ (i) D = \begin{bmatrix} 9 & 0 & 0 \\ 0 & -9 & 0 \\ 0 & 0 & -9 \end{bmatrix}, P = \begin{bmatrix} \frac{4}{\sqrt{8}} & 0 & \frac{1}{3} \\ \frac{1}{\sqrt{18}} & \frac{1}{\sqrt{2}} & -\frac{2}{3} \\ -\frac{1}{\sqrt{18}} & \frac{1}{\sqrt{2}} & \frac{2}{3} \end{bmatrix} \\ (ii) D = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 9 \end{bmatrix}, P = \begin{bmatrix} \frac{1}{3} & \frac{2}{3} & \frac{2}{3} \\ \frac{2}{3} & -\frac{2}{3} & \frac{1}{3} \\ \frac{2}{3} & \frac{1}{3} & -\frac{2}{3} \end{bmatrix} \end{array} \right]$$

3. Find A^{11} , where $A = \begin{bmatrix} -1 & 7 & -1 \\ 0 & 1 & 0 \\ 0 & 15 & -2 \end{bmatrix}$

$$\left[\text{Ans. : } \begin{bmatrix} -1 & 10237 & -2047 \\ 0 & 1 & 0 \\ 0 & 10245 & -2048 \end{bmatrix} \right]$$

Multiple Choice Questions

Select the most appropriate response out of the various alternatives given in each of the following questions:

1. If A is an orthogonal matrix, then A^{-1} equals
 (a) A (b) A^T (c) A^2 (d) none of these
2. If A is a symmetric matrix and $n \in \mathbb{N}$, then A^n is
 (a) symmetric (b) skew symmetric
 (c) diagonal matrix (d) none of these
3. If $A = \begin{bmatrix} 1 & -5 & 7 \\ 0 & 7 & 9 \\ 11 & 8 & 9 \end{bmatrix}$, then trace of the matrix A is
 (a) 17 (b) 25 (c) 3 (d) 12
4. All the four entries of the 2×2 matrix $P = \begin{bmatrix} P_{11} & P_{12} \\ P_{21} & P_{22} \end{bmatrix}$ are non-zero, and one of its eigenvalues is zero. Which of the following statements are true?
 (a) $P_{11}P_{22} - P_{12}P_{21} = 1$ (b) $P_{11}P_{22} - P_{12}P_{21} = -1$
 (c) $P_{11}P_{22} - P_{12}P_{21} = 0$ (d) $P_{11}P_{22} + P_{12}P_{21} = 0$
5. The system of linear equations $4x + 2y = 7$, $2x + y = 6$ has
 (a) a unique solution
 (b) no solution
 (c) an infinite number of solution
 (d) exactly two distinct solutions
6. The rank of the matrix $\begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$ is
 (a) 0 (b) 1 (c) 2 (d) 3
7. The eigenvalues and the corresponding eigenvectors of a 2×2 matrix are given by

eigenvalue	eigenvector
$\lambda_1 = 8$	$x_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$
$\lambda_2 = 4$	$x_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

The matrix is

 (a) $\begin{bmatrix} 6 & 2 \\ 2 & 6 \end{bmatrix}$ (b) $\begin{bmatrix} 4 & 6 \\ 6 & 4 \end{bmatrix}$ (c) $\begin{bmatrix} 2 & 4 \\ 4 & 2 \end{bmatrix}$ (d) $\begin{bmatrix} 4 & 8 \\ 8 & 4 \end{bmatrix}$

8. Let $A = \begin{bmatrix} 2 & -0.1 \\ 0 & 3 \end{bmatrix}$ and $A^{-1} = \begin{bmatrix} \frac{1}{2} & a \\ 0 & b \end{bmatrix}$. Then $(a + b)$ is

- (a) $\frac{7}{20}$ (b) $\frac{3}{20}$ (c) $\frac{19}{60}$ (d) $\frac{11}{20}$

9. Given an orthogonal matrix $A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix}$, $[AA^{-1}]$ is

- (a) $\begin{bmatrix} \frac{1}{4} & 0 & 0 & 0 \\ 0 & \frac{1}{4} & 0 & 0 \\ 0 & 0 & \frac{1}{2} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ (b) $\begin{bmatrix} \frac{1}{2} & 0 & 0 & 0 \\ 0 & \frac{1}{2} & 0 & 0 \\ 0 & 0 & \frac{1}{2} & 0 \\ 0 & 0 & 0 & \frac{1}{2} \end{bmatrix}$
- (c) $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ (d) $\begin{bmatrix} \frac{1}{4} & 0 & 0 & 0 \\ 0 & \frac{1}{4} & 0 & 0 \\ 0 & 0 & \frac{1}{4} & 0 \\ 0 & 0 & 0 & \frac{1}{4} \end{bmatrix}$

10. The characteristic equation of a 3×3 matrix P is defined as $\alpha(\lambda) = |\lambda I - P| = \lambda^3 + \lambda^2 + 2\lambda + 1 = 0$. If I denotes identity matrix, then the inverse of matrix P will be

- (a) $P^2 + P + 2I$ (b) $P^2 + P + I$
(c) $-(P^2 + P + I)$ (d) $-(P^2 + P + 2I)$

11. For the matrix $P = \begin{bmatrix} 3 & -2 & 2 \\ 0 & -2 & 1 \\ 0 & 0 & 1 \end{bmatrix}$, one of the eigenvalue is equal to -2 . Which of

the following is an eigenvector?

- (a) $\begin{bmatrix} 3 \\ -2 \\ 1 \end{bmatrix}$ (b) $\begin{bmatrix} -3 \\ 2 \\ -1 \end{bmatrix}$ (c) $\begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix}$ (d) $\begin{bmatrix} 2 \\ 5 \\ 0 \end{bmatrix}$

12. Consider the following system of equations in three real variables x_1 , x_2 and x_3 .

$$2x_1 - x_2 + 3x_3 = 1$$

$$3x_1 + 2x_2 + 5x_3 = 2$$

$$-x_1 + 4x_2 + x_3 = 3$$

The system equation has

- (a) no solution
 (b) a unique solution
 (c) more than one but a finite number of solutions
 (d) an infinite number of solutions
13. Eigenvalues of a matrix $S = \begin{bmatrix} 3 & 2 \\ 2 & 3 \end{bmatrix}$ are 5 and 1. What are the eigenvalues of the

matrix S^2 ?

- (a) 1 and 25 (b) 6 and 4 (c) 5 and 1 (d) 2 and 10
14. Match the items in columns I and II
- | I | II |
|---------------------------|-----------------------------------|
| (P) Singular matrix | (1) Determinant is not defined |
| (Q) Non-square matrix | (2) Determinant is always ± 1 |
| (R) Real symmetric matrix | (3) Determinant is zero |
| (S) Orthogonal matrix | (4) Eigenvalues are always real |
| | (5) Eigenvalues are not defined |
| (a) P-3, Q-1, R-4, S-2 | (b) P-2, Q-3, R-4, S-1 |
| (c) P-3, Q-2, R-5, S-4 | (d) P-3, Q-4, R-2, S-1 |

15. A is a 3×4 real matrix and $Ax = b$ is an inconsistent system of equations. The highest possible rank of A is

- (a) 1 (b) 2 (c) 3 (d) 4

16. The sum of eigenvalues of the matrix $\begin{bmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{bmatrix}$ is

- (a) 5 (b) 7 (c) 9 (d) 18

17. Let A and B be real symmetric matrices of size $n \times n$. Then which one of the following is true?

- (a) $AA' = I$ (b) $A = A^{-1}$ (c) $AB = BA$ (d) $(AB)' = BA$

18. Among the following, the pair of vectors orthogonal to each other is

- (a) $[3, 4, 7]$, $[3, 4, 7]$ (b) $[1, 0, 0]$, $[1, 1, 0]$
 (c) $[1, 0, 2]$, $[0, 5, 0]$ (d) $[1, 1, 1]$, $[-1, -1, -1]$

19. The eigenvalues of the matrix $\begin{bmatrix} 2 & -1 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & -2 & 0 \\ 0 & 0 & 1 & 4 \end{bmatrix}$ are

- (a) 2, -2, 1, -1 (b) 2, 3, -2, 4
(c) 2, 3, 1, 4 (d) none of these

20. The eigenvalues of $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$ are

- (a) 0, 0, 0 (b) 0, 0, 1 (c) 0, 0, 3 (d) 1, 1, 1

21. The minimum and maximum eigenvalues of the matrix $\begin{bmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{bmatrix}$

are -2 and 6 respectively. What is the other eigenvalue?

- (a) 5 (b) 3 (c) 1 (d) -1

22. The inverse of the matrix $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$ is

(a) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{2} & 0 \\ 0 & 0 & \frac{1}{3} \end{bmatrix}$

(b) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{3} & 0 \\ 0 & 0 & \frac{1}{2} \end{bmatrix}$

(c) $\begin{bmatrix} \frac{1}{2} & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & \frac{1}{3} \end{bmatrix}$

(d) none of these

23. Are the following vectors linearly dependent?

$X_1 = [3, 2, 7]$, $X_2 = [2, 4, 1]$ and $X_3 = [1, -2, 6]$

- (a) Dependent (b) Independent
(c) Cannot say (d) None of these

24. If $A = \begin{bmatrix} 1 & 0 \\ 1 & 7 \end{bmatrix}$, then the value of k for which $A^2 = 8A + kI$ is
 (a) 5 (b) -5 (c) 7 (d) -7
25. $A = \begin{bmatrix} a+ic & -b+id \\ b+id & a-ic \end{bmatrix}$ is unitary matrix if and only if
 (a) $a^2 + b^2 + c^2 = 0$ (b) $b^2 + c^2 + d^2 = 0$
 (c) $a^2 + b^2 + c^2 + d^2 = 1$ (d) $a^2 + b^2 + c^2 + d^2 = 0$
26. Which of the following matrices is not diagonalizable?
 (a) $\begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & 0 \\ 3 & 2 \end{bmatrix}$ (c) $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ (d) $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$
27. The sum of the eigenvalues of the matrix $\begin{bmatrix} 3 & 4 \\ x & 1 \end{bmatrix}$ for real and negative value of x is
 (a) greater than zero (b) less than zero
 (c) zero (d) dependent values of x .
28. A is a singular matrix of order 3 with eigenvalues 2 and 3. The third eigenvalue is
 (a) 1 (b) 0 (c) 4 (d) -1
29. The eigenvalues and the corresponding eigenvectors of a 2×2 matrix are given by
 $\lambda_1 = 8 \quad \mathbf{x}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \lambda_2 = 4 \quad \mathbf{x}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$
 The matrix is
 (a) $\begin{bmatrix} 6 & 2 \\ 2 & 6 \end{bmatrix}$ (b) $\begin{bmatrix} 4 & 6 \\ 6 & 4 \end{bmatrix}$ (c) $\begin{bmatrix} 2 & 4 \\ 4 & 2 \end{bmatrix}$ (d) $\begin{bmatrix} 4 & 8 \\ 8 & 4 \end{bmatrix}$

Answers

1. (b) 2. (a) 3. (a) 4. (c) 5. (b) 6. (c) 7. (a) 8. (a)
 9. (c) 10. (d) 11. (d) 12. (b) 13. (a) 14. (a) 15. (b) 16. (b)
 17. (d) 18. (c) 19. (b) 20. (b) 21. (b) 22. (a) 23. (a) 24. (d)
 25. (c) 26. (d) 27. (a) 28. (b) 29. (a)

Index

A

Absolute Convergence of a Series 5.94
Alternating Series 5.87
Area as Double Integral 9.141
Area in Cartesian Coordinates 9.141
Area in Polar Coordinates 9.154
Area of Surface of Solid of Revolution 4.46
Area of Surface of Solid of Revolution in Cartesian Form 4.46
Area of Surface of Solid of Revolution in Parametric Form 4.53
Area of Surface of Solid of Revolution in Polar Form 4.60

B

Beta Function 3.11
Beta Function as Improper Integral 3.28
Bounded Sequence 5.3

C

Cauchy's Integral Test 5.82
Cauchy's Root Test 5.73
Cayley-Hamilton Theorem 10.108
Chain Rule 8.59
Change of Order of Integration 9.31
Change of Variables from Cartesian to Other Coordinates 9.95
Change of Variables from Cartesian to Polar Coordinates 9.77
Characteristic Equation 10.65
Characteristic Matrix 10.64
Characteristic Polynomial 10.65
Coefficient Matrix 10.23

Cofactor of an Element of a Determinant 10.6
Column Matrix 10.2
Combining Series 5.18
Comparison Test 5.20
Composite Function of One Variable 8.59
Composite Function of Two Variables 8.66
Conditional Convergence of a Series 5.94
Conjugate of a Matrix 10.4
Continuous Function Theorem for Sequences 5.2
Convergence and Divergence of Improper Integrals 2.17
Convergence, Divergence and Oscillation of Infinite Series 5.9

D

Determinant of a Matrix 10.5
Diagonal Matrix 10.3
Diagonalization of a Matrix 10.119
Direct Comparison Test 2.17
Directional Derivative 8.103
Double Integrals Over Rectangles 9.1 and General Regions 9.2
Double Integrals as Volume 9.3
Double Integrals in Polar Coordinates 9.66
Duplication Formula 3.13
D'Alembert's Ratio Test 5.40

E

Echelon Form of a Matrix 10.7

Eigenspace 10.65
 Eigenvalues and Eigenvectors 10.64
 Elementary Matrices 10.7
 Elementary Transformations 10.6
 Equivalence of Matrices 10.7
 Euler's formulae 7.8
 Extreme Values with Constrained Variables 8.134

F

Fourier Coefficients 7.8
 Fourier Series 7.5
 Fourier Series of Even and Odd Functions 7.66
 Fourier Series of Functions of Any Period 7.7
 Functions of Two or More Variables 8.2

G

Gamma Function 3.1
 Gauss Elimination Method 10.24
 Gauss–Jordan Method 10.24
 Geometric Series 5.10
 Gradient 8.103

H

Half-Range Cosine Series 7.93
 Half-Range Fourier Series 7.93
 Half-Range Sine Series 7.93
 Harmonic Analysis 7.1
 Harmonic Series 5.19
 Hermitian Matrix 10.5
 Higher-Order Partial Derivatives 8.11

I

Implicit Differentiation 8.94
 Improper Integral of the Third Kind 2.16
 Improper Integrals 2.1
 Improper Integrals of the First Kind 2.2
 Improper Integrals of the Second Kind 2.9
 Infinite Series 5.8
 Infinite Solutions 10.49
 Interval and Radius of

Convergence 5.101
 Inverse of a Matrix 10.6
 Inverse of a Matrix by Gauss–Jordan Method 10.18

L

Leibnitz's Test for Alternating Series 5.87
 Length of Arc in Cartesian Form 4.6
 Length of Arc in Parametric Form 4.20
 Length of Arc in Polar Form 4.33
 Length of Plane Curves 4.6
 Limit Comparison Test 2.18, 2.23
 Limit and Continuity of Functions of Several Variables 8.2
 Limit of a Sequence 5.2
 Linear Dependence and Independence of Eigenvectors 10.76
 Local Extreme Values (Maximum and Minimum Values) 8.116
 Lower Triangular Matrix 10.3
 L'Hospital's Rule 1.1

M

Maclaurin's Series 6.27
 Matrix 10.2
 Method of Lagrange Multipliers 8.145
 Minor of an Element of a Determinant 10.6
 Mixed Derivative Theorem 8.11
 Monotonic Sequence 5.3
 Multiple Integrals by Substitution 9.77

N

Non-Trivial Solution 10.49
 n^{th} Term Test for Divergence 5.9

O

Orthogonal Matrix 10.5
 Orthogonality of Functions 7.2
 Orthogonality of Trigonometric Functions 7.2
 Orthogonally Similar Matrices 10.119

P

p -Series 5.20
 Partial Derivatives 8.10
 Periodic Functions 7.1
 Power Series 5.101
 Properties of Beta Functions 3.12
 Properties of Double Integrals 9.3
 Properties of Eigenvalues 10.65
 Properties of Eigenvectors 10.76
 Properties of Even and Odd Functions 7.66
 Properties of Gamma Function 3.2
 Properties of Infinite Series 5.9

R

Rank of a Matrix 10.13
 Raabe's Test (Higher Ratio Test) 5.67
 Rank of a Matrix by Echelon Form 10.14
 Relation between Beta and Gamma Functions 3.12
 Row Matrix 10.2

S

Sandwich Theorem for Sequences 5.7
 Scalar Matrix 10.3
 Sequence 5.2
 Similarity Transformation 10.119
 Singular and Non-Singular Matrices 10.6
 Skew-Hermitian Matrix 10.5
 Skew-Symmetric Matrix 10.4
 Solutions of System of Linear Equations 10.23
 Solutions of a System of Linear Equations 10.49
 Spectrum 10.65
 Square Matrix 10.2
 Symmetric Matrix 10.4

System of Homogeneous Linear Equations 10.48
 System of Non-Homogeneous Linear Equations 10.22

T

Tangent Plane and Normal to a Surface 8.107
 Taylor's Series 6.1
 Telescoping Series 5.15
 Test for Non-existence of a Limit 8.2
 Total Derivatives 8.59
 Trace of a Matrix 10.4
 Transpose of a Matrix 10.4
 Transposed Conjugate of a Matrix 10.5
 Trigonometric Form of Beta Function 3.11
 Trigonometric Fourier Series 7.6
 Triple Integrals 9.109
 Triple Integrals in Cartesian Coordinates 9.109
 Triple Integrals in Cylindrical Coordinates 9.109
 Triple Integrals in Spherical Coordinates 9.110
 Trivial Solution 10.49

U

Unit or Identity Matrix 10.3
 Unitary Matrix 10.5
 Upper Triangular Matrix 10.3

V

Volume Using Cross-sections 4.1

Z

Zero or Null Matrix 10.2

